# A Deep BSDE Approach to Time-Consistent Mean-Variance Portfolio Optimisation with Transaction Costs

Jack Adams
MSc Financial Mathematics
MATH5312M: Dissertation in Financial Mathematics

Word count: 9997

Supervised by Dr Khoa Lê

September 2025

# Abstract

The non-additivity of variance poses a major challenge for mean–variance optimisation. Basak and Chabakauri (2010) derived a time-consistent allocation strategy in continuous time, but their frictionless framework lacks practical relevance. Once frictions such as transaction costs are introduced, the problem becomes nonlinear and analytically intractable, motivating numerical methods. We extend the deep BSDE methodology of Han et al. (2018); Beck et al. (2019); Huré et al. (2020) to multi-asset mean–variance optimisation with transaction costs, learning optimal time-consistent strategies in continuous time. In historical backtests, the approach outperforms both the frictionless benchmark and a passive buy-and-hold strategy, achieving a Sharpe ratio of 1.47 with statistically significant improvements in risk-adjusted performance. Robustness is confirmed through parametric bootstrap analysis, which shows consistent Sharpe gains across simulated market paths.

# Intellectual Property

The candidate confirms that the work submitted is his/her own and that appropriate credit has been given where reference has been made to the work of others.



Signed ____________________

# Acknowledgements

I would like to thank my supervisor, Dr Khoa Lê, for his guidance and support throughout this project. I am also grateful to the School of Mathematics at the University of Leeds for providing a stimulating environment during my studies, and to my friends, colleagues, and family for their encouragement, patience, and constant support.

# Contents

# List of Figures

# List of Tables

# Chapter 1

# Introduction

The foundations of portfolio theory were laid by Markowitz (1952), who introduced mean–variance optimisation to balance expected return against risk. Early extensions, such as adding a risk-free asset (Tobin, 1958), enriched the framework but retained the limitation of a single-period horizon. This was later addressed by multi-period discrete-time models (Hakansson, 1971; Li and Ng, 2000) and by continuous-time counterparts (Merton, 1972; Basak and Chabakauri, 2010), which better reflect the stochastic nature of asset prices.

A central challenge in the literature is the non-additivity of variance through time, which complicates dynamic programming. Li and Ng (2000) reformulated the mean–variance objective into an auxiliary quadratic problem with an analytical solution, but this was pre-committed: the optimal policy is fixed at time zero and not optimal at later times. Investors, however, require time-consistent strategies that adapt to changing conditions and remain optimal throughout the investment horizon. Basak and Chabakauri (2010) achieved the first time-consistent continuous-time solution via a hedge-neutral probability measure, but in a frictionless market. Once transaction costs are introduced, the Hamilton–Jacobi–Bellman (HJB) equation becomes nonlinear and analytically intractable.

This motivates the use of numerical methods. Transaction costs are a key friction in practice, as they can substantially erode profits, yet their inclusion renders closed-form optimisation infeasible. Deep learning methods, particularly the deep BSDE approach of Han et al. (2018), exploit the link between partial differential equations (PDEs) and backward stochastic differential equations (BSDEs). Neural networks approximate the gradients needed to solve the HJB, while Huré et al. (2020) provide convergence guarantees for this loss-based approximation.

To our knowledge, this thesis is the first to apply deep BSDEs to continuous-time mean–variance optimisation and to extend them to incorporate transaction costs. Existing time-consistent approaches, such as Basak and Chabakauri (2010), apply only in frictionless markets. Grid-based numerical methods offer further insight but are computationally expensive and scale poorly with dimension. By contrast, deep BSDE methods (Han et al., 2018; Beck et al., 2019; Huré et al., 2020) scale efficiently to higher dimensions and naturally accommodate frictions.

The contributions of this thesis are threefold. First, the continuous-time mean–variance framework is extended to include transaction costs. Second, the deep BSDE scheme of Huré et al. (2020) is adapted to the forward–backward SDE system associated with this formulation, further extended by inclusion of neural networks to approximate the value function curvature. Third, the framework is evaluated on real financial data with historical backtests and bootstrap analysis. In doing so, this thesis provides the first scalable, time-consistent solution to mean–variance portfolio optimisation under transaction costs in continuous time.

# Chapter 2

# Literature Review

In this chapter, we review the key literature on mean–variance portfolio optimisation and examine recent advances, including deep BSDE methods. These modern approaches provide flexible tools for solving stochastic control problems under market frictions, such as transaction costs, where classical continuous-time techniques often become intractable.

## 2.1 Historical Background and Single-Period Mean-Variance Optimisation

Markowitz (1952) shifted portfolio choice from maximising expected return to a trade-off between return and risk, measured by variance, yielding the efficient frontier of portfolios that maximise return for a given level of risk. His key insight was that diversification depends not only on the number of assets but also on their covariances, making portfolio construction a problem of joint risk allocation.

The efficient frontier is the upper boundary of feasible risky portfolios, with points below suboptimal. Figure 2.1 illustrates this trade-off between risk and return.

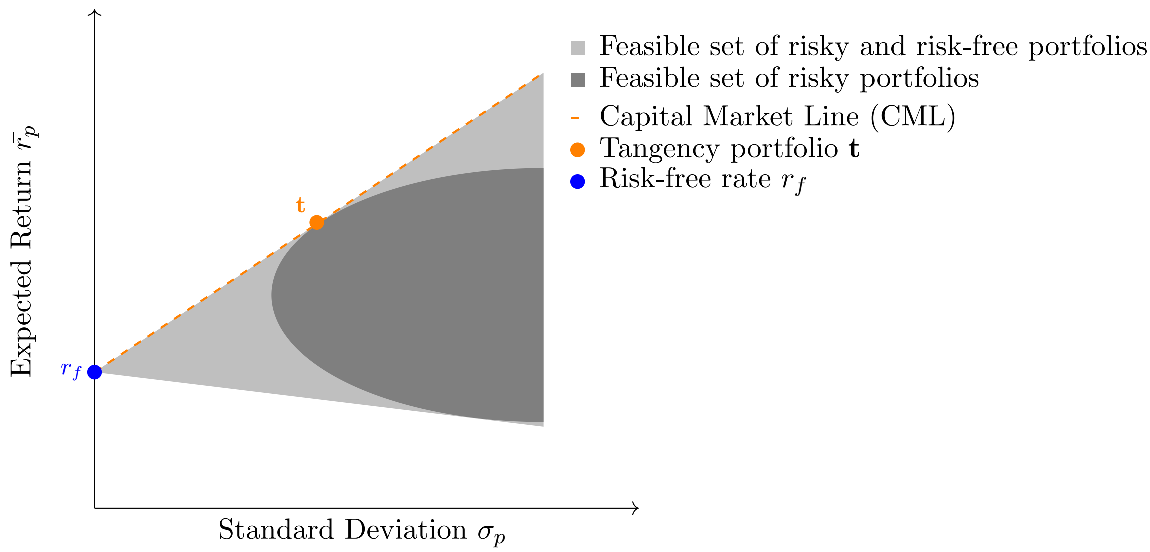



Figure 2.1: Efficient frontier with risk-free asset and tangency portfolio.

Markowitz's framework assumes a single period, known expected returns and covariances, and frictionless markets. Risk is represented solely by variance, which treats gains and losses symmetrically. We retain this assumption for consistency with the classical framework, but revisit its limitations later.

Tobin (1958) extended the model by adding a risk-free asset, leading to the one-fund separation theorem. All investors hold the same tangency portfolio of risky assets and adjust between it and the risk-free asset according to risk tolerance. This produces the Capital Market Line (CML) in Figure 2.1, with the tangency portfolio maximising the Sharpe ratio. The result formalised a two-step process: first select the optimal risky portfolio, then mix it with the risk-free asset.

While elegant, Tobin's extension inherits Markowitz's limitations: frictionless markets, equal borrowing and lending rates, and homogeneous expectations. These assumptions imply all investors hold the same tangency portfolio, which is unrealistic given differences in beliefs and constraints. Later work relaxes these assumptions through heterogeneous beliefs and state-dependent policies. Our framework builds on this evolution by modelling dynamic decision-making under frictions.

In summary, the single-period models of Markowitz (1952) and Tobin (1958) founded modern portfolio theory but are constrained by their static, frictionless setting. Their limitations motivate the shift to multi-period and continuous-time frameworks, which allow dynamic rebalancing and a richer treatment of real-world investment problems.

## 2.2 Multi-Period Mean-Variance Optimisation

While single-period models capture only a static trade-off, multi-period frameworks allow dynamic rebalancing and adaptation, making them more realistic for long-horizon investors.

Samuelson (1969) showed that under CRRA utility and i.i.d. returns, the myopic single-period strategy remains optimal when repeated each period. By contrast, Hakansson (1971) demonstrated that in mean–variance settings such repetition is generally suboptimal, because variance is not additive: wealth dynamics introduce intertemporal covariances that reshape the risk–return trade-off. Optimal strategies must therefore depend on horizon length and evolving distributions, which can shift the efficient frontier and tangency portfolio over time.

Hakansson used dynamic programming to characterise such policies, but the approach was analytically intractable and relied on precommitment: the plan is fixed at $t = 0$ with no revision as information unfolds, creating time inconsistency.

A major breakthrough came with Li and Ng (2000), who reformulated the discrete-time multi-period mean–variance problem as an auxiliary quadratic problem. This recasting resolves the non-separability of variance and yields closed-form (analytical) precommitment policies. The resulting feedback control is affine in wealth and delivers a dynamic one-fund theorem: under homogeneous expectations, all investors combine a common tangency portfolio with the risk-free asset, scaled by risk aversion and current wealth. This formulation remains the canonical benchmark in discrete time.

Subsequent work incorporated constraints such as short-sale limits, transaction costs, and return predictability (e.g. Li et al., 2009; Cui et al., 2012), improving realism but at the cost of tractability. All these models remain precommitment-based. The deeper issue of time inconsistency — designing equilibrium, time-consistent mean–variance policies — has been addressed more fruitfully in continuous time, the focus of the next section.

## 2.3 Continuous-Time Mean-Variance Optimisation

Discrete-time models provide important insights but face challenges in tractability and realism, since financial markets evolve continuously. Continuous-time formulations address these issues, offering a mathematically elegant framework for dynamic trading. Early models are precommitment-based, while later work develops time-consistent strategies.

Merton (1969) provides the first continuous-time formulation of portfolio choice, modelling utility maximisation as a stochastic control problem. The investor's value function satisfies a HJB equation which, under logarithmic and exponential utility, admits closed-form portfolio rules. Although not expressed in mean–variance terms, this work introduced the stochastic control framework that underpins much of the later literature.

Merton (1972) derives the continuous-time mean–variance frontier. He shows that efficient portfolios can be constructed from two mutual funds, providing an analogue of the mutual fund theorem. His open-loop solutions determine portfolio weights at inception as functions of parameters and horizon, but do not adapt to realised wealth paths. The strategy is therefore time-inconsistent. Despite assuming frictionless markets and constant parameters, this work established the foundation for continuous-time mean–variance analysis.

Zhou and Li (2000) extend their earlier discrete-time approach (Li and Ng, 2000) by formulating mean–variance optimisation as a stochastic linear–quadratic control problem. This yields a closed-form feedback law: optimal weights are functions of wealth and time, unlike Merton's open-loop solutions. The feedback form allows dynamic adjustment as wealth evolves, while retaining closed-form tractability. However, because optimisation is fixed at $t = 0$ and not re-optimised as preferences update, it remains a precommitment (time-inconsistent) strategy. Their contribution delivers the standard continuous-time precommitment benchmark, paving the way for time-consistent alternatives.

A time-consistent solution is developed by Basak and Chabakauri (2010), who reformulate the problem using a hedge-neutral measure. The resulting equilibrium policy separates into two components: a myopic demand from the immediate risk–return trade-off, and a hedging demand capturing intertemporal effects of stochastic state variables. Their framework accommodates multiple risky assets and state variables, with closed-form policies even under stochastic volatility, and provides the leading analytical solution to time-consistent mean–variance optimisation in continuous time.

Despite these advances, such models retain simplifying assumptions, excluding frictions such as transaction costs and portfolio constraints. These limitations motivate the use of numerical methods, notably deep BSDE approaches, which overcome these restrictions and form the central focus of this thesis.

## 2.4 Deep Learning and the Deep BSDE Approach

Continuous-time formulations overcame the limitations of discrete-time models; deep BSDE methods similarly address the limitations of classical PDE approaches. Traditional methods for solving HJB equations (see Pham, 2009) — including analytical solutions, finite-difference schemes, and tree methods — quickly become intractable in high dimensions or when frictions such as transaction costs are introduced, reflecting the curse of dimensionality (Bellman, 1957). Neural networks, by contrast, can approximate high-dimensional value functions and control policies with greater flexibility.

To exploit this flexibility, the problem must first be reformulated. The nonlinear Feynman–Kac formula provides the key link, connecting semi-linear PDEs with BSDEs. This representation recasts control problems as forward–backward SDE systems, well suited to simulation-based neural network approximation.

Han et al. (2018) pioneered the deep BSDE method, training neural networks to approximate the process $Z_t$, which represents the gradient of the value function with respect to stochastic factors. By simulating Monte Carlo paths and minimising a terminal loss, they solved nonlinear PDEs in up to 100 dimensions — a practical breakthrough against the curse of dimensionality.

Beck et al. (2019) extended this to fully nonlinear PDEs with the deep 2BSDE method, introducing the second-order process $\Gamma_t$ (the Hessian of the value function). Neural networks approximate the triple $(Y_t, Z_t, \Gamma_t)$, representing the value, gradient, and Hessian. This makes the method relevant for portfolio problems with nonlinearities such as transaction costs, but it suffers from training instability and lacks rigorous convergence guarantees.

Huré et al. (2020) addressed these issues with deep backward schemes, combining backward induction with neural network approximations of the gradient. The value function is learned implicitly through the backward recursion, with variance reduction and tailored loss functions yielding stability and theoretical convergence guarantees. Although developed in a first-order setting, the methodology extends naturally to include the Hessian, required for fully nonlinear problems. This extension underpins the approach adopted in this thesis.

Alternative deep learning approaches have also been proposed. Reinforcement learning methods (e.g. Wu and Li, 2024) and neural network policy approximations (e.g. van Staden et al., 2024; Huh et al., 2025) provide computationally efficient solutions to portfolio optimisation problems without relying on grid-based PDE solvers. However, these frameworks typically address discrete-time or precommitment formulations, and do not target time-consistent mean–variance equilibria in continuous time. This distinction sets apart the deep BSDE methodology, which directly approximates equilibrium solutions of the associated forward–backward SDE system.

So far, applications in finance have focused on nonlinear option pricing (Han et al., 2018) and constrained portfolio problems such as optimal liquidation (Germain et al., 2021). Building on this foundation, we extend the backward schemes of Huré et al. (2020) to the second-order case and apply them to continuous-time mean–variance portfolio optimisation. In particular, we address the unexplored setting of multiple risky assets with nonlinear transaction costs. This constitutes the central contribution of the thesis, showing how modern deep learning frameworks can tackle mean–variance optimisation under realistic market frictions.

## 2.5 Modelling Transaction Costs in Portfolio Optimisation

Transaction costs are a fundamental market feature that materially affect optimal trading and realised performance. Their inclusion, however, often destroys the tractability of classical portfolio models, motivating simulation-based methods such as deep BSDE.

One strand of the literature models proportional costs. Constantinides (1986) showed that even small proportional costs create a "no-trade region": investors tolerate deviations from targets and rebalance only when misalignment grows large. Davis and Norman (1990) gave a continuous-time treatment via impulse control, where discrete costly trades return the portfolio to the boundary of the no-trade region. This paradigm reflects real frictions, but produces non-smooth value functions and free-boundary problems that are analytically demanding and difficult to reconcile with gradient-based neural networks.

A second strand models costs as smooth and convex, most commonly quadratic. Quadratic frictions were popularised in optimal execution models such as Almgren and Chriss (2001), where temporary price impact rises with the square of trading rate. In dynamic portfolio settings, Gârleanu and Pedersen (2013) embed quadratic costs in a linear–quadratic framework, deriving explicit policies that continuously adjust toward a forward-looking "aim portfolio." Unlike proportional costs, quadratic costs do not generate a no-trade region; instead they penalise rapid rebalancing and produce smooth turnover, consistent with the differentiability requirements for neural network training.

Quadratic frictions typically penalise the trading rate (time derivative of allocations). To ensure this is well defined, strategies are assumed absolutely continuous, so allocations admit a trading-rate process. The trading rate then becomes the control variable of the optimisation problem. This integrates naturally with stochastic-control frameworks, as in Gonon et al. (2020), who also interpolate between proportional and quadratic specifications. In practice, however, controlling rates directly can create scaling and stability issues; in our implementation we approximate them via finite differences of allocations (see Section 3.2).

An intermediate specification is the smoothed proportional (Huber-type) penalty: quadratic for small trades, effectively linear for large ones. This retains differentiability while easing penalties on large trades, but introduces a threshold parameter that complicates calibration (Huber, 1964; Boyd and Vandenberghe, 2004). For these reasons we do not adopt it as our baseline, though we return to it in the limitations as a natural extension.

In this thesis we adopt quadratic costs. They promote smooth, realistic trading paths and their convex, differentiable form is well suited to gradient-based neural network training. This avoids the free-boundary complications of proportional costs while remaining straightforward to calibrate and extend.

# Chapter 3

# Mathematical Formulation

In this chapter, we present the mathematical formulation of the models that form the basis of our numerical implementation and backtesting analysis. Our primary focus is on two continuous-time, time-consistent formulations: the closed-form solution of Basak and Chabakauri (2010) and our novel extension incorporating transaction costs, solved via deep BSDE methods. Continuous-time, time-consistent models are prioritised due to their greater real-world relevance and superior performance relative to precommitment strategies.

For completeness, we note that earlier paradigms such as the single-period mean–variance model are already well covered in the literature and are not repeated here. By contrast, the discrete-time multi-period model of Li and Ng (2000) is included in Appendix A with full mathematical formulation, numerical implementation, and results. This provides interested readers with a direct comparison to the continuous-time approach while keeping the main text focused on the models most relevant to our contribution.

## 3.1 Continuous-Time Mean-Variance Optimisation without Transaction Costs (Analytical Approach)

In this section, we follow the analytical framework of Basak and Chabakauri (2010) to derive the time-consistent optimal portfolio policy in the absence of transaction costs, referred to here as the cost-unaware model. We first present the formulation for a single risky asset and a risk-free asset, before extending it to the multi-risky-asset case used in our implementation.

### 3.1.1 Economic Setup

We consider a continuous-time Markovian economy over a finite horizon $[0, T]$, modelled on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$, where $\mathbb{P}$ is the physical (real-world) measure. The filtration is generated by two correlated Brownian motions $\{w_t\}_{t\geq 0}$ and $\{w_{X,t}\}_{t\geq 0}$ with $\rho = \mathrm{corr}(w_t, w_{X,t})\ \forall t \geq 0$.

The investor trades continuously in a risk-free bond accruing at constant rate $r$ and a risky stock with price $S_t$. The stock follows

$$\frac{dS_t}{S_t} = \mu(S_t, X_t, t)\, dt + \sigma(S_t, X_t, t)\, dw_t, \tag{3.1}$$

and a state variable $X_t$, representing a latent factor influencing future returns, evolves as

$$dX_t = m(X_t, t)\, dt + \nu(X_t, t)\, dw_{X,t}. \tag{3.2}$$

For shorthand, we write $\mu_t := \mu(S_t, X_t, t)$, $\sigma_t := \sigma(S_t, X_t, t)$, $m_t := m(X_t, t)$, and $\nu_t := \nu(X_t, t)$.

Under standard regularity conditions, (3.1)–(3.2) admit a unique strong solution, and $(S_t, X_t)$ forms a Markov process. Since $X_t$ is untradable, the market is incomplete unless $\rho = \pm 1$, which induces a non-myopic hedging demand analysed subsequently.

Let $W_t$ denote the investor's wealth and $\theta_t$ the monetary allocation to the risky asset (the remainder $W_t - \theta_t$ is in the risk-free asset). The wealth dynamics follow the self-financing process

$$dW_t = \big[rW_t + \theta_t(\mu_t - r)\big]\,dt + \theta_t \sigma_t\,dw_t. \tag{3.3}$$

with no intermediate consumption.

The investor seeks to maximise a mean–variance objective over terminal wealth $W_T$:

$$\max_{\theta} \quad \mathbb{E}[W_T] - \frac{\gamma}{2}\,\mathrm{Var}[W_T], \tag{3.4}$$

where $\gamma > 0$ is the risk-aversion parameter.

### 3.1.2 Determination of the Optimal Dynamic Investment Policy

A key difficulty in dynamic mean–variance problems is the time–inconsistency of the variance term, which is non–separable over time and prevents the direct use of standard dynamic programming techniques. Following Basak and Chabakauri (2010), we overcome this by expressing the problem recursively, allowing the use of backward induction to obtain the time–consistent policy.

By the law of total variance (Weiss, 2005), terminal wealth satisfies:

$$\mathrm{Var}_t[W_T] = \mathbb{E}_t\left[\mathrm{Var}_{t+\tau}(W_T)\right] + \mathrm{Var}_t\left[\mathbb{E}_{t+\tau}(W_T)\right], \quad \tau > 0. \tag{3.5}$$

The first term is the expected future variance, while the second captures uncertainty in future expected wealth. This decomposition shows why decisions made at time $t$ differ from those re–evaluated later.

The investor's time–$t$ objective is:

$$U_t = \mathbb{E}_t[W_T] - \frac{\gamma}{2}\,\mathrm{Var}_t[W_T], \tag{3.6}$$

which, using (3.5), admits the recursive form

$$U_t = \mathbb{E}_t[U_{t+\tau}] - \frac{\gamma}{2}\,\mathrm{Var}_t\left[\mathbb{E}_{t+\tau}(W_T)\right]. \tag{3.7}$$

Here, the first term is the expected continuation value, while the second quantifies the incentive to deviate from future selves' preferences.

Define the value function under the optimal policy process $\{\theta_s^*\}_{s\in[t,T]}$ as

$$J(W_t, S_t, X_t, t) = \mathbb{E}_t[W_T^*] - \frac{\gamma}{2}\,\mathrm{Var}_t[W_T^*], \tag{3.8}$$

where $W_T^*$ denotes terminal wealth under this policy. Anticipating re–optimisation at $t+\tau$, the investor selects a policy over $[t, t+\tau]$ such that

$$J_t = \max_{\theta_s,\, s\in[t,t+\tau]} \mathbb{E}_t[J_{t+\tau}] - \frac{\gamma}{2}\,\mathrm{Var}_t\left[\mathbb{E}_{t+\tau}(W_T^*)\right], \tag{3.9}$$

with $J_t$ shorthand for $J(W_t, S_t, X_t, t)$.

For the forthcoming HJB derivation, define expected gains from investment as

$$f(W_t, S_t, X_t, t) \equiv \mathbb{E}_t[W_T^*] - W_t e^{r(T-t)}, \tag{3.10}$$

denoted $f_t$ henceforth. Substitution gives

$$
\begin{aligned}
J_t = \max_{\theta_s,\, s\in[t,t+\tau]} & \mathbb{E}_t[J_{t+\tau}] \\
& - \frac{\gamma}{2}\operatorname{Var}_t\left[f_{t+\tau} - f_t + W_{t+\tau}e^{r(T-t-\tau)} - W_t e^{r(T-t)}\right],
\end{aligned}
\tag{3.11}
$$

subject to the budget constraint in Equation (3.3) and terminal condition $J_T = W_T$.

Integrating (3.3) gives

$$
W_T^* = W_t e^{r(T-t)} + \int_t^T \theta_s^*(\mu_s - r)e^{r(T-s)}\,\mathrm{d}s + \int_t^T \theta_s^*\sigma_s e^{r(T-s)}\,\mathrm{d}w_s, \tag{3.12}
$$

so that

$$
\mathbb{E}_t[W_T^*] = W_t e^{r(T-t)} + \mathbb{E}_t\left[\int_t^T \theta_s^*(\mu_s - r)e^{r(T-s)}ds\right]. \tag{3.13}
$$

Hence

$$
f(W_t, S_t, X_t, t) = \mathbb{E}_t\left[\int_t^T \theta_s^*(\mu_s - r)e^{r(T-s)}ds\right]. \tag{3.14}
$$

Restating (3.11), we write

$$
0 = \max_{\theta_t}\mathbb{E}_t[dJ_t] - \frac{\gamma}{2}\operatorname{Var}_t\left[df_t + d\left(W_t e^{r(T-t)}\right)\right]. \tag{3.15}
$$

As in Basak and Chabakauri (2010) (see algebraic steps in Appendix C.1), the value function is separable in wealth:

$$
J(W_t, S_t, X_t, t) = W_t e^{r(T-t)} + \tilde{J}(S_t, X_t, t), \tag{3.16}
$$

where $\tilde{J}(S_t, X_t, t) \equiv \mathbb{E}_t[f_t] - \frac{\gamma}{2}\operatorname{Var}_t[f_t]$ is independent of wealth, implying that $\theta_t^*$ does not depend on $W_t$.

The conditional variance in (3.15) becomes

$$
\begin{aligned}
\operatorname{Var}_t\left[df_t + d(W_t e^{r(T-t)})\right] = & \left(\sigma_t S_t \frac{\partial f_t}{\partial S_t} + \theta_t \sigma_t e^{r(T-t)}\right)^2 \\
& + \left(\frac{\partial f_t}{\partial X_t}\right)^2 \nu_t^2 + 2\rho\sigma_t\nu_t\left(S_t\frac{\partial f_t}{\partial S_t} + \theta_t e^{r(T-t)}\right)\frac{\partial f_t}{\partial X_t},
\end{aligned}
\tag{3.17}
$$

while the drift term is

$$
\mathbb{E}_t[dJ_t] = \mathcal{D}\tilde{J}_t\,dt + \theta_t(\mu_t - r)e^{r(T-t)}dt, \tag{3.18}
$$

with Dynkin operator

$$
\mathcal{D}\tilde{J}_t = \frac{\partial \tilde{J}_t}{\partial t} + \mu_t S_t\frac{\partial \tilde{J}_t}{\partial S_t} + m_t\frac{\partial \tilde{J}_t}{\partial X_t} + \frac{1}{2}\left[\sigma_t^2 S_t^2\frac{\partial^2 \tilde{J}_t}{\partial S_t^2} + \nu_t^2\frac{\partial^2 \tilde{J}_t}{\partial X_t^2} + 2\rho\nu_t\sigma_t S_t\frac{\partial^2 \tilde{J}_t}{\partial S_t \partial X_t}\right]. \tag{3.19}
$$

Substitution yields the HJB equation (full algebra in Appendix C.2):

$$
\begin{aligned}
0 = \max_{\theta_t} \Bigg\{ & \mathcal{D}\tilde{J}_t + \theta_t(\mu_t - r)e^{r(T-t)} \\
& - \frac{\gamma}{2}\Bigg[\sigma_t^2 S_t^2\left(\frac{\partial f_t}{\partial S_t}\right)^2 + \nu_t^2\left(\frac{\partial f_t}{\partial X_t}\right)^2 + 2\rho\sigma_t\nu_t S_t\frac{\partial f_t}{\partial S_t}\frac{\partial f_t}{\partial X_t} \\
& + \theta_t^2\sigma_t^2 e^{2r(T-t)} + 2\theta_t\sigma_t e^{r(T-t)}\left(\sigma_t S_t\frac{\partial f_t}{\partial S_t} + \rho\nu_t\frac{\partial f_t}{\partial X_t}\right)\Bigg]\Bigg\},
\end{aligned}
\tag{3.20}
$$

with terminal condition $\tilde{J}(T) = 0$. This quadratic optimisation in $\theta_t$ admits a closed–form solution (derivation in Appendix C.3):

$$\theta_t^* = \frac{\mu_t - r}{\gamma \sigma_t^2} e^{-r(T-t)} - \left( S_t \frac{\partial f_t}{\partial S_t} + \frac{\rho \nu_t}{\sigma_t} \frac{\partial f_t}{\partial X_t} \right) e^{-r(T-t)}. \tag{3.21}$$

The first term is the myopic demand, while the second is the intertemporal hedging demand ($\theta_{Ht}$), adjusting exposure according to sensitivities of $f_t$ to $S_t$ and $X_t$. Equivalently (see Appendix C.3),

$$\theta_{Ht} = -\frac{\mathrm{cov}_t\left(\frac{\mathrm{d}S_t}{S_t}, \mathrm{d}f_t\right)}{\sigma_t^2 \, \mathrm{d}t} e^{-r(T-t)}, \tag{3.22}$$

so negative covariance implies an increased risky allocation (effective hedge), while positive covariance reduces it.

Since (3.21) is implicit in $f$, we introduce the hedge–neutral measure $\mathbb{P}^*$ via Girsanov's theorem, which eliminates the hedging term (see Appendix C.4 for full derivation). Under $\mathbb{P}^*$,

$$\mathrm{d}w_t^* := \mathrm{d}w_t + \frac{\mu_t - r}{\sigma_t} \mathrm{d}t, \quad \mathrm{d}w_{X,t}^* := \mathrm{d}w_{X,t} + \rho \frac{\mu_t - r}{\sigma_t} \mathrm{d}t, \tag{3.23}$$

and the dynamics become

$$\frac{\mathrm{d}S_t}{S_t} = r \, \mathrm{d}t + \sigma_t \, \mathrm{d}w_t^*, \quad \mathrm{d}X_t = \left( m_t - \rho \nu_t \frac{\mu_t - r}{\sigma_t} \right) \mathrm{d}t + \nu_t \, \mathrm{d}w_{X,t}^*. \tag{3.24}$$

The Radon–Nikodym derivative is

$$\left.\frac{\mathrm{d}\mathbb{P}^*}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \exp\left( -\int_0^t \frac{\mu_s - r}{\sigma_s} \mathrm{d}w_s - \frac{1}{2} \int_0^t \left( \frac{\mu_s - r}{\sigma_s} \right)^2 \mathrm{d}s \right). \tag{3.25}$$

Under $\mathbb{P}^*$, the intertemporal hedging demand vanishes and $f$ can be written as

$$f(S_t, X_t, t) = \mathbb{E}_t^{\mathbb{P}^*} \left[ \int_t^T \frac{1}{\gamma} \left( \frac{\mu_s - r}{\sigma_s} \right)^2 \mathrm{d}s \right]. \tag{3.26}$$

By the Feynman–Kac theorem, this follows from a linear PDE with terminal condition $f(S_T, X_T, T) = 0$ (see Appendix C.4).

Substituting (3.26) into (3.21) yields

$$\begin{aligned} \theta_t^* = {} & \frac{\mu_t - r}{\gamma \sigma_t^2} e^{-r(T-t)} \\ & - \frac{1}{\gamma} \left( S_t \frac{\partial}{\partial S_t} \mathbb{E}_t^{\mathbb{P}^*} \left[ \int_t^T \left( \frac{\mu_s - r}{\sigma_s} \right)^2 \mathrm{d}s \right] + \rho \frac{\nu_t}{\sigma_t} \frac{\partial}{\partial X_t} \mathbb{E}_t^{\mathbb{P}^*} \left[ \int_t^T \left( \frac{\mu_s - r}{\sigma_s} \right)^2 \mathrm{d}s \right] \right) e^{-r(T-t)}. \end{aligned} \tag{3.27}$$

This separates myopic and hedging demands, both expressed as expectations under $\mathbb{P}^*$. While the integrand in (3.26) is deterministic, its dependence on $S_t$ and $X_t$ is implicit via $\mu_s$ and $\sigma_s$, so the derivatives in (3.27) require numerical methods. In practice, we use Monte Carlo under $\mathbb{P}^*$ with Malliavin calculus to approximate these sensitivities (Section 4.2.3), yielding a tractable and flexible implementation.

### 3.1.3 Multiple-Risky-Asset Formulation

We now extend the setup to $D$ risky assets and $K$ state variables. Let $\mathbf{S}_t = (S_{1t}, \ldots, S_{Dt})^\top \in \mathbb{R}^D$ and $\mathbf{X}_t = (X_{1t}, \ldots, X_{Kt})^\top \in \mathbb{R}^K$. The risky assets evolve as

$$\frac{dS_{it}}{S_{it}} = \mu_i(\mathbf{S}_t, \mathbf{X}_t, t)\, dt + \boldsymbol{\sigma}_i(\mathbf{S}_t, \mathbf{X}_t, t)^\top d\mathbf{w}_t, \quad i = 1, \ldots, D, \tag{3.28}$$

where $\boldsymbol{\mu} \equiv (\mu_1, \ldots, \mu_D)^\top$ and $\boldsymbol{\sigma} \equiv (\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_D)^\top \in \mathbb{R}^{D\times D}$ denote the vector of drifts and the volatility matrix of the asset returns, respectively; $\mathbf{w} \equiv (w_1, \ldots, w_D)^\top$ is a $D$-dimensional Brownian motion.

State variables follow

$$dX_{jt} = m_j(\mathbf{X}_t, t)\, dt + \boldsymbol{\nu}_j(\mathbf{X}_t, t)^\top d\mathbf{w}_{X,t}, \quad j = 1, \ldots, K, \tag{3.29}$$

with drift vector $\mathbf{m} = (m_1, \ldots, m_K)^\top$, volatility matrix $\boldsymbol{\nu} = (\boldsymbol{\nu}_1, \ldots, \boldsymbol{\nu}_K)^\top$, and $K$-dimensional Brownian motion $\mathbf{w}_{X,t} = (w_{X1}, \ldots, w_{XK})^\top$. The correlation between asset and state-variable shocks is captured by $\boldsymbol{\rho} = \{\rho_{dk}\} \in \mathbb{R}^{D\times K}$, where $\rho_{dk} = \mathrm{Corr}(w_d, w_{Xk})$.

Wealth $W_t$ is allocated across risky assets via $\boldsymbol{\theta}_t \in \mathbb{R}^D$ (monetary positions), with the remainder $W_t - \mathbf{1}^\top \boldsymbol{\theta}_t$ in the risk-free asset. Its dynamics are

$$dW_t = \Big[ rW_t + \boldsymbol{\theta}_t^\top (\boldsymbol{\mu}_t - r\mathbf{1}) \Big]\, dt + \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t\, d\mathbf{w}_t. \tag{3.30}$$

The hedge-neutral measure $\mathbb{P}^*$ has Radon–Nikodym derivative

$$\frac{d\mathbb{P}^*}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = \exp\left( -\int_0^T \big(\boldsymbol{\sigma}_s^{-1}(\boldsymbol{\mu}_s - r\mathbf{1})\big)^\top d\mathbf{w}_s - \tfrac{1}{2}\int_0^T \big\|\boldsymbol{\sigma}_s^{-1}(\boldsymbol{\mu}_s - r\mathbf{1})\big\|^2 ds \right), \tag{3.31}$$

under which

$$d\mathbf{w}_t^* = d\mathbf{w}_t + \boldsymbol{\sigma}_t^{-1}(\boldsymbol{\mu}_t - r\mathbf{1})\, dt, \tag{3.32}$$

$$d\mathbf{w}_{X,t}^* = d\mathbf{w}_{X,t} + \boldsymbol{\rho}^\top \boldsymbol{\sigma}_t^{-1}(\boldsymbol{\mu}_t - r\mathbf{1})\, dt. \tag{3.33}$$

The optimal portfolio policy is

$$\boldsymbol{\theta}_t^* = \frac{1}{\gamma}(\boldsymbol{\sigma}_t\boldsymbol{\sigma}_t^\top)^{-1}(\boldsymbol{\mu}_t - r\mathbf{1})e^{-r(T-t)} - \left( \mathbf{I}_{\mathbf{S}_t}\frac{\partial f_t}{\partial \mathbf{S}_t^\top} + (\boldsymbol{\nu}_t\boldsymbol{\rho}^\top\boldsymbol{\sigma}_t^{-1})^\top \frac{\partial f_t}{\partial \mathbf{X}_t^\top} \right) e^{-r(T-t)}, \tag{3.34}$$

where $\mathbf{I}_{\mathbf{S}_t}$ is a diagonal $D \times D$ matrix with entries $S_{1t}, \ldots, S_{Dt}$ on the main diagonal, $\frac{\partial f_t}{\partial \mathbf{S}_t^\top}$ and $\frac{\partial f_t}{\partial \mathbf{X}_t^\top}$ are row gradients.

The anticipated portfolio gains satisfy

$$f(\mathbf{S}_t, \mathbf{X}_t, t) = \mathbb{E}_t^{\mathbb{P}^*}\left[ \int_t^T \frac{1}{\gamma}(\boldsymbol{\mu}_s - r\mathbf{1})^\top(\boldsymbol{\sigma}_s\boldsymbol{\sigma}_s^\top)^{-1}(\boldsymbol{\mu}_s - r\mathbf{1})ds \right]. \tag{3.35}$$

This formulation extends the single-asset case: myopic and intertemporal hedging demands remain distinct, but increasing the number of assets ($D$) or state variables ($K$) quickly makes analytic evaluation of the hedging terms infeasible. Moreover, once transaction costs are introduced, the tractable structure of the optimal policy breaks down entirely. These challenges motivate the deep BSDE approach developed in the next section, where we extend the framework to handle high-dimensional settings and additional frictions.

## 3.2 Continuous-Time Mean-Variance Optimisation with Transaction Costs (Deep BSDE Approach)

In this section, we present a novel extension of the continuous-time framework of Basak and Chabakauri (2010) to incorporate quadratic transaction costs, referred to as the cost-aware model. These frictions render closed-form solutions intractable, so we adopt deep BSDE methods: the 2BSDE framework of Beck et al. (2019) and its refinement by Huré et al. (2020), which provides convergence guarantees and improved training stability.

### 3.2.1 Economic Setup

We retain the multiple-risky-asset framework from Section 3.1.3, based on the generalisation of Basak and Chabakauri (2010) with $D$ risky assets and $K$ state variables. The dynamics of $\mathbf{S}_t$ and $\mathbf{X}_t$ remain as in Equations (3.28)–(3.29). We let $\boldsymbol{\theta}_t \in \mathbb{R}^D$ denote the monetary allocation to risky assets at time $t$, which is our control variable.

The baseline model assumed frictionless markets. We now introduce quadratic transaction costs, penalising rapid changes in allocations. Two formulations are common in the literature. The first, which we adopt, approximates the trading speed via a backward finite difference,

$$\dot{\boldsymbol{\theta}}_t \approx \frac{\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-\Delta t}}{\Delta t}, \tag{3.36}$$

where $\Delta t$ is the numerical step.

The wealth process, assumed self-financing with no intermediate consumption, becomes

$$dW_t = \Big[rW_t + \boldsymbol{\theta}_t^\top(\boldsymbol{\mu}_t - r\mathbf{1})\Big]\,dt + \boldsymbol{\theta}_t^\top\boldsymbol{\sigma}_t\,d\mathbf{w}_t - \lambda\left\|\frac{\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-\Delta t}}{\Delta t}\right\|^2 dt, \tag{3.37}$$

with $\lambda > 0$ the cost severity parameter.

Quadratic costs are smooth and differentiable, making them well suited to the gradient-based training of neural networks in deep BSDE methods. They also provide a natural way to penalise reallocation intensity. By contrast, proportional costs introduce non-differentiabilities into the HJB equation and are more challenging to handle (see Section 2.5).

The second approach, following Gonon et al. (2020), assumes $\boldsymbol{\theta}_t$ is absolutely continuous:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_0 + \int_0^t \dot{\boldsymbol{\theta}}_u\,du, \quad \dot{\boldsymbol{\theta}} \in L^2([0,T];\mathbb{R}^D), \tag{3.38}$$

so that the trading speed $\dot{\boldsymbol{\theta}}_t$ is well defined and becomes the control variable. This yields the wealth dynamics

$$dW_t = \Big[rW_t + \boldsymbol{\theta}_t^\top(\boldsymbol{\mu}_t - r\mathbf{1})\Big]\,dt + \boldsymbol{\theta}_t^\top\boldsymbol{\sigma}_t\,d\mathbf{w}_t - \lambda\|\dot{\boldsymbol{\theta}}_t\|^2 dt. \tag{3.39}$$

While mathematically elegant, the trade-speed formulation is impractical in our deep BSDE setting: $\dot{\boldsymbol{\theta}}_t$ must be recovered from neural network outputs via first-order conditions, which is difficult to scale into currency per unit time and leads to unstable wealth paths. The finite-difference version instead keeps $\boldsymbol{\theta}_t$ as the control, enabling direct scaling by wealth and yielding more stable backtests. We therefore adopt this formulation in our main implementation, with the trade-speed case derived in Appendix D.

### 3.2.2 Value Function and HJB Derivation

We define the value function as

$$J_t(W_t, \mathbf{S}_t, \mathbf{X}_t, t) = \mathbb{E}[W_T \mid \mathcal{F}_t] - \frac{\gamma}{2}\mathrm{Var}[W_T \mid \mathcal{F}_t], \tag{3.40}$$

a time-consistent dynamic mean–variance criterion under the information set $\mathcal{F}_t$. This conditional formulation preserves time consistency, allowing application of dynamic programming via the associated HJB equation.

Expanding explicitly,

$$J_t = \mathbb{E}[W_T \mid \mathcal{F}_t] - \frac{\gamma}{2}\Big(\mathbb{E}[W_T^2 \mid \mathcal{F}_t] - \big(\mathbb{E}[W_T \mid \mathcal{F}_t]\big)^2\Big). \tag{3.41}$$

Applying Itô's lemma to $J(W_t, \mathbf{S}_t, \mathbf{X}_t, t)$, using the wealth and state dynamics in Equations (3.37) and (3.29), we note that although $J$ formally depends on $\mathbf{S}_t$, asset prices influence the objective only via $W_t$. Since all gains and costs are aggregated in wealth, we may therefore treat $J$ as a function of $(W_t, \mathbf{X}_t, t)$.

Assuming sufficient smoothness, Itô's lemma gives

$$\begin{aligned} dJ = {} & \frac{\partial J}{\partial t}\,dt + \frac{\partial J}{\partial W}\,dW_t + \sum_{j=1}^{K}\frac{\partial J}{\partial X_j}\,dX_{jt} \\ & + \frac{1}{2}\frac{\partial^2 J}{\partial W^2}\,\langle dW_t, dW_t\rangle + \frac{1}{2}\sum_{i,j=1}^{K}\frac{\partial^2 J}{\partial X_i \partial X_j}\,\langle dX_{it}, dX_{jt}\rangle + \sum_{j=1}^{K}\frac{\partial^2 J}{\partial W \partial X_j}\,\langle dW_t, dX_{jt}\rangle. \end{aligned} \tag{3.42}$$

The quadratic variations are

$$\langle dW_t, dW_t\rangle = \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top \boldsymbol{\theta}_t\,dt, \tag{3.43}$$

$$\langle dX_{it}, dX_{jt}\rangle = \boldsymbol{\nu}_i^\top \boldsymbol{\nu}_j\,dt, \quad i, j = 1, \ldots, K, \tag{3.44}$$

$$\langle dW_t, dX_{jt}\rangle = \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t\,\boldsymbol{\rho}_j\,\boldsymbol{\nu}_j\,dt, \tag{3.45}$$

where $\boldsymbol{\rho}_j \in \mathbb{R}^D$ denotes correlations between the $D$ risky assets and the Brownian motion driving the $j$-th state variable.

Substituting these terms, together with the wealth and state dynamics (Equations (3.37) and (3.29)), into (3.42), the drift becomes

$$\begin{aligned} \mathbb{E}[dJ \mid \mathcal{F}_t] = {} & \frac{\partial J}{\partial t} + \left[rW_t + \boldsymbol{\theta}_t^\top(\boldsymbol{\mu}_t - r\mathbf{1}) - \lambda\left\|\frac{\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-\Delta t}}{\Delta t}\right\|^2\right]\frac{\partial J}{\partial W} + \sum_{j=1}^{K} m_j(\mathbf{X}_t, t)\frac{\partial J}{\partial X_j} \\ & + \frac{1}{2}\boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top \boldsymbol{\theta}_t \frac{\partial^2 J}{\partial W^2} + \frac{1}{2}\sum_{i,j=1}^{K}\boldsymbol{\nu}_i^\top \boldsymbol{\nu}_j \frac{\partial^2 J}{\partial X_i \partial X_j} + \sum_{j=1}^{K}\boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t\,\boldsymbol{\rho}_j\,\boldsymbol{\nu}_j \frac{\partial^2 J}{\partial W \partial X_j}. \end{aligned} \tag{3.46}$$

By the dynamic programming principle, the drift vanishes under the optimal control $\boldsymbol{\theta}_t^*$:

$$\sup_{\boldsymbol{\theta}_t}\mathbb{E}[dJ \mid \mathcal{F}_t] = 0, \tag{3.47}$$

leading to the HJB equation

$$
\begin{aligned}
0 = \frac{\partial J}{\partial t} + \sup_{\boldsymbol{\theta}_t} \Bigg\{ & \left[ rW_t + \boldsymbol{\theta}_t^\top (\boldsymbol{\mu}_t - r\mathbf{1}) - \lambda \left\| \frac{\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-\Delta t}}{\Delta t} \right\|^2 \right] \frac{\partial J}{\partial W} + \sum_{j=1}^{K} m_j(\mathbf{X}_t, t) \frac{\partial J}{\partial X_j} \\
& + \frac{1}{2} \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top \boldsymbol{\theta}_t \frac{\partial^2 J}{\partial W^2} + \frac{1}{2} \sum_{i,j=1}^{K} \left( \boldsymbol{\nu}_i^\top \boldsymbol{\nu}_j \right) \frac{\partial^2 J}{\partial X_i \partial X_j} \\
& + \sum_{j=1}^{K} \left( \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \, \boldsymbol{\rho}_j \, \boldsymbol{\nu}_j \right) \frac{\partial^2 J}{\partial W \partial X_j} \Bigg\},
\end{aligned} \tag{3.48}
$$

with terminal condition

$$J_T(W_T) = W_T - \frac{\gamma}{2} W_T^2. \tag{3.49}$$

This HJB equation characterises the time-consistent optimal control problem under the finite-difference quadratic transaction cost specification, extending the Basak and Chabakauri (2010) framework to frictional markets. It also underpins the associated BSDE reformulation developed in the next section, which will later form the basis for our numerical implementation.

### 3.2.3 Deep 2BSDE Framework

#### Control Policy Parameterisation

The control variable $\boldsymbol{\theta}_t$ specifies the portfolio allocation at each time $t$. Rather than prescribing $\boldsymbol{\theta}_t$ exogenously or solving for it analytically, we parameterise it as a neural network $\mathrm{NN}_\phi(\cdot)$, following Beck et al. (2019), with trainable parameters $\phi$, mapping the current state to allocations:

$$\boldsymbol{\theta}_t = \mathrm{NN}_\phi(W_t, \mathbf{S}_t, \mathbf{X}_t, t), \tag{3.50}$$

where $(W_t, \mathbf{S}_t, \mathbf{X}_t, t)$ denote current wealth, asset prices, state variables, and time. This parameterisation enables end-to-end training of the control policy from simulated trajectories, in conjunction with the value function approximation defined via the 2BSDE framework. The learned policy remains time-consistent, as it depends only on the current state rather than past trading paths or pre-committed plans. In practice, the network outputs an allocation vector that is rescaled to monetary units during numerical implementation (see Section 4.3.2).

#### Forward–Backward 2BSDE System

The output of the policy network, $\boldsymbol{\theta}_t = \mathrm{NN}_\phi(W_t, \mathbf{S}_t, \mathbf{X}_t, t)$, is substituted into both the forward and backward components of the 2BSDE system, ensuring each control decision propagates through wealth dynamics and optimisation.

**Forward Dynamics.** The control $\boldsymbol{\theta}_t$ directly influences wealth via the forward SDE in Equation (3.37).

**Backward Dynamics.** To approximate the value function $J(W_t, \mathbf{S}_t, \mathbf{X}_t, t)$, we follow the 2BSDE structure of Beck et al. (2019):

$$Y_t := J(W_t, \mathbf{S}_t, \mathbf{X}_t, t), \quad Z_t := \frac{\partial J(W_t, \mathbf{S}_t, \mathbf{X}_t, t)}{\partial W}, \quad \Gamma_t := \frac{\partial^2 J(W_t, \mathbf{S}_t, \mathbf{X}_t, t)}{\partial W^2}, \tag{3.51}$$

where $Z_t$ and $\Gamma_t$ are scalars representing the first and second derivatives of $J$ with respect to $W_t$.

The 2BSDE system is written as:

$$-dY_t = g(W_t, \mathbf{S}_t, \mathbf{X}_t, \boldsymbol{\theta}_t, Z_t, \Gamma_t, t)\, dt - Z_t\, \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t\, d\mathbf{w}_t{}^{1}, \tag{3.52}$$

$$dZ_t = -\Gamma_t\, \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t\, d\mathbf{w}_t, \tag{3.53}$$

with terminal condition $Y_T = J_T(W_T)$. The generator $g$ corresponds to the HJB drift (Equation (3.48)) restricted to the wealth dimension:

$$g(W_t, \mathbf{S}_t, \mathbf{X}_t, \boldsymbol{\theta}_t, Z_t, \Gamma_t, t) = \left[ rW_t + \boldsymbol{\theta}_t^\top (\boldsymbol{\mu}_t - r\mathbf{1}) - \lambda \left\| \frac{\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-\Delta t}}{\Delta t} \right\|^2 \right] Z_t + \frac{1}{2} \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top \boldsymbol{\theta}_t\, \Gamma_t. \tag{3.54}$$

Equation (3.54) includes only wealth-related terms, since $Z_t$ and $\Gamma_t$ are defined w.r.t. $W_t$. The effect of state-variable dynamics enters implicitly through the forward simulation of $(W_t, \mathbf{X}_t)$ and the network-predicted $\boldsymbol{\theta}_t$, preserving correct coupling while simplifying training.

This formulation mirrors the HJB structure but is tailored to the deep 2BSDE method, where $Z_t$ and $\Gamma_t$ are approximated via neural networks trained on simulated trajectories. The finite-difference transaction cost term matches the forward simulation scheme, ensuring alignment between numerical implementation and the continuous-time formulation.

### Loss Formulation and Training Strategy

We consider two established training schemes for solving the 2BSDE associated with our control problem: the global loss method of Han et al. (2018); Beck et al. (2019) and the stepwise loss method of Huré et al. (2020).

**Global loss.** This approach minimises the squared deviation between the simulated terminal value $Y_T$ and the prescribed terminal condition:

$$\mathcal{L}_{\text{global}} = \mathbb{E}\left[ \left| Y_T - \left( W_T - \frac{\gamma}{2} W_T^2 \right) \right|^2 \right], \tag{3.55}$$

where $Y_T$ is obtained by simulating the forward–backward 2BSDE with the learned control policy $\boldsymbol{\theta}_t$, alongside neural approximations for the gradient $Z_t = \frac{\partial V}{\partial W}$ and curvature $\Gamma_t = \frac{\partial^2 V}{\partial W^2}$. The terminal condition corresponds to quadratic utility $V_T(W_T) = W_T - \frac{\gamma}{2} W_T^2$, consistent with mean–variance preferences.

**Stepwise loss (DBDP1).** The stepwise scheme, introduced by Huré et al. (2020) as Deep Backward Dynamic Programming (DBDP1), enforces the discretised Bellman relation at each grid point $\{t_n\}_{n=0}^N$:

$$\mathcal{L}_{\text{stepwise}} = \sum_{n=0}^{N-1} \mathbb{E}\left[ \left| Y_{t_n} - \left( Y_{t_{n+1}} + g_{t_n} \Delta t - Z_{t_n}\, \boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n}\, \Delta \mathbf{w}_{t_n} \right) \right|^2 \right], \tag{3.56}$$

where $g_{t_n}$ is the discretised generator of the HJB equation and $\Delta \mathbf{w}_{t_n} \sim \mathcal{N}(0, \Delta t)$ are the Brownian increments driving the risky asset dynamics. By enforcing local dynamic consistency rather than a single terminal constraint, this scheme improves gradient flow and controls temporal error accumulation.

We adopt the DBDP1-style stepwise scheme due to its theoretical convergence guarantees (see Huré et al. (2020)) and supporting empirical evidence (see Section 4.3.3), which make it well

[1] In Beck et al. (2019), the BSDE is expressed in backward form $Y_t = V_T(W_T) - \int_t^T g_s\, ds - \int_t^T Z_s^\top d\mathbf{w}_s$. Our Equation (3.52) adopts the equivalent forward form $Y_0 = V_T(W_T) - \int_0^T g_t\, dt + \int_0^T Z_t^\top d\mathbf{w}_t$, which differs only in sign due to integration direction, chosen here for consistency with our forward simulation procedure.

suited to long-horizon stochastic control. In their original formulation, only the first derivative $Z_t$ is approximated by a neural network. Our setting, however, stems from a fully non-linear HJB equation requiring the curvature term $\Gamma_t$. We therefore extend DBDP1 with an additional neural network for $\Gamma_t$, enabling accurate treatment of second-order terms in the 2BSDE. To our knowledge, this combination of the stepwise scheme with a dedicated curvature network has not previously appeared in the literature.

# Chapter 4

# Numerical Implementation

In this chapter we translate the theoretical models of Chapter 3 into a practical computational framework. We describe the market setup and data sources, before detailing how portfolio allocations are obtained and backtested for both the cost-unaware benchmark of Basak and Chabakauri (2010) and our cost-aware extension.

## 4.1 Overview of Data and Market Setup

We consider a market with three risky assets, one risk-free asset, and a single state variable. While simplified relative to institutional portfolios with many risky assets, this reduced setup keeps computation manageable while preserving sufficient richness to evaluate model behaviour and comparative performance.

Daily asset and state variable data are obtained via the `yfinance` Python library over a 10-year horizon (2015–2025). The risky assets — Apple Inc. (AAPL), JPMorgan Chase & Co. (JPM), and Exxon Mobil Corp. (XOM) — provide sectoral diversification and heterogeneous risk–return profiles. The state variable is the credit spread between high-yield and investment-grade bonds, proxied by ETF prices (HYG vs. LQD), which captures credit sentiment and evolves smoothly, supporting its modelling as an arithmetic Brownian motion (ABM).

The risk-free rate is fixed at 2.5%, reflecting a long-run average and avoiding the complexity of time-varying rates. Initial wealth is set to £100 across all strategies to ensure comparability.

Quadratic transaction costs are included in all backtests. We set the annualised coefficient to $\lambda = 1 \times 10^{-5}$, implying a daily value $\lambda_{\text{day}} = 0.0003$ (with $\Delta t = 1/30$). This calibration yields costs of approximately 5 basis points (bps) for small trades, consistent with the 3–6 bps range reported by Frazzini et al. (2018) for institutional-sized equity trades. In a proportional model, the same 5 bps would require $\lambda_{\text{day}} \approx 0.0005$, but quadratic costs scale with $\|\Delta\boldsymbol{\theta}\|^2$ and penalise large rebalancing moves more strongly. Since our strategies operate at much smaller wealth levels than those institutional benchmarks, realised costs are likely lower in practice. The chosen $\lambda_{\text{day}}$ therefore provides a conservative benchmark while remaining scalable. The same value is applied across all models for comparability.

## 4.2 Cost-Unaware Model Implementation (Analytical Approach)

We implement the cost-unaware continuous-time mean–variance strategy of Basak and Chabakauri (2010) using real financial data. The model is discretised with Euler–Maruyama, parameters are re-estimated on a rolling window, and portfolio allocations are re-optimised daily. The procedure follows the theoretical formulation in Section 3.1, adapted to empirical inputs.

### 4.2.1 Rolling Parameter Estimation and Path Simulation

At each backtest date $t_n$, parameters are estimated from a 252-day rolling window: asset drifts $\boldsymbol{\mu}_{t_n} \in \mathbb{R}^D$ as historical mean arithmetic returns, state drifts $\mathbf{m}_{t_n} \in \mathbb{R}^K$ as the historical mean of state changes, asset volatilities $\boldsymbol{\sigma}_{t_n} \in \mathbb{R}^D$ and state volatilities $\boldsymbol{\nu}_{t_n} \in \mathbb{R}^K$ as sample standard deviations, the asset correlation matrix $\boldsymbol{\Sigma}_{t_n} \in \mathbb{R}^{D \times D}$, and the asset–state correlation matrix $\boldsymbol{\rho}_{t_n} \in \mathbb{R}^{K \times D}$ as sample correlations. This reflects an investor updating beliefs solely from past information.

The trading horizon is $T = 1$ year, discretised into $N = 30$ steps (matching the later deep BSDE framework). Daily steps ($N = 252$) offered negligible performance gains. We simulate $M = 30$ independent market trajectories. Initial prices $\mathbf{S}_0 \in \mathbb{R}^D$ and state values $\mathbf{X}_0 \in \mathbb{R}^K$ are taken from market data on 2 January 2015.

On the time grid $\{t_0, \ldots, t_N = T\}$ with step size $\Delta t = T/N$, trajectories are simulated via Euler–Maruyama applied to the asset and state dynamics in Equations (3.28)–(3.29). For $D$ risky assets and $K$ state variables:

$$\mathbf{S}_{t_{n+1}} = \mathbf{S}_{t_n} + \operatorname{diag}(\mathbf{S}_{t_n}) \left[ \boldsymbol{\mu}(\mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n) \, \Delta t + \boldsymbol{\sigma}(\mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n) \, \Delta \mathbf{w}_n \right], \tag{4.1}$$

$$\mathbf{X}_{t_{n+1}} = \mathbf{X}_{t_n} + \mathbf{m}(\mathbf{X}_{t_n}, t_n) \, \Delta t + \boldsymbol{\nu}(\mathbf{X}_{t_n}, t_n) \, \Delta \mathbf{w}_{X,n}, \tag{4.2}$$

To incorporate contemporaneous correlation between asset and state shocks, we compute the Cholesky factor $\mathbf{L}_{t_n}$ of $\boldsymbol{\Sigma}_{t_n}$ such that:

$$\boldsymbol{\Sigma}_{t_n} = \mathbf{L}_{t_n} \mathbf{L}_{t_n}^\top. \tag{4.3}$$

The correlated Brownian increments for asset prices are then:

$$\Delta \mathbf{w}_{t_n} = \sqrt{\Delta t} \, \mathbf{L}_{t_n} \mathbf{z}_{t_n}, \quad \mathbf{z}_{t_n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D), \tag{4.4}$$

and for the state variables:

$$\Delta \mathbf{w}_{X,t_n} = \boldsymbol{\rho}_{t_n} \, \Delta \mathbf{w}_{t_n} + \sqrt{\Delta t} \, \boldsymbol{\xi}_{t_n}, \quad \boldsymbol{\xi}_{t_n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K), \tag{4.5}$$

This Cholesky-based correlation construction is standard in financial simulations; see Glasserman (2004, Ch. 3). Algorithm 1 summarises the full rolling simulation procedure.

**Algorithm 1** Rolling Simulation of Market Paths

**Input:** Initial prices $\mathbf{S}_0$, state variable $\mathbf{X}_0$, market data, rolling window size, parameters $(r, T, N, M)$
Initialise time grid with step size $\Delta t = T/N$
**for** $m = 1$ to $M$ **do**
  Set $\mathbf{S}_{t_0}^{(m)} = \mathbf{S}_0$, $\mathbf{X}_{t_0}^{(m)} = \mathbf{X}_0$
  **for** $n = 0$ to $N - 1$ **do**
    Estimate $\boldsymbol{\mu}_{t_n}, \boldsymbol{\sigma}_{t_n}, \mathbf{m}_{t_n}, \boldsymbol{\nu}_{t_n}, \boldsymbol{\rho}_{t_n}, \boldsymbol{\Sigma}_{t_n}$ from market data
    Compute Cholesky factor $\mathbf{L}_{t_n}$ of $\boldsymbol{\Sigma}_{t_n}$
    Sample $\mathbf{z}_{t_n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, $\boldsymbol{\xi}_{t_n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_K)$
    Generate correlated increments $\Delta \mathbf{w}_{t_n}$, $\Delta \mathbf{w}_{X,t_n}$ via Equations (4.4)-(4.5)
    Update $\mathbf{S}_{t_{n+1}}^{(m)}$, $\mathbf{X}_{t_{n+1}}^{(m)}$ via Equations (4.1)–(4.2)
  **end for**
**end for**
**Output:** $\{\mathbf{S}_{t_n}^{(m)}, \mathbf{X}_{t_n}^{(m)}\}_{n=0}^N$, $m = 1, \ldots, M$; $\{\boldsymbol{\mu}_{t_n}, \boldsymbol{\sigma}_{t_n}, \mathbf{m}_{t_n}, \boldsymbol{\nu}_{t_n}, \boldsymbol{\Sigma}_{t_n}, \boldsymbol{\rho}_{t_n}\}_{n=0}^{N-1}$

### 4.2.2 Anticipated Gains Computation

Given the simulated asset and state trajectories $\{\mathbf{S}_t, \mathbf{X}_t\}_{t\in[0,T]}$ under the real-world measure $\mathbb{P}$, we approximate the anticipated portfolio gains $f(\mathbf{S}_t, \mathbf{X}_t, t)$, which are required to compute the optimal portfolio policy $\boldsymbol{\theta}_t^*$.

Following Equation (3.35), the analytical expression for $f$ is derived under the hedge-neutral measure $\mathbb{P}^*$. In practice, we evaluate this formula along simulated paths, using the time-varying drifts and volatilities obtained from the asset trajectories. For each simulation path $m$ and time step $t_n$, we compute:

$$f_{t_n}^{(m)} = \sum_{j=n}^{N-1} \frac{1}{\gamma} \left(\boldsymbol{\mu}_{t_j}^{(m)} - r\mathbf{1}\right)^\top \left(\boldsymbol{\sigma}_{t_j}^{(m)} \boldsymbol{\sigma}_{t_j}^{(m)\top}\right)^{-1} \left(\boldsymbol{\mu}_{t_j}^{(m)} - r\mathbf{1}\right) \Delta t, \tag{4.6}$$

where $\gamma$ is the risk aversion parameter. In all experiments, we set $\gamma = 3$ to reflect a moderate level of risk aversion. The full procedure is summarised in Algorithm 2.

**Algorithm 2** Computation of Anticipated Gains $f_{t_n}^{(m)}$

1: **Input:** Drifts $\boldsymbol{\mu}_{t_j}^{(m)}$, volatilities $\boldsymbol{\sigma}_{t_j}^{(m)}$, parameters ($r$, $\gamma$, $\Delta t$, $N$, $M$)
2: **for** $m = 1$ to $M$ **do**
3:     **for** $n = 0$ to $N-1$ **do**
4:         Initialise $f_{t_n}^{(m)} \leftarrow 0$
5:         **for** $j = n$ to $N-1$ **do**
6:             Update:

$$f_{t_n}^{(m)} \mathrel{+}= \frac{1}{\gamma} \left(\boldsymbol{\mu}_{t_j}^{(m)} - r\mathbf{1}\right)^\top \left(\boldsymbol{\sigma}_{t_j}^{(m)} \boldsymbol{\sigma}_{t_j}^{(m)\top}\right)^{-1} \left(\boldsymbol{\mu}_{t_j}^{(m)} - r\mathbf{1}\right) \Delta t$$

7:         **end for**
8:     **end for**
9: **end for**
10: **Output:** $\{f_{t_n}^{(m)}\}_{n=0}^N$, $m = 1, \ldots, M$

### 4.2.3 Gradient Estimation and Optimal Policy Computation

The anticipated gains $f(\mathbf{S}_t, \mathbf{X}_t, t)$ quantify expected reward over the horizon $[t, T]$. Computing the optimal allocation $\boldsymbol{\theta}_t^*$ requires sensitivities with respect to asset prices and state variables, $\partial f/\partial \mathbf{S}_t$ and $\partial f/\partial \mathbf{X}_t$, which enter Equation (4.9).

Since $f$ depends implicitly on the stochastic processes $(\mathbf{S}_t, \mathbf{X}_t)$, direct differentiation is intractable. Using Malliavin calculus, we reformulate gradients as conditional expectations of Brownian motion terms via the Clark–Ocone formula[1]. This amounts to an integration-by-parts argument on Wiener space, replacing state derivatives with expectations involving realised Brownian shocks.

The resulting likelihood ratio estimators, applied per asset $i$ and state variable $j$, are:

$$\frac{\partial f_{t_n}^{(m)}}{\partial S_{it_n}} \approx \frac{f_{t_n}^{(m)}}{\sigma_{it_n} S_{it_n}\, \Delta t} \Delta w_{it_n}^{(m)}, \quad i = 1, \ldots, D, \tag{4.7}$$

$$\frac{\partial f_{t_n}^{(m)}}{\partial X_{jt_n}} \approx \frac{f_{t_n}^{(m)}}{\nu_{jt_n}\, \Delta t} \Delta w_{X,jt_n}^{(m)}, \quad j = 1, \ldots, K, \tag{4.8}$$

[1]The Clark–Ocone formula expresses any square-integrable functional of Brownian motion as its conditional expectation plus the time-integral of its Malliavin derivative; see Detemple (2003); Nualart (1995).

where $\sigma_{it_n}$ and $\nu_{jt_n}$ denote the time-varying volatilities, and $\Delta w_{it_n}^{(m)}$, $\Delta w_{X,jt_n}^{(m)}$ are the simulated Brownian increments.

Gradients are clipped to $[-10, 10]$ to mitigate extreme shocks and small denominators. We use $M = 30$ simulation paths per rebalancing date: larger $M$ reduces variance but is computationally prohibitive over a 10-year daily horizon, while smaller $M$ yields excessive noise. Empirically, $M = 30$ balances stability and cost.

**Algorithm 3** Gradient Estimation via Likelihood Ratio Method

**Input:** $f_{t_n}^{(m)}, \mathbf{S}_{t_n}^{(m)}, \mathbf{X}_{t_n}^{(m)}, \Delta \mathbf{w}_{t_n}^{(m)}, \Delta \mathbf{w}_{X,t_n}^{(m)}, \boldsymbol{\sigma}_{t_n}^{(m)}, \boldsymbol{\nu}_{t_n}^{(m)}, \Delta t, c, M$
**for** $m = 1$ to $M$ **do**
    **for** $n = 0$ to $N-1$ **do**
        $(\nabla_{\mathbf{S}} f)_{t_n}^{(m)} \leftarrow \frac{f_{t_n}^{(m)}}{\boldsymbol{\sigma}_{t_n}^{(m)} \circ \mathbf{S}_{t_n}^{(m)} \Delta t} \circ \Delta \mathbf{w}_{t_n}^{(m)}$
        $(\nabla_{\mathbf{X}} f)_{t_n}^{(m)} \leftarrow \frac{f_{t_n}^{(m)}}{\boldsymbol{\nu}_{t_n}^{(m)} \Delta t} \circ \Delta \mathbf{w}_{X,t_n}^{(m)}$
        Clip to $[-c, c]$
    **end for**
**end for**
**Output:** $\{\nabla_{\mathbf{S}} f_{t_n}, \nabla_{\mathbf{X}} f_{t_n}\}_{n=0}^{N}$, where $\nabla_{\mathbf{S}} f_{t_n} = \frac{1}{M} \sum_m (\nabla_{\mathbf{S}} f)_{t_n}^{(m)}$, $\nabla_{\mathbf{X}} f_{t_n} = \frac{1}{M} \sum_m (\nabla_{\mathbf{X}} f)_{t_n}^{(m)}$.

The optimal monetary allocation vector $\boldsymbol{\theta}_{t_n}^*$ is:

$$\boldsymbol{\theta}_{t_n}^* = \frac{1}{\gamma} \left( \boldsymbol{\sigma}_{t_n} \boldsymbol{\sigma}_{t_n}^\top \right)^{-1} (\boldsymbol{\mu}_{t_n} - r\mathbf{1})\, e^{-r(T-t_n)} - \left[ \mathbf{I}_{\mathbf{S}_{t_n}} \nabla_{\mathbf{S}} f_{t_n} + \left( \boldsymbol{\nu}_{t_n} \boldsymbol{\rho}_{t_n} \boldsymbol{\sigma}_{t_n}^{-1} \right)^\top \nabla_{\mathbf{X}} f_{t_n} \right] e^{-r(T-t_n)}, \tag{4.9}$$

where $\mathbf{I}_{\mathbf{S}_{t_n}}$ is a diagonal $D \times D$ matrix with entries $S_{1t_n}, \ldots, S_{Dt_n}$ on the main diagonal.

Two alternative gradient strategies were explored. The first precomputes $\nabla f$ on a grid of asset prices and state variables, then interpolates. While feasible in low dimensions, this quickly suffers from the curse of dimensionality: even a coarse 20-point grid in three assets and one state variable requires $> 8 \times 10^5$ evaluations, each involving full simulation of $f$. Smaller grids reduce cost but cause frequent interpolation failures. The second trains neural networks to learn gradient surfaces, avoiding dimensionality but embedding a data-driven component into what is intended as an analytical framework, blurring separation from the deep BSDE approach. We therefore retain the Malliavin estimator as the most faithful and transparent method for the cost-unaware model of Basak and Chabakauri (2010).

### 4.2.4 Backtesting Framework

We implement a rolling window backtesting procedure for the cost-unaware model. On each rebalancing day, parameters are estimated from a trailing window of historical data, forward simulations are generated to compute anticipated gains and their sensitivities, and the optimal portfolio allocations $\boldsymbol{\theta}_t^*$ are obtained via Equation (4.9). The portfolio is then reallocated accordingly. The procedure is summarised in Algorithm 4.

Transaction costs are included in all backtests for comparability, modelled as a quadratic penalty $\lambda_{\text{day}} \|\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}\|^2$ on daily changes in monetary allocations. Since $\lambda_{\text{day}}$ is defined per day, no scaling by $\Delta t$ is applied, ensuring cost magnitudes are directly comparable across all daily-rebalanced strategies.

**Algorithm 4** Backtesting the Cost-Unaware Model

**Input:** Market data $(\mathbf{S}_{t_n}, \mathbf{X}_{t_n})$, initial wealth $W_{t_0}$, rolling window size, parameters $(r, \gamma, \lambda_{\text{day}}, T, N, M)$
Initialise wealth $W_{t_0}$, allocation $\boldsymbol{\theta}_{t_0} \leftarrow \mathbf{0}$
Define rebalancing indices from rolling window and $\Delta t$
**for** each rebalancing index $i$ **do**
    Estimate parameters from rolling window and simulate paths (Algorithm 1)
    Compute anticipated gains and gradients (Algorithms 2–3)
    Compute optimal allocation $\boldsymbol{\theta}^*_{t_n}$ via Equation (4.9)
    Update wealth:

$$W_{t_{n+1}} \leftarrow \left(\frac{\boldsymbol{\theta}_{t_n}}{\mathbf{S}_{t_n}}\right)^{\top} \mathbf{S}_{t_{n+1}} + \left(W_{t_n} - \boldsymbol{\theta}_{t_n}^{\top}\mathbf{1}\right)(1 + r\Delta t) - \lambda_{\text{day}} \|\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}\|^2$$

    Set $\boldsymbol{\theta}_{t_{n-1}} \leftarrow \boldsymbol{\theta}_{t_n}$
**end for**
**Output:** $\{W_{t_n}, \boldsymbol{\theta}_{t_n}\}_{n=0}^{N}$

## 4.3 Cost-Aware Model Implementation (Deep BSDE Approach)

We implement the cost-aware continuous-time mean–variance extension of Basak and Chabakauri (2010) using the deep BSDE framework introduced in Section 3.2.3. Neural networks are trained on simulated trajectories to approximate the optimal investment policy as a function of current market and portfolio states. Portfolio allocations are then updated daily during backtesting, ensuring consistency with the rolling framework of the cost-unaware model. The implementation mirrors the theoretical formulation while adapting it to empirical inputs.

### 4.3.1 Simulation Framework

Forward trajectories of asset prices and state variables are simulated using the discretised dynamics from Section 4.2.1 (Equations (4.1)–(4.5)), ensuring consistency with the analytical implementation.

Initial values $\mathbf{S}_0 \in \mathbb{R}^D$ and $\mathbf{X}_0 \in \mathbb{R}^K$ are taken from the first observation in the historical dataset.

The horizon is set to $T = 1$ year with $N = 30$ steps. Compared with daily discretisation in backtesting, this coarser grid yields smoother controls and lower turnover by reducing sensitivity to high-frequency noise.

Each training epoch uses $M = 64$ simulated paths, whereas the analytical implementation relied on $M = 30$ paths to approximate expectations. In the neural network setting, a larger sample improves data diversity for stable learning rather than merely reducing Monte Carlo error, while keeping computational cost reasonable. Unlike the analytical case, where parameters remain fixed at their historical estimates, we apply Gaussian jitter to the estimated parameters at the start of each path to prevent overfitting to a single parameter set:

$$\text{parameter}^{\text{jittered}} = \text{parameter} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \delta^2), \tag{4.10}$$

with perturbations $\delta_\mu = 0.005$, $\delta_\sigma = 0.002$, $\delta_m = \delta_\nu = 0.001$, $\delta_\rho = 0.02$. Values are clipped to preserve feasibility (e.g. positive volatilities), and the correlation matrix $\boldsymbol{\Sigma}$ is left unchanged to maintain positive semi-definiteness. Parameters remain fixed within each epoch and are re-estimated only between retraining periods (see Section 4.3.3).

Wealth then evolves according to the discretised transaction-cost recursion:

$$W_{t_{n+1}} = W_{t_n} + \left[ rW_{t_n} + \boldsymbol{\theta}_{t_n}^\top (\boldsymbol{\mu}_{t_n} - r\mathbf{1}) \right] \Delta t + \boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n} \Delta \mathbf{w}_n - \lambda \frac{\|\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}\|^2}{\Delta t}, \quad n \geq 1, \quad (4.11)$$

with $\boldsymbol{\theta}_{t_{-1}} = \mathbf{0}$ at $t_0$.

Here $\lambda$ denotes the annualised transaction cost coefficient. With $N = 30$ steps per year ($\Delta t = 1/30$), setting $\lambda = 1 \times 10^{-5}$ implies a daily cost coefficient of $\lambda_{\text{day}} = 0.0003$, consistent with the backtesting setup.

This framework generates the synthetic trajectories used to train the neural networks within the deep BSDE approach with the learned policy $\boldsymbol{\theta}_t$ applied dynamically during simulation. The procedure is summarised in Algorithm 5.

---

**Algorithm 5** Forward Simulation under Policy Network (Deep BSDE)

---

1: **Input:** Initial values $W_0, \mathbf{S}_0, \mathbf{X}_0$; parameters $(\boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{m}, \boldsymbol{\nu}, \boldsymbol{\rho}, \boldsymbol{\Sigma}, r, \lambda, T, N, M)$; policy network $\text{NN}_\phi$
2: Set $\Delta t = T/N$, precompute $\mathbf{L} = \text{Cholesky}(\boldsymbol{\Sigma})$
3: **for** $m = 1$ to $M$ **do**
4: Initialise $W_0^{(m)} = W_0, \mathbf{S}_0^{(m)} = \mathbf{S}_0, \mathbf{X}_0^{(m)} = \mathbf{X}_0, \boldsymbol{\theta}_{t_{-1}}^{(m)} = \mathbf{0}$
5: Apply Gaussian jitter to $\boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{m}, \boldsymbol{\nu}, \boldsymbol{\rho}$
6: **for** $n = 0$ to $N - 1$ **do**
7: Update policy: $\boldsymbol{\theta}_{t_n}^{(m)} \leftarrow W_{t_n}^{(m)} \cdot \text{NN}_\phi(t_n, W_{t_n}^{(m)}, \mathbf{S}_{t_n}^{(m)}, \mathbf{X}_{t_n}^{(m)})$
8: Sample $\mathbf{z}_{t_n} \sim \mathcal{N}(0, I_D)$, $\boldsymbol{\xi}_{t_n} \sim \mathcal{N}(0, I_K)$
9: Compute increments: $\Delta \mathbf{w}_{t_n} = \sqrt{\Delta t}\, \mathbf{L} \mathbf{z}_{t_n}$, $\Delta \mathbf{w}_{X,t_n} = \boldsymbol{\rho} \Delta \mathbf{w}_{t_n} + \sqrt{\Delta t}\, \boldsymbol{\xi}_{t_n}$
10: Update $\mathbf{S}_{t_{n+1}}^{(m)}, \mathbf{X}_{t_{n+1}}^{(m)}$ via Equations (4.1)–(4.2)
11: Update $W_{t_{n+1}}^{(m)}$ via Equation (4.11)
12: **end for**
13: **end for**
14: **Output:** $\{W_{t_n}^{(m)}, \mathbf{S}_{t_n}^{(m)}, \mathbf{X}_{t_n}^{(m)}, \boldsymbol{\theta}_{t_n}^{(m)}\}_{n=0}^N$, $m = 1, \ldots, M$

---

### 4.3.2 Neural Network Architecture

#### Control Network (Policy)

The control policy $\boldsymbol{\theta}_{t_n}$, introduced in Section 3.2.3, is parameterised by a neural network $\text{NN}_\phi$ with trainable weights $\phi$. At each time step, the network takes the current state $(W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n)$, where $W_{t_n}$ is wealth, $\mathbf{S}_{t_n} \in \mathbb{R}^D$ are risky asset prices, $\mathbf{X}_{t_n} \in \mathbb{R}^K$ are state variables, and $t_n$ is time. All inputs are standardised to zero mean and unit variance using statistics from the training window.

The network outputs a dimensionless allocation vector

$$\tilde{\boldsymbol{\theta}}_{t_n} = \text{NN}_\phi(W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n), \quad (4.12)$$

with $\tilde{\boldsymbol{\theta}}_{t_n} \in [-1, 1]^D$ due to the final Tanh activation. This represents scaled risky-asset allocations. To obtain monetary allocations consistent with the model, outputs are rescaled by current wealth:

$$\boldsymbol{\theta}_{t_n} = W_{t_n} \cdot \tilde{\boldsymbol{\theta}}_{t_n}, \quad (4.13)$$

The policy network is a fully connected feedforward architecture with one hidden layer. A summary is given in Table 4.1.

Table 4.1: Control network architecture for policy function $\mathrm{NN}_\phi$.

| **Component** | **Description** |
|---|---|
| Input | $W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n \in \mathbb{R}^{1+D+K+1}$ |
| Normalisation | Standardisation (zero mean, unit variance) |
| Hidden layer | Fully connected, 64 units, Tanh activation |
| Output layer | $\tilde{\boldsymbol{\theta}}_{t_n} \in [-1,1]^D$ (Tanh) |

The Tanh activation bounds outputs, stabilising training. A shallow design proved sufficient for convergence and balances capacity with interpretability.

#### Gradient and Curvature Networks

Two auxiliary networks approximate the gradient $Z_{t_n}$ and curvature $\Gamma_{t_n}$ required by the BSDE solver, denoted $\mathrm{NN}_\psi$ and $\mathrm{NN}_\xi$.

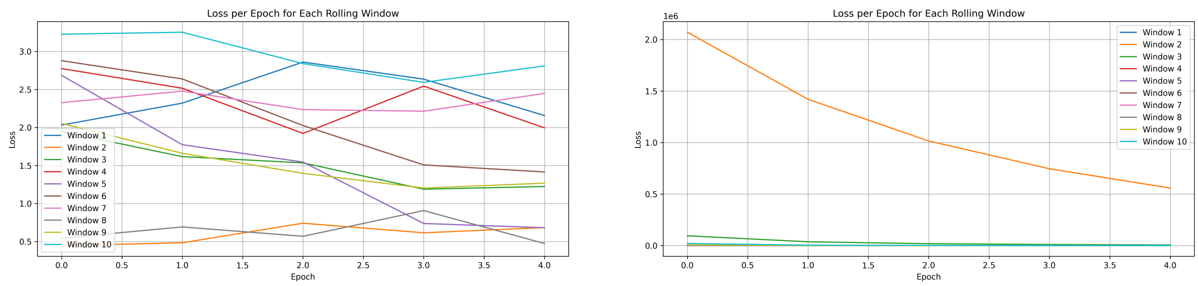
Both take the same standardised inputs $(W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}, t_n)$ and mirror the architecture in Table 4.1. Each is trained independently, with output layers adapted to produce scalar $Z_{t_n}$ and $\Gamma_{t_n}$, respectively.

This shared design ensures consistent state encoding and efficient implementation. For stability, pathological outputs (NaNs, extreme values) are clipped or replaced during training.

### 4.3.3 Training Methodology

We implement the two training schemes introduced in Section 3.2.3, where their theoretical formulation was detailed: the global loss of Beck et al. (2019) and the stepwise loss of Huré et al. (2020). Both integrate naturally into our deep 2BSDE framework but differ in how the value function is propagated backward in time.

While both schemes are theoretically valid, the stepwise formulation converges faster and more reliably, both in theory (Huré et al., 2020) and in practice. In our rolling experiments it typically stabilises within five epochs per window, whereas the global loss often stagnates (Figure 4.1). We therefore adopt the stepwise loss with a fixed budget of five epochs per window, while the global approach is summarised in Appendix G.




(a) Global loss (b) Stepwise loss

Figure 4.1: Comparison of training loss trajectories.

#### Stepwise Loss (DBDP)

The stepwise scheme follows the DBDP1 principle of Huré et al. (2020), enforcing the one-step Bellman relation locally. At time $t_n$, the networks output the control $\tilde{\boldsymbol{\theta}}_{t_n}$, the derivative $Z_{t_n}$, and, in our extended setting, the curvature $\Gamma_{t_n}$. These define a BSDE-consistent target

$\hat{Y}_{t_n}$, constructed from the continuation value $Y_{t_{n+1}}$ together with generator terms and Brownian increments. Training minimises the discrepancy between $\hat{Y}_{t_n}$ and $Y_{t_{n+1}}$, ensuring dynamic consistency at each grid point (Algorithm 6).

**Algorithm 6** Stepwise Loss Training (DBDP-Inspired)

Initialise neural networks $\mathrm{NN}_\phi, \mathrm{NN}_\psi, \mathrm{NN}_\xi$
**for** each epoch **do**
  Simulate forward paths $\{W_t, \mathbf{S}_t, \mathbf{X}_t, \Delta \mathbf{w}_t\}$ using Algorithm 5
  Set terminal condition: $Y_T = J_T(W_T) = W_T - \frac{\gamma}{2} W_T^2$
  **for** each $t_n = t_{N-1}, \dots, t_0$ **do**
    Evaluate neural networks and scale allocation:

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_{t_n} &\leftarrow \mathrm{NN}_\phi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}) \\
\boldsymbol{\theta}_{t_n} &\leftarrow W_{t_n} \tilde{\boldsymbol{\theta}}_{t_n} \\
Z_{t_n} &\leftarrow \mathrm{NN}_\psi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}) \\
\Gamma_{t_n} &\leftarrow \mathrm{NN}_\xi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n})
\end{aligned}$$

    Compute BSDE-consistent target:

$$\begin{aligned}
\hat{Y}_{t_n} = Y_{t_{n+1}} + \Bigg( \bigg[ r W_{t_n} + \boldsymbol{\theta}_{t_n}^\top (\boldsymbol{\mu}_{t_n} - r\mathbf{1}) - \lambda \bigg\| \frac{\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}}{\Delta t} \bigg\|^2 \bigg] Z_{t_n} \\
+ \frac{1}{2} \Gamma_{t_n} \boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n} \boldsymbol{\sigma}_{t_n}^\top \boldsymbol{\theta}_{t_n} \Bigg) \Delta t - Z_{t_n} \boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n} \Delta \mathbf{w}_{t_n}.
\end{aligned}$$

    Compute local loss: $\mathcal{L}_n = |\hat{Y}_{t_n} - Y_{t_{n+1}}|^2$
    Update parameters via backpropagation
  **end for**
**end for**

This formulation embeds wealth dynamics, transaction costs, curvature effects, and stochastic shocks directly into the learning signal. By enforcing the recursion step by step rather than only at maturity, it yields richer gradients and greater stability over long horizons. The sequential structure also aligns with our rolling backtest, where allocations are updated through time, ensuring methodological coherence between training and evaluation. Training uses the Adam optimiser with learning rate $10^{-3}$ and risk aversion parameter $\gamma = 3$, consistent with Section 4.2.

### Rolling Window Training

A natural baseline is fixed-window training, where parameters such as drift, volatility, and correlation are estimated from a single historical period and neural networks are trained once using simulated paths. While computationally efficient, this approach assumes stationarity of market dynamics and consistently underperformed in our experiments, as it failed to adapt to time-varying regimes.

To address this limitation, we adopt a rolling window framework. At each retraining date, parameters are re-estimated from the most recent $L$ days of data, and the policy network is updated accordingly. This produces a sequence of models tailored to prevailing conditions.

We use a one-year (252-day) window with matching retraining frequency, balancing statistical reliability with computational cost. More frequent updates yielded little additional benefit, whereas less frequent updates produced stale policies.

To avoid abrupt allocation shifts between windows, which would translate into excessive transaction costs, we add an anchor penalty aligning the new policy's initial output with the previous policy's final allocation:

$$\mathcal{L}_{\text{anchor}} = \eta \big\| W_{t_0} \boldsymbol{\theta}_{t_0} - W_{t_0} \hat{\boldsymbol{\theta}}_T^{\text{prev}} \big\|^2, \tag{4.14}$$

where $\eta > 0$ controls its strength. Networks are also warm-started from the previous window to stabilise convergence, complementing the anchor penalty. We set $\eta = 10$, which promotes smooth transitions while still allowing adaptation to updated parameters. Larger values risk over-smoothing, while smaller ones permit abrupt allocation jumps and inflated transaction costs.

This rolling scheme (summarised in Algorithm 7) therefore promotes stable yet adaptive portfolio policies, capturing time-varying dynamics while controlling transaction costs, and forms the basis of our backtesting experiments.

**Algorithm 7** Rolling Window Training

```
Define window size L, retraining frequency f
for each retraining date t_n do
    Extract data window {t_{n-L+1}, ..., t_n}
    Estimate parameters μ, σ, ρ, m, ν, Σ
    Initialise training with these parameters and warm-start networks
    Apply anchor penalty from previous allocation
    Train on simulated paths
    Store networks for backtesting
end for
```

### 4.3.4 Backtesting Framework

We implement a backtesting framework to evaluate the cost-aware portfolio strategy on historical market data. At each trading day, the trained policy network generates portfolio allocations from current market inputs, which are scaled into monetary allocations and used to update wealth under realistic trading frictions (Algorithm 8).

**Algorithm 8** Backtesting the Cost-Aware Model

1: **Input:** Market data $(\mathbf{S}_{t_n}, \mathbf{X}_{t_n})$, trained neural networks $\{\text{model}_k\}$, retraining dates $\{t_k\}$, initial wealth $W_{t_0}$, parameters $(r, \lambda_{\text{day}}, T, N)$
2: Initialise $W_{t_0} \leftarrow$ initial wealth, model index $k \leftarrow 0$
3: **for** each time step $t_n$ in trading horizon **do**
4: **if** $k+1 < |\{t_k\}|$ and $t_n \geq t_{k+1}$ **then** update $k \leftarrow k+1$
5: **end if**
6: Evaluate policy network:

$$\tilde{\boldsymbol{\theta}}_{t_n} \leftarrow \text{NN}_\phi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n})$$

7: Scale to monetary allocation:

$$\boldsymbol{\theta}_{t_n} = W_{t_n} \cdot \tilde{\boldsymbol{\theta}}_{t_n}$$

8: Update wealth:

$$W_{t_{n+1}} \leftarrow \left(\frac{\boldsymbol{\theta}_{t_n}}{\mathbf{S}_{t_n}}\right)^\top \mathbf{S}_{t_{n+1}} + (W_{t_n} - \boldsymbol{\theta}_{t_n}^\top \mathbf{S}_{t_n})(1 + r\Delta t) - \lambda_{\text{day}} \|\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}\|^2$$

9: **end for**
10: **Output:** $\{W_{t_n}, \boldsymbol{\theta}_{t_n}\}_{n=0}^N$

Different trained models $\{\text{model}_k\}$ are applied at their corresponding retraining dates $\{t_k\}$, consistent with our rolling training setup. This structure ensures that the policy adapts over time as new market information becomes available.

## 4.4 Parametric Bootstrap Backtesting Framework

To assess statistical robustness, we implement a parametric bootstrap backtesting framework based on simulated market paths. Paths follow the dynamics in Section 4.2.1: risky assets evolve as geometric Brownian motion (GBM) and the state variable as ABM correlated with returns. This preserves key empirical features while generating plausible scenarios, consistent with simulation-based robustness tests in the literature (de Prado and Zoonekynd, 2025).

We generate $B = 100$ synthetic market paths, balancing statistical resolution and runtime. Larger $B$ did not materially alter performance distributions but substantially increased cost, especially for the analytical approach, which requires repeated gradient-based optimisation.

Each path spans 2015–2025 with parameters re-estimated annually from a 252-day trailing window (e.g. 2014 estimates simulate 2015). Within each block, $\mathbf{S}_0$ and $\mathbf{X}_0$ are initialised from the previous year's final values, ensuring temporal continuity. This rolling design mirrors neural network training in our deep BSDE-based approach, captures regime shifts and persistence in volatility across yearly blocks, and generates diverse alternative histories.

Each synthetic path is evaluated with the relevant backtesting algorithm. For the deep BSDE approach, we reuse policy networks trained on historical data rather than retraining on simulated paths. This avoids both the cost of repeated training and degraded signals: simplified dynamics weaken dependencies between assets and state variables, reducing training informativeness. Reusing historical policies ensures comparability and realism.

# Chapter 5

# Results and Discussion

This chapter introduces the performance metrics and statistical tests used to evaluate portfolio strategies. As a baseline, we employ a buy-and-hold (B&H) portfolio with equal weights in the three risky assets and the risk-free asset at the start of the backtest period. We then present results for our optimisation models: the cost-unaware case, solved analytically, and the cost-aware case, implemented via the deep BSDE approach. Each is compared against B&H and, where relevant, against one another. Results are reported for both historical and bootstrap backtests.

## 5.1 Evaluation Metrics and Statistical Methodology

This section introduces the performance metrics and hypothesis tests used to evaluate portfolio strategies in historical and bootstrap backtests. The metrics capture portfolio performance, risk, and trading activity, providing a comprehensive view of profitability and robustness. Hypothesis tests formally compare strategies against benchmarks, ensuring that observed differences are statistically significant rather than due to sampling variation. Throughout, the benchmark $B$ is the B&H strategy.

**Return and Risk Metrics**
Terminal wealth $W_T$ is the final portfolio value. The compound annual growth rate (CAGR) is

$$\mu^{\text{ann}} = \left(\frac{W_T}{W_0}\right)^{1/T} - 1, \tag{5.1}$$

where $W_0$ is initial wealth and $T$ the backtest length (years).

Annualised volatility is

$$\sigma^{\text{ann}} = \sqrt{252}\,\sigma, \tag{5.2}$$

with $\sigma$ the standard deviation of daily arithmetic returns $R_t$.

Downside risk is measured by 95% Value-at-Risk (VaR) and Expected Shortfall (ES):

$$\text{VaR}_{95} = -\text{quantile}_{0.05}(R_t), \qquad \text{ES}_{95} = -\mathbb{E}[R_t \mid R_t \leq -\text{VaR}_{95}]. \tag{5.3}$$

VaR gives the worst expected daily loss at 95% confidence, while ES captures the average loss in such extreme events, making it a more coherent measure of tail risk. Together, VaR and ES summarise both the threshold of losses and the behaviour of the tail.

**Risk-Adjusted Performance Metrics**
The Sharpe ratio

$$SR = \frac{\mu^{\text{ann}} - r}{\sigma^{\text{ann}}}, \qquad r = 2.5\%, \tag{5.4}$$

summarises the return–volatility trade-off central to mean–variance theory, and is therefore the most direct measure of whether our strategy improves on the classical benchmarks.

The Sortino ratio penalises only downside volatility,

$$\text{Sortino} = \frac{\mu^{\text{ann}} - r}{\sigma^{-\,\text{ann}}}, \qquad \sigma^{-\,\text{ann}} = \sqrt{252} \cdot \sqrt{\tfrac{1}{N} \sum_{R_t<0} R_t^2}, \tag{5.5}$$

and the Calmar ratio scales return by maximum drawdown (MDD):

$$\text{Calmar} = \frac{\mu^{\text{ann}}}{\text{MDD}}. \tag{5.6}$$

Unlike the Sharpe or Sortino ratios, the Calmar ratio is particularly relevant for long-horizon strategies, as it penalises prolonged drawdowns rather than short-lived volatility spikes. This highlights the ability of a strategy to recover from sustained losses.

**Drawdown and Recovery Metrics**
MDD is the largest peak-to-trough loss:

$$\text{MDD} = \max_{t\in[0,T]} \left( \frac{\max_{s\leq t} W_s - W_t}{\max_{s\leq t} W_s} \right). \tag{5.7}$$

Average drawdown is the mean depth across all drawdowns; average recovery time is the mean number of days to return to a prior peak. These metrics emphasise downside risk in cumulative wealth paths, complementing volatility-based measures.

**Turnover, Costs, and Exposure Metrics**
Portfolio weights are $\boldsymbol{\pi}_t = \boldsymbol{\theta}_t / W_t$.

Turnover is trading activity per step relative to wealth:

$$\text{Turnover}_t = \frac{\sum_{i=1}^{d} |\theta_{t,i} - \theta_{t-1,i}|}{W_{t-1}}. \tag{5.8}$$

Average turnover is $\frac{1}{N}\sum_{t=1}^{N} \text{Turnover}_t$; annual turnover multiplies this by the trading frequency $1/\Delta t$. Even B&H strategies exhibit non-zero turnover due to relative price movements. Thus, turnover captures both active trading and passive weight drift, with only the former generating realised costs.

Transaction costs are aggregated across trades. Profit per turnover reports return per unit of trading activity:

$$\text{Profit per Turnover} = \frac{\text{Total Profit}}{\sum_{t=1}^{N} \sum_{i=1}^{d} |\theta_{t,i} - \theta_{t-1,i}|}. \tag{5.9}$$

Average leverage is the mean risky exposure relative to wealth:

$$\overline{L} = \frac{1}{N+1} \sum_{t=0}^{N} \frac{\sum_{i=1}^{d} |\theta_{t,i}|}{W_t}, \tag{5.10}$$

and concentration is measured by maximum position size:

$$\max_{t,i} |\theta_{t,i}|. \tag{5.11}$$

**Portfolio Stability Metrics**
These metrics are computed on risky assets only, with the cash position excluded.

Weight volatility is the cross-asset average of time-series standard deviations:

$$\sigma_\pi = \frac{1}{d}\sum_{i=1}^{d} \mathrm{sd}(\pi_{t,i}). \tag{5.12}$$

Average absolute change in weights (AAC) is

$$\mathrm{AAC}(\pi) = \frac{1}{N}\sum_{t=1}^{N}\sum_{i=1}^{d} |\pi_{t,i} - \pi_{t-1,i}|. \tag{5.13}$$

Both capture the smoothness of allocation paths, which affects transaction costs and practical implementability.

**Statistical Hypothesis Testing**

Two tests compare strategy $A$ to $B$: Sharpe ratio differences and return differences. Each is applied in the historical and bootstrap backtests. We focus on Sharpe and CAGR for formal testing, as they capture the two core questions: whether strategies improve mean–variance efficiency and whether they deliver higher long-run growth.

(i) **Sharpe ratio differences.**

$$H_0 : SR_A \leq SR_B, \qquad H_1 : SR_A > SR_B, \tag{5.14}$$

with statistic

$$z = \frac{SR_A - SR_B}{\hat{\sigma}_{SR_A - SR_B}}, \qquad p\text{-value} = 1 - \Phi(z). \tag{5.15}$$

The historical test follows the Jobson and Korkie (1981) statistic with the finite-sample correction of Memmel (2003), which assumes i.i.d. returns.[1] The bootstrap test instead evaluates Sharpe ratio differences nonparametrically:

$$D_i = SR_A^{(i)} - SR_B^{(i)}, \qquad i = 1, \ldots, B, \tag{5.16}$$

with

$$\bar{D} = \tfrac{1}{B}\sum_{i=1}^{B} D_i, \qquad \hat{\sigma}_D^2 = \tfrac{1}{B-1}\sum_{i=1}^{B}(D_i - \bar{D})^2, \tag{5.17}$$

$$t = \frac{\bar{D}}{\hat{\sigma}_D/\sqrt{B}}, \qquad p\text{-value} = 1 - \Phi(t). \tag{5.18}$$

By dropping the i.i.d. assumption, the bootstrap provides a more robust check of Sharpe outperformance. We also report the outperformance probability

$$\widehat{\mathbb{P}}(SR_A > SR_B) = \tfrac{1}{B}\sum_{i=1}^{B} \mathbf{1}\{D_i > 0\}, \tag{5.19}$$

which gives an intuitive measure of the frequency with which a strategy beats the benchmark.

(ii) **Return differences.**

$$H_0 : \mathbb{E}[\mu_A^{\mathrm{ann}} - \mu_B^{\mathrm{ann}}] \leq 0, \qquad H_1 : \mathbb{E}[\mu_A^{\mathrm{ann}} - \mu_B^{\mathrm{ann}}] > 0. \tag{5.20}$$

Historical backtest (paired $t$-test across $n$ yearly blocks).

$$D_i = \mu_{A,i}^{\mathrm{ann}} - \mu_{B,i}^{\mathrm{ann}}, \qquad i = 1, \ldots, n, \tag{5.21}$$

[1] The variance estimator is $\hat{\sigma}^2_{SR_A - SR_B} = \frac{1}{T}\left(1 + \frac{1}{2}SR_A^2\right) + \frac{1}{T}\left(1 + \frac{1}{2}SR_B^2\right)$, with $T$ the sample size.

$$\bar{D} = \tfrac{1}{n}\sum_{i=1}^{n} D_i, \qquad s_D^2 = \tfrac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2, \tag{5.22}$$

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}, \qquad p\text{-value} = 1 - F_t(t;\, n-1). \tag{5.23}$$

This test assumes block-level returns are i.i.d., which may be restrictive in financial data. Bootstrap backtest (normal approximation for large $B$).

$$D_i = \mu_A^{\mathrm{ann},(i)} - \mu_B^{\mathrm{ann},(i)}, \qquad i = 1, \dots, B, \tag{5.24}$$

$$\bar{D} = \tfrac{1}{B}\sum_{i=1}^{B} D_i, \qquad \hat{\sigma}_D^2 = \tfrac{1}{B-1}\sum_{i=1}^{B}(D_i - \bar{D})^2, \tag{5.25}$$

$$t = \frac{\bar{D}}{\hat{\sigma}_D/\sqrt{B}}, \qquad p\text{-value} = 1 - \Phi(t). \tag{5.26}$$

By dropping the i.i.d. assumption, the bootstrap provides a more robust check of return outperformance. We also report the outperformance probability

$$\widehat{\mathbb{P}}(\mu_A^{\mathrm{ann}} > \mu_B^{\mathrm{ann}}) = \tfrac{1}{B}\sum_{i=1}^{B} \mathbf{1}\{D_i > 0\}, \tag{5.27}$$

which gives the fraction of bootstrap replications in which strategy $A$ exceeds B&H, offering a practical measure of relative dominance.

**Bootstrap Diagnostics**

Any historical metric $M_{\mathrm{hist}}$ is compared to its bootstrap distribution $\{M^{(i)}\}_{i=1}^{B}$:

$$z = \frac{M_{\mathrm{hist}} - \tilde{m}}{s}, \qquad p\text{-value} = 2\big(1 - \Phi(|z|)\big), \tag{5.28}$$

$$p_{\mathrm{hist}} = \tfrac{1}{B}\sum_{i=1}^{B} \mathbf{1}\{M^{(i)} \geq M_{\mathrm{hist}}\}, \tag{5.29}$$

where $\tilde{m}$ is the bootstrap median and $s$ the bootstrap standard deviation. The $p$-value tests whether the historical value is extreme relative to the bootstrap distribution, while $p_{\mathrm{hist}}$ reports the empirical probability of exceeding it. Together, these distinguish statistical significance (extremeness) from empirical ranking (relative performance). Tables additionally report the bootstrap median and 5th/95th percentiles to summarise the distribution.

## 5.2 Benchmark Buy-and-Hold Strategy: Results

In this section, we present the performance of the B&H strategy over the 10-year backtest period in both historical and bootstrap settings. As the simplest passive allocation with no rebalancing or transaction costs, it provides a natural baseline for assessing whether our active strategies add value beyond a naïve passive investment.

### 5.2.1 Historical Backtest

Table 5.1 reports performance of the B&H strategy, with Figures 5.1–5.5 illustrating the underlying dynamics. Terminal wealth reached £459.62, a CAGR of 16.51%, reflecting strong absolute growth over the ten-year horizon, as shown in Figure 5.1. Annual volatility of 18.74% is consistent with equity-heavy portfolios, while downside risk was substantial: the portfolio suffered several sharp drawdowns, the largest being 32.26% (Figure 5.2), and tail-loss measures

(95% VaR of 1.76%, ES of 2.76%) confirm exposure to severe equity market corrections. This underscores B&H's role as a natural benchmark: it fully reflects equity risk premia but also leaves the portfolio exposed to large drawdowns without any mechanism for risk control.

Risk-adjusted returns were moderate: a Sharpe ratio of 0.75 and Sortino ratio of 0.97 indicate that returns compensated for risk but without exceptional efficiency, while the Calmar ratio of 0.51 underscores the drag of drawdowns on long-run performance. Daily returns were centred close to zero with a roughly symmetric distribution (Figure 5.3), and the rolling Sharpe ratio (Figure 5.4) highlights substantial instability, including extended periods of negative values despite a positive full-sample average. This instability highlights the limitations of a passive allocation, providing a useful baseline for assessing whether active models can deliver more consistent performance.

Turnover averaged 1.01% per step (253.51% annually) despite no active rebalancing, arising solely from shifting relative prices. Figure 5.5 shows how this translated into gradually shifting weights: Apple's appreciation led to a maximum position size of £265.97, illustrating the concentration risk that can accumulate under a passive strategy. As expected, leverage and transaction costs were absent.

| Metric | Value |
|---|---|
| Terminal Wealth (£) | 459.62 |
| CAGR (%) | 16.51 |
| Annual Volatility (%) | 18.74 |
| Sharpe Ratio | 0.75 |
| Sortino Ratio | 0.97 |
| Calmar Ratio | 0.51 |
| Maximum Drawdown (%) | 32.26 |
| Average Drawdown (%) | 1.11 |
| Average Recovery Time (days) | 8.18 |
| VaR 95% (%) | 1.76 |
| Expected Shortfall 95% (%) | 2.76 |
| Average Leverage | 1.00 |
| Maximum Position Size (£) | 265.97 |
| Average Turnover per Step (%) | 1.01 |
| Annual Turnover (%) | 253.51 |
| Standard Deviation of Portfolio Weights (%) | 7.37 |
| Average Absolute Change in Portfolio Weights (%) | 0.62 |
| Profit per Turnover (£) | 0.07 |
| Total Transaction Costs (£) | 0.00 |

Table 5.1: Performance metrics (B&H, historical).

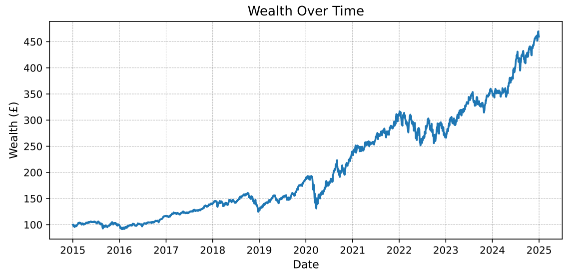



Figure 5.1: Wealth (B&H, historical).

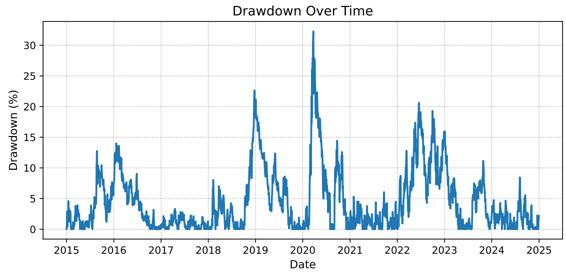



Figure 5.2: Drawdowns (B&H, historical).

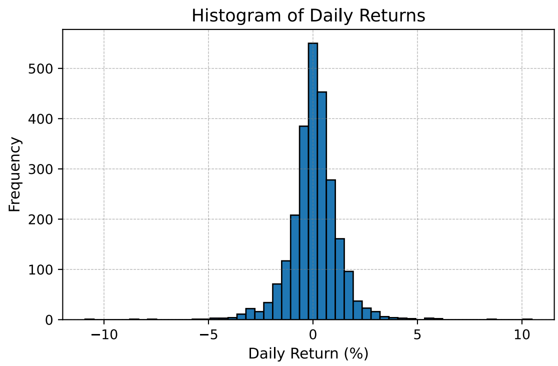



Figure 5.3: Return distribution (B&H, historical).

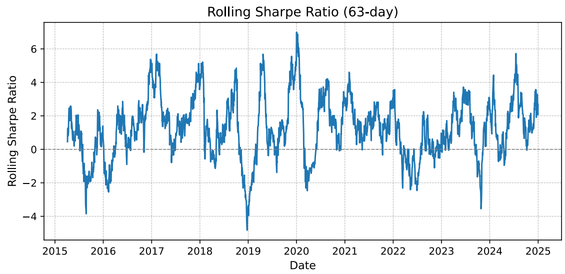



Figure 5.4: Rolling 63-day Sharpe ratio (B&H, historical).

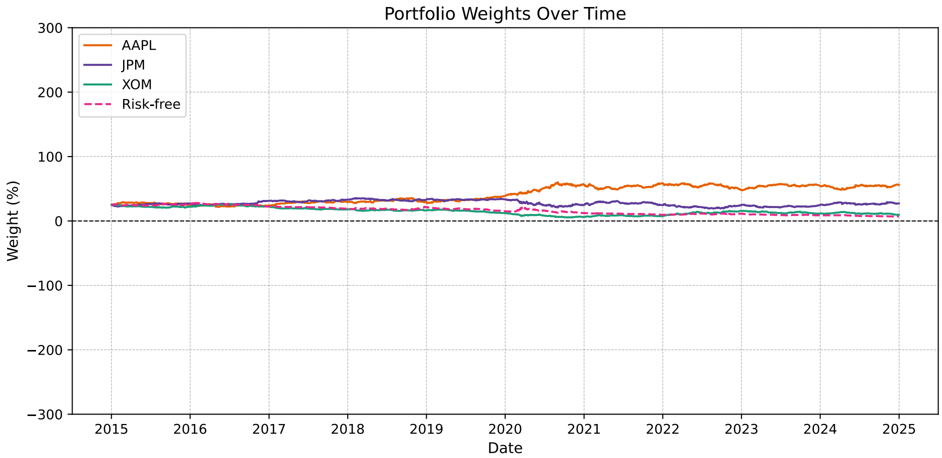



Figure 5.5: Portfolio weights (B&H, historical).

### 5.2.2 Parametric Bootstrap Backtest

Table 5.2 compares historical results with their bootstrap distributions. Since B&H serves as a baseline, the bootstrap provides a check on whether the historical path is representative of alternative simulated histories, rather than a test of outperformance. Across core performance metrics, bootstrap medians are close to historical values and $p$-values are large, indicating no statistically meaningful deviation. The only exception is annual turnover, where the bootstrap median is modestly higher ($p \approx 0.03$). This reflects that turnover in a passive B&H strategy is mechanically driven by relative price movements: Apple's dominance in the historical path dampened turnover, while bootstrap paths with alternative relative winners induced more frequent shifts. Overall, the bootstrap confirms that B&H outcomes are representative under the assumed dynamics, with no evidence of systematic bias, making it a reliable reference point against which to judge active strategies.

| Metric | Historical | Bootstrap Median | 5th %ile | 95th %ile | p-value | Prob. of Outperformance (%) |
|---|---|---|---|---|---|---|
| Terminal Wealth (£) | 459.62 | 542.87 | 234.75 | 1,693.45 | 0.911 | 55.00 |
| CAGR (%) | 16.51 | 17.77 | 8.60 | 31.43 | 0.873 | 54.00 |
| Annual Volatility (%) | 18.74 | 20.07 | 17.75 | 21.97 | 0.305 | 83.00 |
| Sharpe Ratio | 0.75 | 0.75 | 0.33 | 1.40 | 0.993 | 51.00 |
| Sortino Ratio | 0.97 | 1.06 | 0.49 | 2.03 | 0.862 | 56.00 |
| Calmar Ratio | 0.51 | 0.47 | 0.18 | 1.07 | 0.879 | 43.00 |
| Maximum Drawdown (%) | 32.26 | 36.54 | 26.48 | 57.24 | 0.682 | 70.00 |
| Average Drawdown (%) | 1.11 | 1.80 | 1.55 | 2.43 | 0.097 | 100.00 |
| VaR 95% (%) | 1.76 | 1.88 | 1.70 | 2.09 | 0.336 | 83.00 |
| Expected Shortfall 95% (%) | 2.76 | 2.81 | 2.53 | 3.08 | 0.771 | 61.00 |
| Annual Turnover (%) | 253.51 | 280.81 | 260.03 | 298.38 | 0.034 | 97.00 |
| Average Absolute Change in Portfolio Weights (%) | 0.62 | 0.62 | 0.52 | 0.67 | 0.971 | 52.00 |
| Total Transaction Costs (£) | 0.00 | 0.00 | 0.00 | 0.00 | N/A | 100.00 |

Table 5.2: Performance metrics (B&H, bootstrap).

Figure 5.6 shows the evolution of wealth across bootstrap trials with a 90% confidence interval. Dispersion widens markedly over time, reflecting the strong path-dependence of B&H: without rebalancing, portfolio outcomes diverge as relative asset prices evolve. This highlights the lack of stable long-run wealth trajectories under a passive allocation. Figure 5.7 shows that terminal wealth is right-skewed: most paths end in the low–mid range, while a small number of extreme runs produce very large values. Figure 5.8 shows that Sharpe ratios are centred around 0.6–0.8, close to the historical value of 0.75, with moderate variation, while Figure 5.9 shows annual turnover tightly clustered in the high-200% range (roughly 270–300%), consistent with Table 5.2.

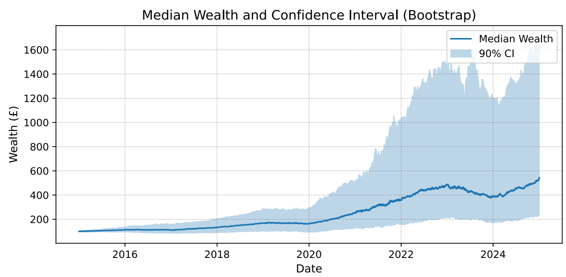



Figure 5.6: Wealth paths with 90% confidence interval (B&H, bootstrap).

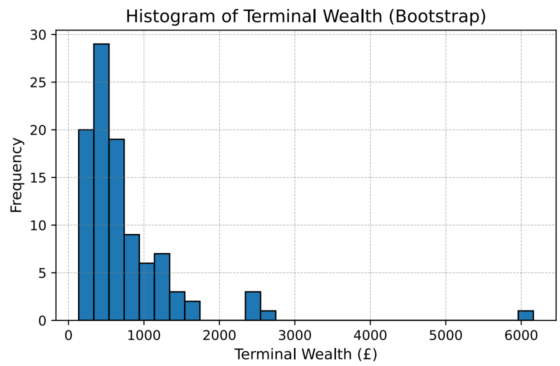



Figure 5.7: Distribution of terminal wealth (B&H, bootstrap).

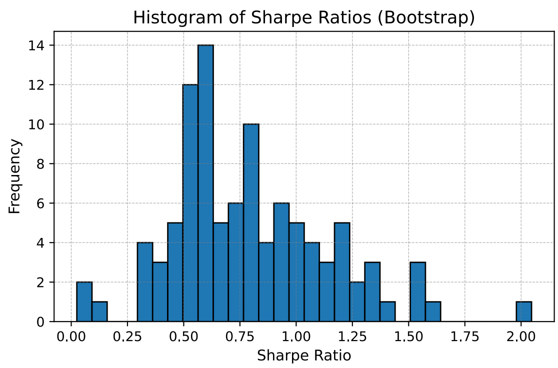



Figure 5.8: Distribution of Sharpe ratios (B&H, bootstrap).

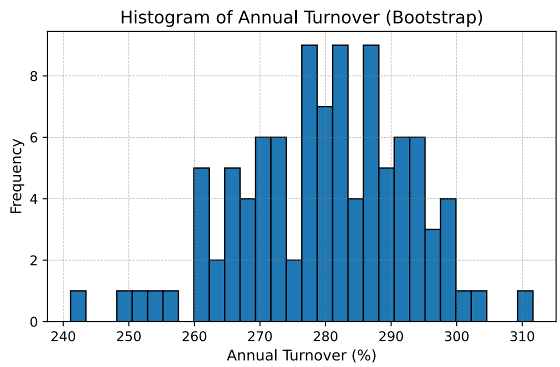



Figure 5.9: Distribution of annual turnover (B&H, bootstrap).

## 5.3 Cost-Unaware Model (Analytical Approach): Results

In this section, we evaluate the performance of the continuous-time cost-unaware model of Basak and Chabakauri (2010) over the 10-year historical backtest. Bootstrap results are omitted from the main text, as they are statistically indistinguishable from the historical outcomes; full details are provided in Appendix H.3.

### 5.3.1 Historical Backtest

Table 5.3 summarises the historical backtest results for the cost-unaware model. The strategy achieves a terminal wealth of only £136.15, corresponding to a CAGR of 3.18%, far below the benchmark B&H portfolio. Annual volatility is just 3.35%, substantially lower than the benchmark level of 18.74%. However, this apparent risk reduction stems not from effective diversification but from the portfolio's very low average leverage (0.13), which kept risky exposure minimal and prevented meaningful return generation. Consequently, risk-adjusted performance is extremely weak: the Sharpe ratio is 0.20 and the Sortino ratio 0.25. Formal inference confirms no outperformance of the B&H at any confidence level, with both Sharpe and returns-difference tests yielding $p > 0.95$.

| Metric | Value |
|---|---|
| Terminal Wealth (£) | 136.15 |
| CAGR (%) | 3.18 |
| Annual Volatility (%) | 3.35 |
| Sharpe Ratio | 0.20 |
| Sortino Ratio | 0.25 |
| Calmar Ratio | 0.48 |
| Maximum Drawdown (%) | 6.64 |
| Average Drawdown (%) | 0.20 |
| Average Recovery Time (days) | 7.05 |
| VaR 95% (%) | 0.23 |
| Expected Shortfall 95% (%) | 0.46 |
| Sharpe Difference t-stat | -18.73 |
| Sharpe Difference p-value | 1.0000 |
| Returns Difference t-stat | -2.25 |
| Returns Difference p-value | 0.9877 |
| Average Leverage | 0.13 |
| Maximum Position Size (£) | 82.49 |
| Average Turnover per Step (%) | 18.54 |
| Annual Turnover (%) | 4671.89 |
| Profit per Turnover (£) | 0.00 |
| Standard Deviation of Portfolio Weights (%) | 7.52 |
| Average Absolute Change in Portfolio Weights (%) | 18.53 |
| Total Transaction Costs (£) | 321.25 |

Table 5.3: Performance metrics (cost-unaware model, historical).

Figures 5.10 and 5.11 show sluggish wealth growth and frequent small drawdowns. Drawdowns are shallower than B&H (maximum 6.64%, average 0.20%), but this simply reflects the model's very low leverage and correspondingly subdued return variance. Likewise, downside risk measures (VaR 0.23%, ES 0.46%) appear small only because of the muted risk exposure, not because of genuine risk control. Recovery times are almost identical to B&H (7 vs. 8 days), offering no resilience benefit.

The return distribution in Figure 5.12 is tightly concentrated around zero, with nearly all returns in $[-1\%, 1\%]$. By contrast, the benchmark shows heavier tails in $[-5\%, 5\%]$, reflecting meaningful exposure to asset risk premia. Rolling Sharpe ratios (Figure 5.13) oscillate but remain negative for extended periods, consistent with the poor long-run Sharpe ratio.

Turnover is the most striking feature. As trading frictions are ignored, the portfolio rebalances aggressively, with average turnover of 18.54% per step and annual turnover 4,671.89% (Table 5.3). Figures 5.14 and 5.15 show daily turnover often exceeding 100% of wealth, producing cumulative costs of £321.25 despite negligible gains (profit per turnover $\approx 0$). The weight dynamics in Figure 5.16 confirm this instability: allocations swing violently between assets and the risk-free account, with an average absolute change in weights of 18.53% per step, quantifying the lack of stability.

In sum, the cost-unaware model performs substantially worse than B&H. Shallow drawdowns and low downside risk are artefacts of minimal leverage, not genuine robustness, while extreme turnover and trading frictions erode returns completely. These results underscore the importance of incorporating transaction costs explicitly, motivating the deep BSDE approach in Section 5.4.

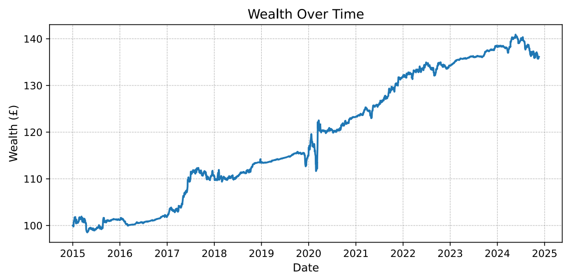



Figure 5.10: Wealth (cost-unaware model, historical).

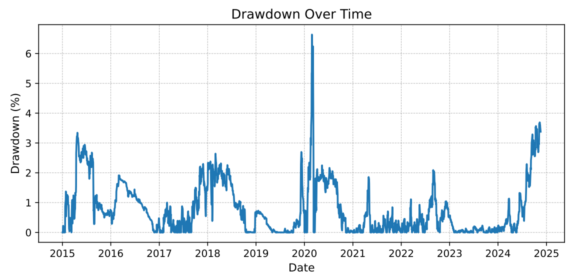



Figure 5.11: Drawdowns (cost-unaware model, historical).

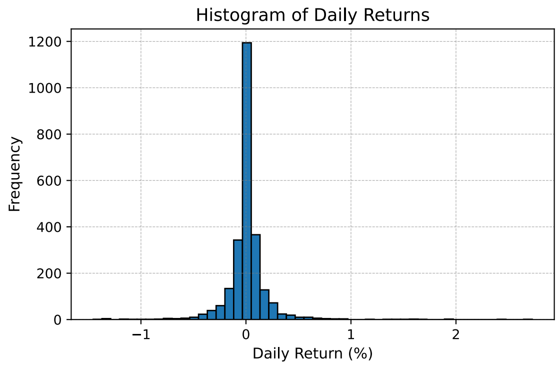



Figure 5.12: Return distribution (cost-unaware model, historical).

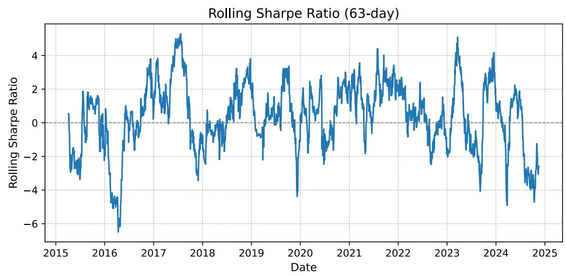



Figure 5.13: Rolling 63-day Sharpe ratio (cost-unaware model, historical).

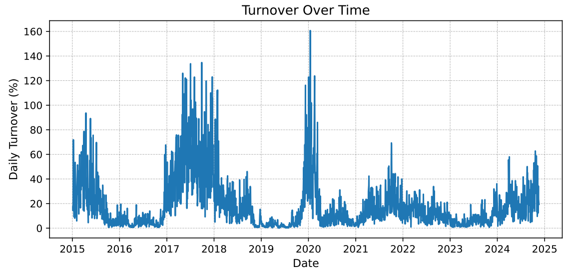



Figure 5.14: Daily turnover (cost-unaware model, historical).

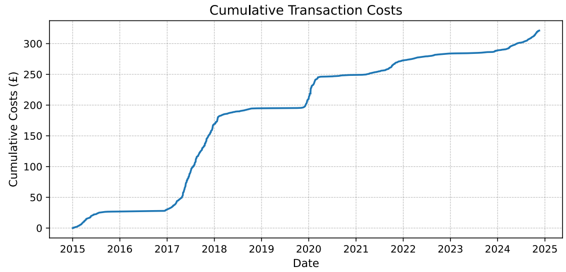



Figure 5.15: Cumulative costs (cost-unaware model, historical).

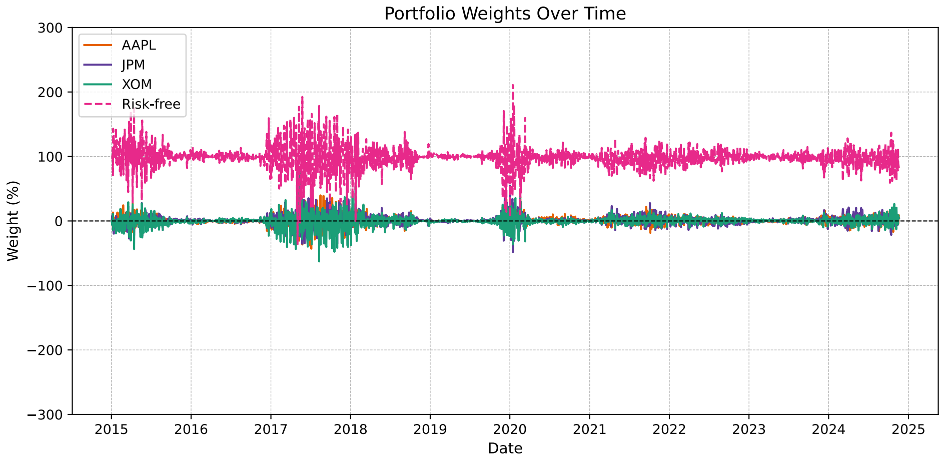



Figure 5.16: Portfolio weights (cost-unaware model, historical).

## 5.4 Cost-Aware Model (Deep BSDE Approach): Results

We now turn to the continuous-time cost-aware model, implemented through our deep BSDE approach. This is the core methodological contribution of the thesis, as transaction costs are embedded directly into the optimisation procedure rather than imposed ex post, in contrast to the cost-unaware formulation.

Results are presented in two parts. First, we examine the historical backtest to assess realised performance relative to the benchmark B&H portfolio. Second, we evaluate robustness using a parametric bootstrap, testing whether the observed improvements hold across simulated sample paths. Together, these analyses provide a comprehensive assessment of whether the deep BSDE approach can deliver superior performance while overcoming the weaknesses of the cost-unaware model.

### 5.4.1 Historical Backtest

Table 5.4 reports the historical backtest results. The cost-aware model achieved a terminal wealth of £919.53, more than doubling the B&H outcome (£459.62) and vastly outperforming the cost-unaware model (£136.15). This corresponds to a CAGR of 24.92%, well above the benchmark's 16.51%. At the same time, annual volatility was contained at 15.26%, below B&H's 18.74%, indicating a more efficient risk-return trade-off.

Risk-adjusted performance was excellent, with a Sharpe ratio of 1.47 and Sortino ratio of 1.87, roughly double the benchmark (0.75 and 0.97). The Calmar ratio of 1.09 confirms superior returns relative to drawdowns. Maximum drawdown (22.96%) was lower than B&H's 32.26%, and average drawdowns under 1% highlight stronger downside resilience. Tail-risk measures were also improved (VaR 1.37%, ES 2.24%), reflecting tighter risk control. Statistical tests are consistent with this picture: the Sharpe-difference test indicates a highly significant improvement ($p = 0.0000$), while the returns-difference test finds no significant outperformance in mean returns ($p = 0.2329$). Hence, the advantage of the cost-aware model lies in its stronger risk-adjusted performance and more stable risk control, rather than in higher raw returns.

| Metric | Value |
|---|---|
| Terminal Wealth (£) | 919.53 |
| CAGR (%) | 24.92 |
| Annual Volatility (%) | 15.26 |
| Sharpe Ratio | 1.47 |
| Sortino Ratio | 1.87 |
| Calmar Ratio | 1.09 |
| Maximum Drawdown (%) | 22.96 |
| Average Drawdown (%) | 0.94 |
| Average Recovery Time (days) | 5.07 |
| VaR 95% (%) | 1.37 |
| Expected Shortfall 95% (%) | 2.24 |
| Sharpe Difference t-stat | 18.94 |
| Sharpe Difference p-value | 0.0000 |
| Returns Difference t-stat | 0.73 |
| Returns Difference p-value | 0.2329 |
| Average Leverage | 0.80 |
| Maximum Position Size (£) | 548.05 |
| Average Turnover per Step (%) | 1.41 |
| Annual Turnover (%) | 354.91 |
| Profit per Turnover (£) | 0.07 |
| Standard Deviation of Portfolio Weights (%) | 28.70 |
| Average Absolute Change in Portfolio Weights (%) | 0.91 |
| Total Transaction Costs (£) | 79.42 |

Table 5.4: Performance metrics (cost-aware model, historical).

Figures 5.17 and 5.18 show steady wealth growth and shallower, shorter drawdowns than B&H, with average recovery in 5 vs. 8 days. The return distribution (Figure 5.19) has tighter tails than B&H, consistent with lower volatility, while rolling Sharpe ratios (Figure 5.20) frequently exceeded 2 and remained persistently positive, contrasting with the benchmark's sub-1 values and prolonged negative stretches.

Turnover dynamics (Figure 5.21) show that trading was active but far more controlled than the cost-unaware model. Average turnover was 1.41% per step (354.91% annually), slightly above B&H but vastly below 4,671.89% for the cost-unaware case. Cumulative transaction costs were £79.42 (Figure 5.22), modest relative to wealth growth and far below the £321.25 in the cost-unaware model. Profit per turnover (£0.07) matched the benchmark, but at far greater scale.

Structural features also highlight the advantages of the cost-aware model. Average leverage was 0.80, far higher than the cost-unaware model's 0.13 but below the full-risk exposure of B&H (1.0). This intermediate positioning enabled meaningful participation in risky assets while maintaining headroom for risk control, supporting superior risk-adjusted performance. The maximum position size reached £548.05, above B&H's £265.97, but reflected intentional active allocation rather than passive concentration. Average absolute change in weights was 0.91%, lower than the instability of the cost-unaware model (18.53%) and only modestly above B&H (0.62%). Figure 5.23 confirms this pattern: weights evolved smoothly over time, avoiding both the violent swings of the cost-unaware model and the one-sided concentration of B&H.

Overall, the cost-aware model overcame the pitfalls of the cost-unaware formulation and outperformed B&H in historical tests. By embedding trading frictions directly into the optimisation, it avoided excessive rebalancing while retaining flexibility to adjust exposures. We next test robustness in the bootstrap setting.

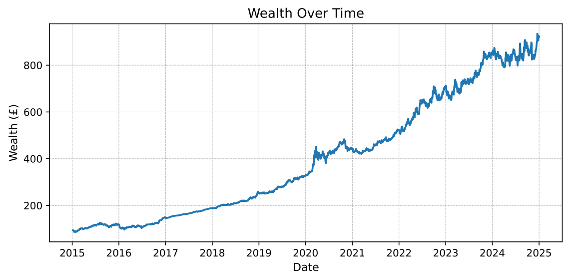



Figure 5.17: Wealth (cost-aware model, historical).

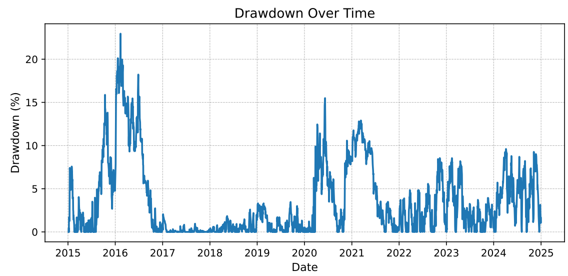



Figure 5.18: Drawdowns (cost-aware model, historical).

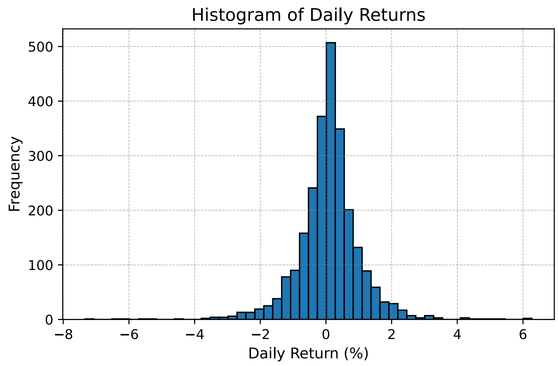



Figure 5.19: Return distribution (cost-aware model, historical).

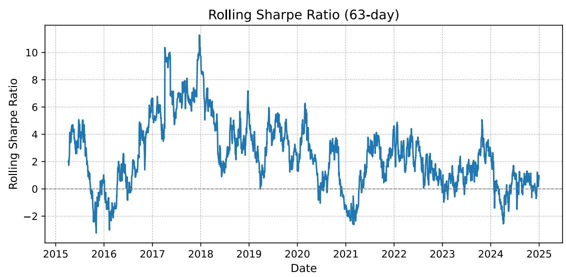



Figure 5.20: Rolling 63-day Sharpe ratio (cost-aware model, historical).

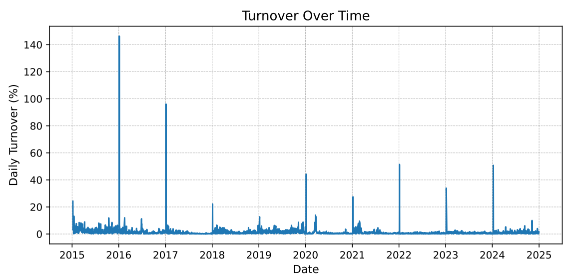



Figure 5.21: Daily turnover (cost-aware model, historical).

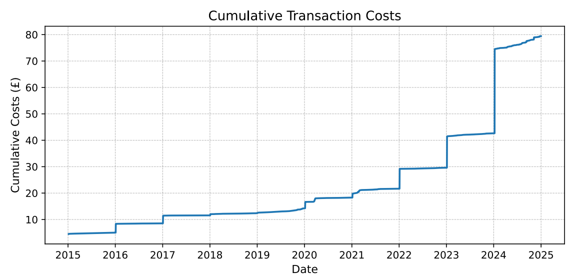



Figure 5.22: Cumulative costs (cost-aware model, historical).

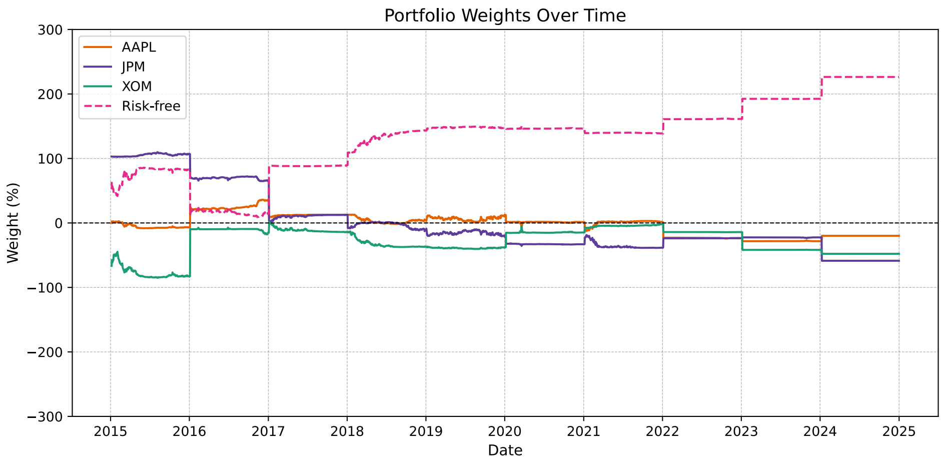



Figure 5.23: Portfolio weights (cost-aware model, historical).

### 5.4.2 Parametric Bootstrap Backtest

Table 5.5 reports the bootstrap results for the cost-aware model. Most core metrics show bootstrap medians close to their historical values, with large $p$-values indicating no systematic deviation. For example, while terminal wealth and CAGR are somewhat lower in the bootstrap distribution, the associated $p$-values ($p = 0.276$ and $p = 0.249$) suggest these differences are not statistically meaningful. This supports the conclusion that historical performance is broadly representative.

Some metrics do diverge. Annual turnover declines from 354.91% historically to a bootstrap median of 285.10%, and the average absolute change in weights falls from 0.91% to 0.60%. By contrast, 95% VaR rises slightly from 1.37% to 1.58%. These shifts indicate that rebalancing intensity and tail-risk exposures can vary under simulated dynamics. This partly reflects the limitations of the bootstrap environment, which cannot fully replicate real-world dependencies, but also illustrates how the model adapts under alternative price paths, providing additional evidence of its robustness.

| Metric | Historical | Bootstrap Median | 5th %ile | 95th %ile | p-value | Prob. of Outperformance (%) |
|---|---|---|---|---|---|---|
| Terminal Wealth (£) | 919.53 | 599.14 | 273.66 | 1,218.04 | 0.276 | 16.00 |
| CAGR (%) | 24.92 | 18.92 | 10.24 | 27.38 | 0.249 | 12.00 |
| Annual Volatility (%) | 15.26 | 15.37 | 14.79 | 16.15 | 0.789 | 59.00 |
| Sharpe Ratio | 1.47 | 1.07 | 0.51 | 1.66 | 0.238 | 12.00 |
| Sortino Ratio | 1.87 | 1.45 | 0.69 | 2.38 | 0.401 | 20.00 |
| Calmar Ratio | 1.09 | 0.71 | 0.34 | 1.48 | 0.323 | 18.00 |
| Maximum Drawdown (%) | 22.96 | 25.56 | 17.56 | 39.91 | 0.727 | 64.00 |
| Average Drawdown (%) | 0.94 | 0.97 | 0.83 | 1.20 | 0.831 | 59.00 |
| VaR 95% (%) | 1.37 | 1.58 | 1.49 | 1.68 | 0.003 | 100.00 |
| Expected Shortfall 95% (%) | 2.24 | 2.21 | 2.08 | 2.36 | 0.741 | 39.00 |
| Annual Turnover (%) | 354.91 | 285.10 | 271.25 | 317.03 | 0.000 | 0.00 |
| Average Absolute Change in Portfolio Weights (%) | 0.91 | 0.60 | 0.49 | 0.80 | 0.005 | 3.00 |
| Total Transaction Costs (£) | 79.42 | 38.18 | 17.41 | 78.21 | 0.042 | 5.00 |

Table 5.5: Performance metrics (cost-aware model, bootstrap).

Figures 5.24–5.27 illustrate the distribution of outcomes. The 90% confidence band for wealth is narrower than for B&H, indicating greater robustness of compounding. The terminal wealth histogram (Figure 5.25) shows a more symmetric distribution than B&H, while Sharpe ratios (Figure 5.26) cluster around higher values. Turnover (Figure 5.27) remains stable in the 280–310% range, well below the extreme levels of the cost-unaware model and consistent with manageable trading costs.

Formal inference aligns with these findings (Table 5.6). The Sharpe-difference test confirms a highly significant improvement over B&H ($p = 0.0000$), while the returns-difference test detects

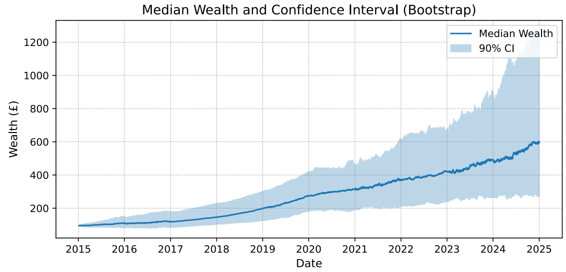



Figure 5.24: Wealth paths with 90% confidence interval (cost-aware model, bootstrap).

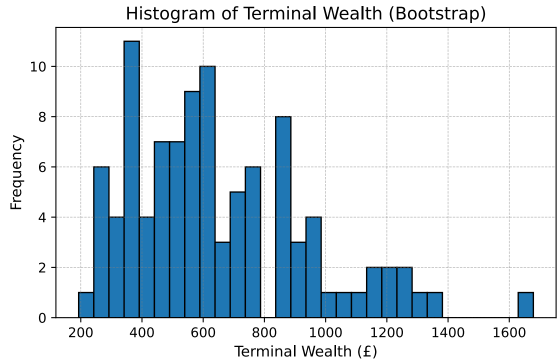



Figure 5.25: Distribution of terminal wealth (cost-aware model, bootstrap).

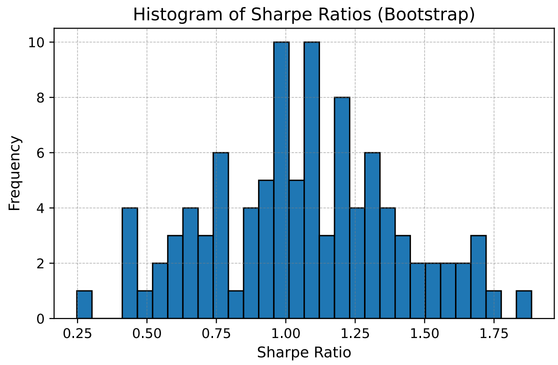



Figure 5.26: Distribution of Sharpe ratios (cost-aware model, bootstrap).

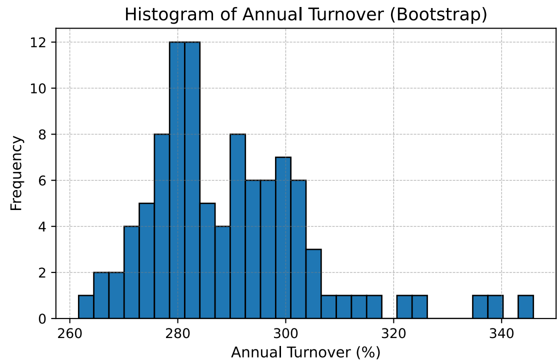



Figure 5.27: Distribution of annual turnover (cost-aware model, bootstrap).

no significant outperformance ($p = 0.4386$). Thus, the bootstrap reinforces the conclusion that the cost-aware model's edge lies in robustly superior risk-adjusted performance rather than higher raw returns.

| Metric | Mean Diff | t-stat | p-value | Probability > Benchmark (%) |
|---|---|---|---|---|
| Sharpe Ratio Difference | 0.2766 | 4.77 | 0.0000 | 71.00 |
| Returns Difference (%) | 0.17 | 0.15 | 0.4386 | 57.00 |

Table 5.6: Statistical performance metrics (cost-aware model, bootstrap).

Some divergence from historical results remains. The bootstrap assumes simplified GBM/ABM dynamics and retrains only on historical parameters, which may understate richer asset–state dependencies or adaptive market behaviour. These limitations help explain differences in turnover, weight adjustments, and tail-risk metrics. However, they do not overturn the central finding: the cost-aware model delivers consistently stronger risk-adjusted returns across simulated paths, confirming that the historical improvements are robust. The results also underline the central role of trading frictions in portfolio optimisation: when explicitly accounted for, strategies gain stability and efficiency.

# Chapter 6

# Limitations and Extensions

In this chapter, we discuss the main limitations of our approach and outline avenues for extension.

A first limitation is the choice of risk measure. Variance penalises both upside and downside moves and does not capture tail risk. Alternative measures such as semi-variance (Markowitz, 1959), VaR, or ES (Rockafellar and Uryasev, 2000, 2002) are widely used in practice, particularly under Basel regulations. Replacing variance with ES would alter the HJB equation but remains implementable within the deep BSDE framework, since ES can be formulated as a convex optimisation problem and embedded into stochastic control (Ruszczyński and Shapiro, 2006).

A second limitation is the restriction to quadratic transaction costs. Real markets also involve short-selling costs, asymmetric funding rates, and margin requirements (Duffie et al., 2002). Such frictions could be introduced via a quadratic shorting penalty,

$$-\lambda\,(\theta_t^-)^2\,dt, \quad \theta_t^- = \min(\theta_t, 0), \tag{6.1}$$

or through differential borrowing and lending rates, with $r^B > r^L$. Quadratic costs ensured differentiability for training (Gonon et al., 2020), but likely exaggerate large-trade penalties, making results conservative. Smoothed proportional (Huber-type) costs (Huber, 1964; Boyd and Vandenberghe, 2004) could provide a more realistic compromise.

A third limitation is the modelling of dynamics and parameters. Assets were assumed to follow GBM and the state variables ABM, with parameters estimated from historical rolling windows. This backward-looking approach may fail under regime shifts. Extensions include forward-looking methods using option-implied volatilities (Bakshi et al., 1997), Bayesian updating for parameter uncertainty (Avramov, 2002), or machine learning forecasts from macro-financial covariates (Gu et al., 2020). Richer dynamics such as stochastic volatility (Heston, 1993), CIR processes (Cox et al., 1985), or stochastic interest rates (Vasicek, 1977) could also be incorporated within the BSDE framework, albeit at higher computational cost.

A fourth constraint is dimensionality. The implementation was restricted to three risky assets and one state variable due to hardware limits. In principle, the method scales to higher dimensions, since deep BSDE avoids spatial discretisation and relies only on time discretisation with neural networks. With larger compute budgets, it could be extended to institutional-scale multi-asset portfolios, which is particularly relevant for industry deployment.

Finally, the backtesting design has limits. Both historical and bootstrap experiments showed robustness, but the bootstrap relied on GBM/ABM dynamics with parameters updated annually. This captures regime shifts at a block level but not within-year stochastic volatility, jumps, or feedback effects. Richer specifications such as Heston for equity variance, CIR for VIX or rates, or jump–diffusion models (Merton, 1976) could provide a stronger bootstrap evaluation. Stress testing offers a complementary extension: whereas the bootstrap explores typical paths implied by estimated dynamics, stress tests impose extreme but plausible scenarios such as crisis-level volatility spikes or sudden crashes (Basel Committee on Banking Supervision, 2017).

Overall, the most important limitations lie in (i) reliance on variance as the risk measure, (ii) the quadratic cost structure, and (iii) restrictive dynamics and dimensionality. Each has feasible extensions within the deep BSDE framework, and addressing them would enhance both theoretical realism and relevance for institutional applications.

# Chapter 7

# Conclusion

The contributions of this thesis are threefold. First, the continuous-time mean–variance formulation of Basak and Chabakauri (2010) was extended to include transaction costs. Second, the deep BSDE scheme of Huré et al. (2020) was adapted to the forward–backward SDE system arising from this formulation. This was further extended by incorporating neural networks to approximate the value function curvature. Third, the framework was implemented and evaluated on real financial data using rolling backtests and bootstrap analysis.

The empirical evaluation demonstrated that incorporating frictions yields substantially smoother allocations and more realistic trading behaviour. Compared with the frictionless analytical benchmark, the deep BSDE approach reduced turnover and trading costs, while improving risk-adjusted performance. Relative to buy-and-hold, it achieved superior Sharpe ratios while maintaining active flexibility, with bootstrap analysis confirming that the improvements were robust across resampled paths.

Taken together, these findings highlight the importance of modelling frictions in dynamic portfolio optimisation. They establish, to our knowledge, the first scalable, time-consistent solution to mean–variance portfolio optimisation under transaction costs in continuous time. In doing so, this thesis demonstrates that deep BSDE methods can be applied effectively in this setting, providing a viable tool for high-dimensional, nonlinear portfolio problems that are analytically intractable. This also opens paths for extensions to richer frictions, alternative risk measures, and more complex asset or state dynamics. In this way, the thesis contributes both a methodological advance and a practical step towards bridging the gap between classical portfolio theory and real-world implementation.

# References


Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–40.

Avramov, D. (2002). Stock return predictability and model uncertainty. *Journal of Financial Economics*, 64(3):423–458.

Bakshi, G., Cao, C., and Chen, Z. (1997). Empirical performance of alternative option pricing models. *The Journal of Finance*, 52(5):2003–2049.

Basak, S. and Chabakauri, G. (2010). Dynamic mean-variance asset allocation. *Review of Financial Studies*, 23(8):2970–3016.

Basel Committee on Banking Supervision (2017). Supervisory and bank stress testing: Range of practices. Technical report, Bank for International Settlements.

Beck, C., E, W., and Jentzen, A. (2019). Machine learning approximation algorithms for high-dimensional fully nonlinear partial differential equations and second-order backward stochastic differential equations. *Journal of Nonlinear Science*, 29(4):1563–1619.

Bellman, R. (1957). *Dynamic Programming*. Princeton University Press, Princeton, NJ.

Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press, Cambridge.

Constantinides, G. M. (1986). Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862.

Cox, J. C., Ingersoll, J. E., and Ross, S. A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407.

Cui, X., Li, D., Li, S., and Zhang, J. (2012). Mean–variance policy under various constraints: Closed-form solutions and applications. *Mathematical Finance*, 22(2):227–254.

Davis, M. H. A. and Norman, A. R. (1990). Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713.

de Prado, M. L. and Zoonekynd, V. (2025). A protocol for causal factor investing. Technical Report 5277078, SSRN. Posted: 2 Jun 2025.

Detemple, J. (2003). A monte carlo method for optimal portfolios. *The Journal of Finance*, 58(1):441–471.

Duffie, D., Gârleanu, N., and Pedersen, L. H. (2002). Securities lending, shorting, and pricing. *Journal of Financial Economics*, 66(2-3):307–339.

Frazzini, A., Israel, R., and Moskowitz, T. J. (2018). Trading costs. Technical report, AQR Capital Management. First draft: April 2018, This draft: August 2018.

Gârleanu, N. and Pedersen, L. H. (2013). Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340.

Germain, M., Pham, H., and Warin, X. (2021). Neural networks-based algorithms for stochastic control and pdes in finance. Preprint, April 19, 2021. Contribution to *Machine Learning for Financial Markets: a guide to contemporary practices*, Cambridge University Press.

Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*, volume 53 of *Stochastic Modelling and Applied Probability*. Springer, New York, NY.

Golub, G. H. and Loan, C. F. V. (2013). *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4th edition.

Gonon, L., Muhle-Karbe, J., and Shi, X. (2020). Asset pricing with general transaction costs: Theory and numerics. arXiv preprint arXiv:1905.03025.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273.

Hakansson, N. H. (1971). Multi-period mean-variance analysis: Toward a general theory of portfolio choice. *The Journal of Finance*, 26(4):857–884.

Han, J., Jentzen, A., and E, W. (2018). Solving high-dimensional partial differential equations using deep learning. *Proceedings of the National Academy of Sciences*, 115(34):8505–8510.

Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343.

Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–101.

Huh, J., Jeon, J., and Koo, H. K. (2025). Breaking the dimensional barrier: A pontryagin-guided direct policy optimization for continuous-time multi-asset portfolio. *arXiv preprint arXiv:2504.11116*.

Huré, C., Pham, H., and Warin, X. (2020). Deep backward schemes for high-dimensional nonlinear pdes. *Mathematics of Computation*, 89(324):1547–1579.

Jobson, J. D. and Korkie, B. (1981). Performance hypothesis testing with the sharpe and treynor measures. *Journal of Finance*, 36(4):889–908.

Karatzas, I. and Shreve, S. E. (1991). *Brownian Motion and Stochastic Calculus*. Springer-Verlag, New York, 2nd edition.

Li, D. and Ng, W.-L. (2000). Optimal dynamic portfolio selection: Multi-period mean-variance formulation. *Mathematical Finance*, 10(3):387–406.

Li, X., Zhou, X. Y., and Lim, A. E. B. (2009). Dynamic mean-variance portfolio selection with no-shorting constraints. *SIAM Journal on Control and Optimization*, 48(3):1156–1178.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Markowitz, H. M. (1959). *Portfolio Selection: Efficient Diversification of Investments*. John Wiley & Sons, New York.

Memmel, C. (2003). Performance hypothesis testing with the sharpe ratio. *Finance Letters*, 1:21–23.

Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The Review of Economics and Statistics*, 51(3):247–257.

Merton, R. C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4):1851–1872.

Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1-2):125–144.

Nualart, D. (1995). *The Malliavin Calculus and Related Topics*. Springer.

Pham, H. (2009). *Continuous-Time Stochastic Control and Optimization with Financial Applications*, volume 61 of *Stochastic Modelling and Applied Probability*. Springer.

Rockafellar, R. T. and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41.

Rockafellar, R. T. and Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471.

Ruszczyński, A. and Shapiro, A. (2006). Optimization of convex risk functions. *Mathematics of Operations Research*, 31(3):433–452.

Samuelson, P. A. (1969). Lifetime portfolio selection by dynamic stochastic programming. *The Review of Economics and Statistics*, 51(3):239–246.

Tobin, J. (1958). Liquidity preference as behavior towards risk. *The Review of Economic Studies*, 25(2):65–86.

van Staden, P. M., Forsyth, P. A., and Li, Y. (2024). A global-in-time neural network approach to dynamic portfolio optimization. *Applied Mathematical Finance*, 31(3):131–163.

Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2):177–188.

Weiss, N. (2005). *A Course in Probability*. Addison-Wesley, Boston.

Wu, B. and Li, L. (2024). Reinforcement learning for continuous-time mean-variance portfolio selection in a regime-switching market. *Journal of Economic Dynamics and Control*, 158:104843.

Zhou, X. Y. and Li, D. (2000). Continuous-time mean-variance portfolio selection: A stochastic lq framework. *Applied Mathematics and Optimization*, 42(1):19–33.

# A Mathematical Formulation of the Discrete-Time Model (Precommitment, Li and Ng (2000))

This appendix presents the discrete-time, multi-period formulation of Li and Ng (2000). The model delivers a closed-form precommitment strategy, which is included here for completeness and for interested readers. Its inclusion serves to contrast with the continuous-time, time-consistent approaches developed in the main text, which are more relevant for real-world portfolio applications.

To remain consistent with Li and Ng (2000), we adopt their notation in this appendix. This differs from the notation used in the main text, reflecting both the discrete-time market setup and the distinct modelling framework relative to the continuous-time case.

Extended derivations are given in Appendix B, with supporting proofs in Appendix E. The numerical implementation and corresponding backtest results are presented in Appendices F and H.

## A.1 Economic Setup

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t=0}^{T}, \mathbb{P})$, where $\mathcal{F}_t$ represents the information available up to time $t$. Consider a finite horizon with $T$ discrete periods, $t = 0, 1, \dots, T-1$. The investor allocates wealth between $n$ risky assets and one risk-free asset. Let $x_t$ denote wealth at time $t$, and let $\mathbf{u}_t = (u_t^1, \dots, u_t^n)^\top \in \mathbb{R}^n$ be the vector of amounts invested in risky assets. The remainder, $x_t - \mathbf{1}^\top \mathbf{u}_t$, is placed in the risk-free asset.

The gross return on the risk-free asset is denoted by the deterministic constant $s_t$, and $\mathbf{e}_t = (e_t^1, \dots, e_t^n)^\top$ is the random vector of gross returns on risky assets. It is assumed that the risk-free return is uncorrelated with the risky returns, i.e. $\operatorname{cov}(s_t, e_t^i) = 0$ for all $i = 1, \dots, n$ and $t = 0, 1, \dots, T-1$. The wealth process then evolves as

$$x_{t+1} = s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t, \quad t = 0, 1, \dots, T-1, \tag{A.1}$$

where the excess returns are

$$\mathbf{P}_t = \mathbf{e}_t - s_t \mathbf{1}. \tag{A.2}$$

A portfolio policy is a sequence of feedback functions

$$\pi = \{\mu_0, \mu_1, \dots, \mu_{T-1}\}, \tag{A.3}$$

with each function mapping wealth to a decision,

$$\mathbf{u}_t = \mu_t(x_t), \quad t = 0, 1, \dots, T-1, \tag{A.4}$$

adapted to $\mathcal{F}_t$. The admissible policy set is

$$\Pi = \Big\{\pi = \{\mu_0, \dots, \mu_{T-1}\} : \mu_t : \mathbb{R} \to \mathbb{R}^n\Big\}. \tag{A.5}$$

A policy $\pi^*$ is efficient if no other $\pi \in \Pi$ satisfies

$$\mathbb{E}[x_T]_\pi \geq \mathbb{E}[x_T]_{\pi^*}, \tag{A.6}$$

$$\operatorname{Var}(x_T)_\pi \leq \operatorname{Var}(x_T)_{\pi^*}, \tag{A.7}$$

with at least one inequality strict.

Following Li and Ng (2000), the optimisation problem is written as

$$\max_{\pi \in \Pi} \mathbb{E}[x_T] - w \, \mathrm{Var}(x_T), \tag{A.8}$$

where $w > 0$ is a risk–aversion parameter, interpretable at the optimum as the marginal rate of substitution between expected return and variance.

Iterating the wealth dynamics (A.1) yields

$$x_T = x_0 \prod_{t=0}^{T-1} s_t + \sum_{t=0}^{T-1} \left( \left( \prod_{k=t+1}^{T-1} s_k \right) \mathbf{P}_t^\top \mathbf{u}_t \right). \tag{A.9}$$

This representation of terminal wealth underlies the subsequent derivations.

## A.2 Auxiliary Quadratic Problem Formulation

Dynamic mean–variance optimisation is complicated by the non–separability of the variance operator, which prevents direct application of dynamic programming. Following Li and Ng (2000), we reformulate the objective (A.8) into a separable form by introducing an auxiliary quadratic stochastic control problem.

The mean–variance objective (A.8) can be rewritten in quadratic form as

$$-w \, \mathbb{E}[x_T^2] + \big(1 + w \, \mathbb{E}[x_T]\big) \, \mathbb{E}[x_T]. \tag{A.10}$$

This suggests introducing an auxiliary problem of the form

$$\mathbb{E}\big[-w x_T^2 + \lambda x_T\big] , \tag{A.11}$$

where $\lambda$ is a Lagrange multiplier capturing the trade-off between mean and variance. For a fixed $\lambda$, solving (A.11) yields an optimal policy $\pi^*(\lambda)$, and varying $\lambda$ generates a family of candidate solutions.

There exists a unique value $\lambda^*$ such that the optimal solution of the auxiliary problem coincides with the original mean–variance problem (A.8). Hence, the auxiliary stochastic control problem we solve is

$$\max_{\pi \in \Pi} \mathbb{E}\big[-w x_T^2 + \lambda^* x_T\big] , \quad \text{s.t.} \quad x_{t+1} = s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t, \quad t = 0, 1, \ldots, T-1. \tag{A.12}$$

The constant $\lambda^*$ is chosen such that the optimal value and policy of the auxiliary problem coincide with those of the original mean–variance formulation. It is derived in Appendix B.1 as

$$\lambda^* = 1 + 2w \, \mathbb{E}[x_T \mid \pi^*]. \tag{A.13}$$

This reformulation converts the original non-separable mean–variance objective into a quadratic objective with linear dynamics, which is tractable via dynamic programming.

## A.3 Dynamic Programming Solution

### Quadratic Value Function

To solve the auxiliary problem (A.11) via dynamic programming, we define the value function at time $t$ as

$$J_t(x_t) = \sup_{\{\mathbf{u}_\tau\}_{\tau=t}^{T-1}} \mathbb{E}_t\big[-w \, x_T^2 + \lambda \, x_T\big] , \tag{A.14}$$

where $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$ is the conditional expectation and the supremum is taken over all future control decisions.

This formulation corresponds to a precommitment policy: although decisions respond to realised wealth $x_t$, the entire policy is computed at time zero and not re-optimised later. Thus, the policy is optimal from the perspective of $t = 0$, but not dynamically time-consistent.

We conjecture that $J_t$ takes a quadratic form:

$$J_t(x_t) = -w\,\alpha_t x_t^2 + \beta_t x_t + \delta_t, \tag{A.15}$$

with deterministic coefficients $\alpha_t, \beta_t, \delta_t$ determined recursively. At the terminal time $T$, we have

$$J_T(x_T) = -w x_T^2 + \lambda x_T, \tag{A.16}$$

which implies the terminal conditions

$$\alpha_T = 1, \qquad \beta_T = \lambda, \qquad \delta_T = 0. \tag{A.17}$$

### Optimal Portfolio Control

By the Bellman principle, the value function (A.14) can be expressed recursively, and the control $\mathbf{u}_t^*$ is chosen to maximise the one-step return plus the continuation value. The optimal policy takes the affine form

$$\mathbf{u}_t^* = -s_t \mathbf{K}_t x_t + \mathbf{v}_t, \tag{A.18}$$

where

$$\mathbf{K}_t = \big(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\,\mathbb{E}_t[\mathbf{P}_t]^\top\big)^{-1}\mathbb{E}_t[\mathbf{P}_t], \tag{A.19}$$

$$\mathbf{v}_t = \frac{\beta_{t+1}}{2w\alpha_{t+1}}\big(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\,\mathbb{E}_t[\mathbf{P}_t]^\top\big)^{-1}\mathbb{E}_t[\mathbf{P}_t]. \tag{A.20}$$

The derivation of this affine form is provided in Appendix B.2.

The coefficients $\alpha_t$ and $\beta_t$ follow recursive relations that capture the quadratic structure:

$$\alpha_t = \alpha_{t+1}\, s_t^2(1 - B_t), \tag{A.21}$$
$$\beta_t = \beta_{t+1}\, s_t(1 - 2B_t), \tag{A.22}$$

where

$$\mathbf{Q}_t = \mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\,\mathbb{E}_t[\mathbf{P}_t]^\top, \tag{A.23}$$

$$B_t = \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{K}_t. \tag{A.24}$$

Full derivations are given in Appendix B.3.

### Definition of $\gamma$

To simplify notation, we define

$$\gamma = \frac{\lambda}{w}. \tag{A.25}$$

Using the optimal $\lambda^*$ from Equation (A.13), the corresponding $\gamma^*$ follows directly. This substitution streamlines the recursion for $\beta_t$ and allows the closed-form policy to be expressed more compactly.

## A.4 Closed-Form Optimal Policy

Iterating the recursions (A.21)–(A.22) with terminal conditions (A.17) gives closed-form expressions.

**Closed form for $\alpha_t$.**

$$\alpha_t = \prod_{k=t}^{T-1} \big( s_k^2 (1 - B_k) \big). \tag{A.26}$$

**Closed form for $\beta_t$.**

$$\beta_t = w\gamma \prod_{k=t}^{T-1} \big( s_k (1 - 2B_k) \big), \tag{A.27}$$

where $\gamma = \lambda / w$ as in (A.25). Full derivations are provided in Appendix B.3.

**Closed form for $\mathbf{v}_t$.** Substituting (A.26)–(A.27) into (A.20) gives

$$\mathbf{v}_t = \frac{\gamma}{2} \prod_{k=t+1}^{T-1} \frac{1 - 2B_k}{s_k (1 - B_k)} \mathbf{K}_t. \tag{A.28}$$

The algebraic steps are provided in Appendix B.4.

**Optimal policy.** The affine policy (A.18) becomes

$$\mathbf{u}_t^* = -s_t \mathbf{K}_t x_t + \frac{\gamma}{2} \prod_{k=t+1}^{T-1} \frac{1 - 2B_k}{s_k (1 - B_k)} \mathbf{K}_t. \tag{A.29}$$

**Closed form for $\gamma^*$.** By definition, $\gamma^* = \lambda^* / w$. Using (A.13), this can be written as

$$\gamma^* = \frac{1}{w} + 2\,\mathbb{E}[x_T \mid \pi^*]. \tag{A.30}$$

Substituting the optimal control (A.29) into the wealth dynamics (A.1) and taking expectations over time yields the closed-form expression

$$\gamma^* = 2 \Big( \prod_{t=0}^{T-1} s_t \Big) x_0 + \frac{1}{w \prod_{t=0}^{T-1} (1 - B_t)}. \tag{A.31}$$

The detailed derivation of (A.31) is provided in Appendix B.5.

**Final explicit policy.** Substituting (A.31) into (A.29) gives the fully explicit solution

$$\mathbf{u}_t^* = -s_t \mathbf{K}_t x_t + \left[ \Big( \prod_{k=0}^{T-1} s_k \Big) x_0 + \frac{1}{2w \prod_{k=0}^{T-1} (1 - B_k)} \right] \Big( \prod_{k=t+1}^{T-1} \frac{1}{s_k} \Big) \mathbf{K}_t. \tag{A.32}$$

This matches the closed-form solution of Li and Ng (2000). Further algebraic details are given in Appendix B.6.

The optimal policy is affine in wealth, combining a myopic term $-s_t \mathbf{K}_t x_t$ proportional to current wealth and a hedging term $\mathbf{v}_t$ that is independent of current wealth. The latter reflects a non-myopic adjustment determined at time zero, consistent with the precommitment nature of the strategy. Figure 1 provides a feedback control representation.

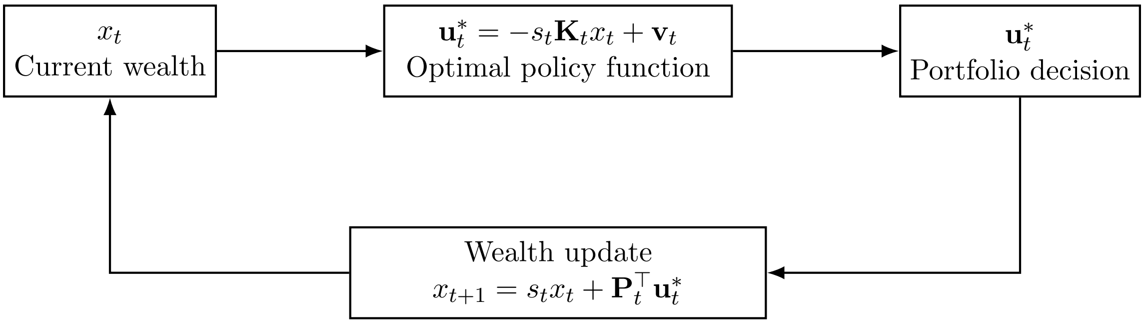



Figure 1: Feedback control representation of the optimal portfolio policy.

# B Extended Derivations for the Discrete-Time Model (Precommitment, Li and Ng (2000))

## B.1 Derivation of Optimal $\lambda^*$ in Auxiliary Problem

We derive the unique $\lambda^*$ such that the auxiliary and original mean–variance problems yield the same optimal solution. For any policy $\pi$, the auxiliary objective is

$$\mathbb{E}\big[-w x_T^2 + \lambda x_T\big] = -w\,\mathbb{E}[x_T^2] + \lambda\,\mathbb{E}[x_T]. \tag{B.33}$$

Meanwhile, the mean–variance objective (A.8) can be written as

$$-w\,\mathbb{E}[x_T^2] + \big(1 + w\,\mathbb{E}[x_T]\big)\,\mathbb{E}[x_T]. \tag{B.34}$$

For consistency, the first-order conditions of the two problems with respect to the optimal policy $\pi^*(\lambda)$ must coincide. The auxiliary objective gives

$$-w\,\frac{\partial}{\partial\lambda}\mathbb{E}[x_T^2] + \lambda\,\frac{\partial}{\partial\lambda}\mathbb{E}[x_T] = 0, \tag{B.35}$$

while the mean–variance objective implies

$$-w\,\frac{\partial}{\partial\lambda}\mathbb{E}[x_T^2] + \big(1 + 2w\,\mathbb{E}[x_T]\big)\frac{\partial}{\partial\lambda}\mathbb{E}[x_T] = 0. \tag{B.36}$$

Comparing (B.35) and (B.36) yields

$$\lambda^* = 1 + 2w\,\mathbb{E}[x_T \mid \pi^*]. \tag{B.37}$$

Thus, choosing $\lambda = \lambda^*$ ensures that the auxiliary and original mean–variance problems have the same optimal value and policy.

## B.2 Derivation of the Bellman Expansion and Affine Optimal Portfolio Control

### Bellman Equation and Expansion

By the principle of dynamic programming, the value function at time $t$ satisfies

$$J_t(x_t) = \max_{\mathbf{u}_t}\ \mathbb{E}_t\big[J_{t+1}(x_{t+1})\big]. \tag{B.38}$$

Substituting the wealth dynamics (A.1) gives

$$J_t(x_t) = \max_{\mathbf{u}_t}\ \mathbb{E}_t\Big[J_{t+1}\Big(s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t\Big)\Big]. \tag{B.39}$$

Substituting the quadratic form (A.15) into Equation (B.39) yields

$$
\begin{aligned}
J_t(x_t) &= \max_{\mathbf{u}_t} \ \mathbb{E}_t\Big[-w\alpha_{t+1}\big(s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t\big)^2 + \beta_{t+1}\big(s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t\big) + \delta_{t+1}\Big] \\
&= \max_{\mathbf{u}_t} \ \mathbb{E}_t\Big[-w\alpha_{t+1}\big(s_t^2 x_t^2 + 2 s_t x_t \mathbf{P}_t^\top \mathbf{u}_t + \mathbf{u}_t^\top \mathbf{P}_t \mathbf{P}_t^\top \mathbf{u}_t\big) + \beta_{t+1}\big(s_t x_t + \mathbf{P}_t^\top \mathbf{u}_t\big) + \delta_{t+1}\Big].
\end{aligned} \tag{B.40}
$$

Taking conditional expectations, we obtain

$$
\begin{aligned}
J_t(x_t) = \max_{\mathbf{u}_t} \Big\{ &-w\alpha_{t+1}\big[s_t^2 x_t^2 + 2 s_t x_t \, \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{u}_t + \mathbf{u}_t^\top \big(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\big)\mathbf{u}_t\big] \\
&+ \beta_{t+1}\big[s_t x_t + \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{u}_t\big] + \delta_{t+1}\Big\}.
\end{aligned} \tag{B.41}
$$

### Optimal Portfolio Control

Equation (B.41) is quadratic in $\mathbf{u}_t$. Differentiating with respect to $\mathbf{u}_t$ and setting the gradient to zero gives

$$
0 = -2w\alpha_{t+1}\big(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\big)\mathbf{u}_t - 2w\alpha_{t+1} s_t x_t \, \mathbb{E}_t[\mathbf{P}_t] + \beta_{t+1}\mathbb{E}_t[\mathbf{P}_t]. \tag{B.42}
$$

Rearranging yields the optimal control

$$
\mathbf{u}_t^* = -s_t x_t \left(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\right)^{-1} \mathbb{E}_t[\mathbf{P}_t] + \frac{\beta_{t+1}}{2w\alpha_{t+1}} \left(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\right)^{-1} \mathbb{E}_t[\mathbf{P}_t]. \tag{B.43}
$$

Hence, the optimal policy is affine in wealth:

$$
\mathbf{u}_t^* = -s_t \mathbf{K}_t x_t + \mathbf{v}_t, \tag{B.44}
$$

with

$$
\mathbf{K}_t = \left(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\right)^{-1} \mathbb{E}_t[\mathbf{P}_t], \tag{B.45}
$$

$$
\mathbf{v}_t = \frac{\beta_{t+1}}{2w\alpha_{t+1}} \left(\mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\mathbb{E}_t[\mathbf{P}_t]^\top\right)^{-1} \mathbb{E}_t[\mathbf{P}_t]. \tag{B.46}
$$

Thus, the Bellman expansion delivers the affine control form (B.44)–(B.46), with coefficients determined by backward recursion.

## B.3 Backward Recursions for the Coefficients $\alpha_t$ and $\beta_t$

The affine control in Equation (B.43) still depends on the coefficients $\alpha_{t+1}$ and $\beta_{t+1}$. To obtain a closed-form solution, we derive backward recursions for these coefficients by substituting $\mathbf{u}_t^*$ into the Bellman equation.

Substituting $\mathbf{u}_t^* = -s_t \mathbf{K}_t x_t + \mathbf{v}_t$ from (B.44) into the Bellman recursion (B.41) yields

$$
\begin{aligned}
J_t(x_t) = &-w\alpha_{t+1}\Big[s_t^2 x_t^2 + 2 s_t x_t \big(-s_t \mathbf{K}_t x_t + \mathbf{v}_t\big)^\top \mathbb{E}_t[\mathbf{P}_t] + \big(-s_t \mathbf{K}_t x_t + \mathbf{v}_t\big)^\top \mathbf{Q}_t \big(-s_t \mathbf{K}_t x_t + \mathbf{v}_t\big)\Big] \\
&+ \beta_{t+1}\Big[s_t x_t + \mathbb{E}_t[\mathbf{P}_t]^\top \big(-s_t \mathbf{K}_t x_t + \mathbf{v}_t\big)\Big] + \delta_{t+1},
\end{aligned} \tag{B.47}
$$

where $\mathbf{Q}_t$ is as in (A.23), $\mathbf{K}_t$ as in (A.19), $\mathbf{v}_t$ as in (A.20), and $B_t$ as in (A.24).

#### Recursion for $\alpha_t$

Collecting the $x_t^2$ terms in (B.47) gives

$$-w\alpha_{t+1}s_t^2\Big(1 - 2\mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{K}_t + \mathbf{K}_t^\top \mathbf{Q}_t K_t\Big). \tag{B.48}$$

Since $\mathbf{K}_t^\top \mathbf{Q}_t \mathbf{K}_t = \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{K}_t = B_t$, the bracketed term reduces to $1 - B_t$, and hence

$$\alpha_t = \alpha_{t+1}s_t^2(1 - B_t). \tag{B.49}$$

#### Recursion for $\beta_t$

The coefficient of $x_t$ in (B.47) is

$$\beta_t = -2w\alpha_{t+1}s_t\,\mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{v}_t + \beta_{t+1}s_t(1 - B_t). \tag{B.50}$$

Using $\mathbf{v}_t = \frac{\beta_{t+1}}{2w\alpha_{t+1}}\mathbf{K}_t$ and $B_t = \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{K}_t$ gives

$$\beta_t = \beta_{t+1}s_t(1 - 2B_t). \tag{B.51}$$

Thus, the coefficients $\alpha_t$ and $\beta_t$ satisfy the backward recursions (B.49)–(B.51), which, together with the terminal conditions (A.17), fully determine the value function and optimal control.

### B.4 Algebraic Derivation of $\mathbf{v}_t$ in Terms of $\gamma$

Recall that the optimal policy is affine in wealth, with a constant term $\mathbf{v}_t$ that captures the investor's risk-return trade-off and hedging motives. From Equation (B.46):

$$\mathbf{v}_t = \frac{\beta_{t+1}}{2w\,\alpha_{t+1}} \cdot \mathbf{K}_t. \tag{B.52}$$

Substituting the closed-form expressions for $\alpha_{t+1}$ and $\beta_{t+1}$ from Equations (A.26) and (A.27) yields:

$$\begin{aligned}\mathbf{v}_t &= \frac{w \cdot \gamma \cdot \displaystyle\prod_{k=t+1}^{T-1} [s_k \cdot (1 - 2B_k)]}{2w \cdot \displaystyle\prod_{k=t+1}^{T-1} [s_k^2 \cdot (1 - B_k)]} \cdot \mathbf{K}_t \\ &= \frac{\gamma}{2} \cdot \prod_{k=t+1}^{T-1} \left[\frac{1 - 2B_k}{s_k \cdot (1 - B_k)}\right] \cdot \mathbf{K}_t.\end{aligned} \tag{B.53}$$

Thus, the constant term $\mathbf{v}_t$ is expressed in terms of $\gamma$, $\mathbf{K}_t$, and the recursion coefficients, as in Equation (A.28).

### B.5 Derivation of Closed-Form $\gamma^*$

Recall from Equation (A.30) that:

$$\gamma^* = \frac{1}{w} + 2 \cdot \mathbb{E}\left[x_T \,\middle|\, \pi^*\right]. \tag{B.54}$$

To express $\gamma^*$ in terms of model parameters and the initial wealth $x_0$, we first derive an explicit formula for $\mathbb{E}[x_T]$ under the optimal policy.

Substituting the optimal control $\mathbf{u}_t^*$ from Equation (B.44) into the wealth dynamics yields:

$$x_{t+1} = s_t\, x_t + \mathbf{P}_t^\top \left(-s_t\, \mathbf{K}_t\, x_t + \mathbf{v}_t\right), \tag{B.55}$$

where $\mathbf{v}_t$ is as defined in Equation (A.28).

Taking conditional expectations gives:

$$\begin{aligned} \mathbb{E}_t[x_{t+1}] &= s_t\, x_t + \mathbb{E}_t[\mathbf{P}_t]^\top \left(-s_t\, \mathbf{K}_t\, x_t + \mathbf{v}_t\right) \\ &= s_t(1 - B_t)\, x_t + \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{v}_t, \end{aligned} \tag{B.56}$$

Applying this recursion forward from $t = 0$ to $T - 1$ yields:

$$\begin{aligned} \mathbb{E}[x_T] = x_0 \cdot \prod_{t=0}^{T-1} \big(s_t(1 - B_t)\big) \\ + \sum_{t=0}^{T-1} \left[ \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{v}_t \cdot \prod_{j=t+1}^{T-1} \big(s_j(1 - B_j)\big) \right]. \end{aligned} \tag{B.57}$$

Substituting the closed-form expression for $\mathbf{v}_t$ from Equation (A.28) into (B.57) gives:

$$\begin{aligned} &\mathbb{E}[x_T] = x_0 \cdot \prod_{t=0}^{T-1} \big(s_t(1 - B_t)\big) \\ &\quad + \sum_{t=0}^{T-1} \left[ \mathbb{E}_t[\mathbf{P}_t]^\top \cdot \left( \frac{\gamma^*}{2} \cdot \prod_{k=t+1}^{T-1} \frac{1 - 2B_k}{s_k(1 - B_k)} \right) \cdot \mathbf{K}_t \cdot \prod_{j=t+1}^{T-1} \big(s_j(1 - B_j)\big) \right]. \end{aligned} \tag{B.58}$$

Observe that:

$$\prod_{j=t+1}^{T-1} \big(s_j(1 - B_j)\big) \cdot \prod_{k=t+1}^{T-1} \frac{1 - 2B_k}{s_k(1 - B_k)} = \prod_{k=t+1}^{T-1} \frac{1 - 2B_k}{1 - B_k} = \prod_{k=t+1}^{T-1} (1 - B_k)^{-1}. \tag{B.59}$$

Hence, the sum simplifies to:

$$\begin{aligned} \sum_{t=0}^{T-1} \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{v}_t \cdot \prod_{j=t+1}^{T-1} \big(s_j(1 - B_j)\big) &= \sum_{t=0}^{T-1} \frac{\gamma^*}{2} \cdot \left( \mathbb{E}_t[\mathbf{P}_t]^\top \mathbf{K}_t \right) \cdot \prod_{k=t+1}^{T-1} (1 - B_k)^{-1} \\ &= \frac{\gamma^*}{2} \cdot \sum_{t=0}^{T-1} B_t \cdot \prod_{k=t+1}^{T-1} (1 - B_k)^{-1}. \end{aligned} \tag{B.60}$$

A direct calculation (see Appendix E.1) shows that

$$\sum_{t=0}^{T-1} B_t \cdot \prod_{k=t+1}^{T-1} (1 - B_k)^{-1} = 1 - \prod_{t=0}^{T-1} (1 - B_t). \tag{B.61}$$

Therefore, the expected terminal wealth is

$$\mathbb{E}[x_T] = x_0 \cdot \prod_{t=0}^{T-1} \big(s_t(1 - B_t)\big) + \frac{\gamma^*}{2} \left[ 1 - \prod_{t=0}^{T-1} (1 - B_t) \right]. \tag{B.62}$$

Substituting (B.62) into the definition of $\gamma^*$ (A.30) and solving gives:

$$\gamma^* = \frac{\frac{1}{w} + 2x_0 \prod_{t=0}^{T-1}(s_t(1-B_t))}{\prod_{t=0}^{T-1}(1-B_t)}. \tag{B.63}$$

Simplifying,

$$\gamma^* = 2\left(\prod_{t=0}^{T-1} s_t\right)x_0 + \frac{1}{w\prod_{t=0}^{T-1}(1-B_t)}. \tag{B.64}$$

Thus, the closed-form solution (B.64) coincides with Equation (A.31).

### B.6 Derivation of Fully Explicit Optimal Policy

Substituting $\gamma^*$ from Equation (A.31) into the optimal policy expressed in terms of $\gamma$ in Equation (A.29), we obtain

$$\begin{aligned}\mathbf{u}_t^* =& -s_t\mathbf{K}_t x_t \\ &+ \left[\left(\prod_{k=0}^{T-1} s_k\right)x_0 + \frac{1}{2w\cdot\prod_{k=0}^{T-1}(1-B_k)}\right]\cdot\prod_{k=t+1}^{T-1}\frac{1-2B_k}{s_k(1-B_k)}\cdot\mathbf{K}_t.\end{aligned} \tag{B.65}$$

One can show (see Appendix E.2) that

$$\prod_{k=t+1}^{T-1}\frac{1-2B_k}{s_k(1-B_k)} = \prod_{k=t+1}^{T-1}\frac{1}{s_k}. \tag{B.66}$$

Hence, the fully explicit optimal policy simplifies to

$$\begin{aligned}\mathbf{u}_t^* =& -s_t\mathbf{K}_t x_t \\ &+ \left[\left(\prod_{k=0}^{T-1} s_k\right)x_0 + \frac{1}{2w\cdot\prod_{k=0}^{T-1}(1-B_k)}\right]\cdot\left(\prod_{k=t+1}^{T-1}\frac{1}{s_k}\right)\cdot\mathbf{K}_t.\end{aligned} \tag{B.67}$$

where $\mathbf{K}_t$ and $B_t$ are as previously defined in Equations (A.19) and (A.24).

## C Extended Derivations for the Cost-Unaware Continuous-Time Model (Time-Consistent, Basak and Chabakauri (2010))

### C.1 Decomposition of the Value Function

Following Basak and Chabakauri (2010), we can show the value function is separable in wealth:

$$J(W_t, S_t, X_t, t) = W_t e^{r(T-t)} + \tilde{J}(S_t, X_t, t). \tag{C.68}$$

This separability arises from the optimal terminal wealth:

$$W_T^* = W_t e^{r(T-t)} + f_t. \tag{C.69}$$

Substituting into the expression for the value function (Equation (3.8)) gives:

$$J(W_t, S_t, X_t, t) = \mathbb{E}_t \left[ W_t e^{r(T-t)} + f_t \right] - \frac{\gamma}{2} \operatorname{Var}_t \left[ W_t e^{r(T-t)} + f_t \right]. \tag{C.70}$$

Since $W_t e^{r(T-t)}$ is deterministic:

$$\mathbb{E}_t \left[ W_t e^{r(T-t)} + f_t \right] = W_t e^{r(T-t)} + \mathbb{E}_t[f_t], \tag{C.71}$$

$$\operatorname{Var}_t \left[ W_t e^{r(T-t)} + f_t \right] = \operatorname{Var}_t[f_t]. \tag{C.72}$$

Thus, the value function simplifies to:

$$J(W_t, S_t, X_t, t) = W_t e^{r(T-t)} + \left( \mathbb{E}_t[f_t] - \frac{\gamma}{2} \operatorname{Var}_t[f_t] \right). \tag{C.73}$$

We define the separable component as:

$$\tilde{J}(S_t, X_t, t) = \mathbb{E}_t[f_t] - \frac{\gamma}{2} \operatorname{Var}_t[f_t]. \tag{C.74}$$

## C.2 Derivation of the HJB Equation

Using the separability in Equation (3.16), and applying Itô's Lemma to $\tilde{J}(S_t, X_t, t)$, we now derive the full HJB equation from (3.15).

First, consider the product rule for the discounted wealth term using Equation (3.3):

$$\mathrm{d}(W_t e^{r(T-t)}) = \theta_t(\mu_t - r) e^{r(T-t)} \,\mathrm{d}t + \theta_t \sigma_t e^{r(T-t)} \,\mathrm{d}w_t. \tag{C.75}$$

Now apply Itô's Lemma to $f_t = f(S_t, X_t, t)$, treating it as a function of two stochastic state variables. The full differential is:

$$\begin{aligned} \mathrm{d}f_t = &\left( \frac{\partial f_t}{\partial t} + \mu_t S_t \frac{\partial f_t}{\partial S_t} + m_t \frac{\partial f_t}{\partial X_t} + \frac{1}{2}\sigma_t^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2} + \frac{1}{2}\nu_t^2 \frac{\partial^2 f_t}{\partial X_t^2} + \rho \nu_t \sigma_t S_t \frac{\partial^2 f_t}{\partial S_t \partial X_t} \right) \mathrm{d}t \\ &+ \sigma_t S_t \frac{\partial f_t}{\partial S_t} \,\mathrm{d}w_t + \nu_t \frac{\partial f_t}{\partial X_t} \,\mathrm{d}w_{X,t}. \end{aligned} \tag{C.76}$$

For the variance calculation below, we focus on the stochastic components of $\mathrm{d}f_t$ and $\mathrm{d}(W_t e^{r(T-t)})$, which determine the diffusion term in the HJB. Combining the stochastic components from Equations (C.76) and (C.75) gives:

$$df_t + d(W_t e^{r(T-t)}) = \left( \sigma_t S_t \frac{\partial f_t}{\partial S_t} + \theta_t \sigma_t e^{r(T-t)} \right) dw_t + \nu_t \frac{\partial f_t}{\partial X_t} dw_{X,t}. \tag{C.77}$$

So the variance becomes:

$$\begin{aligned} \operatorname{Var}_t \left[ df_t + d(W_t e^{r(T-t)}) \right] = &\left( \sigma_t S_t \frac{\partial f_t}{\partial S_t} + \theta_t \sigma_t e^{r(T-t)} \right)^2 \\ &+ \left( \frac{\partial f_t}{\partial X_t} \right)^2 \nu_t^2 \\ &+ 2\rho \sigma_t \nu_t \left( S_t \frac{\partial f_t}{\partial S_t} + \theta_t e^{r(T-t)} \right) \frac{\partial f_t}{\partial X_t}. \end{aligned} \tag{C.78}$$

Now consider the drift term. Using the separability of $J$:

$$\mathbb{E}_t[dJ_t] = \mathbb{E}_t\left[d\tilde{J}_t + d(W_t e^{r(T-t)})\right] = \mathbb{E}_t[d\tilde{J}_t] + \theta_t(\mu_t - r)e^{r(T-t)}dt. \tag{C.79}$$

Define $\mathcal{D}\tilde{J}_t\, dt = \mathbb{E}_t[d\tilde{J}_t]$. Then:

$$\mathbb{E}_t[dJ_t] = \mathcal{D}\tilde{J}_t\, dt + \theta_t(\mu_t - r)e^{r(T-t)}dt, \tag{C.80}$$

where the Dynkin operator $\mathcal{D}$ is given by:

$$\mathcal{D}\tilde{J}_t = \frac{\partial \tilde{J}_t}{\partial t} + \mu_t S_t \frac{\partial \tilde{J}_t}{\partial S_t} + m_t \frac{\partial \tilde{J}_t}{\partial X_t} + \frac{1}{2}\left[\sigma_t^2 S_t^2 \frac{\partial^2 \tilde{J}_t}{\partial S_t^2} + \nu_t^2 \frac{\partial^2 \tilde{J}_t}{\partial X_t^2} + 2\rho\nu_t\sigma_t S_t \frac{\partial^2 \tilde{J}_t}{\partial S_t \partial X_t}\right]. \tag{C.81}$$

Substituting both Equations (C.79) and (C.78) into the recursive HJB Equation (3.15):

$$\begin{aligned} 0 = \max_{\theta_t} \Bigg\{ & \mathcal{D}\tilde{J}_t + \theta_t(\mu_t - r)e^{r(T-t)} \\ & - \frac{\gamma}{2}\Bigg[\sigma_t^2 S_t^2 \left(\frac{\partial f_t}{\partial S_t}\right)^2 + \nu_t^2 \left(\frac{\partial f_t}{\partial X_t}\right)^2 + 2\rho\sigma_t\nu_t S_t \frac{\partial f_t}{\partial S_t}\frac{\partial f_t}{\partial X_t} \\ & + \theta_t^2\sigma_t^2 e^{2r(T-t)} + 2\theta_t\sigma_t e^{r(T-t)}\left(\sigma_t S_t \frac{\partial f_t}{\partial S_t} + \rho\nu_t \frac{\partial f_t}{\partial X_t}\right)\Bigg]\Bigg\}. \end{aligned} \tag{C.82}$$

subject to $\tilde{J}(T) = 0$. This equation is quadratic in $\theta_t$, enabling an explicit solution for the optimal control $\theta_t^*$.

## C.3 Implicit Expression for Optimal Stock Policy and Covariance Calculation

We now use the HJB equation to derive an implicit expression for the optimal stock investment policy $\theta_t^*$. We start by isolating the $\theta_t$-dependent terms from Equation (C.82):

$$\theta_t(\mu_t - r)e^{r(T-t)} - \frac{\gamma}{2}\left[\theta_t^2\sigma_t^2 e^{2r(T-t)} + 2\theta_t\sigma_t e^{r(T-t)}\left(\sigma_t S_t \frac{\partial f_t}{\partial S_t} + \rho\nu_t \frac{\partial f_t}{\partial X_t}\right)\right]. \tag{C.83}$$

This is a quadratic function in $\theta_t$ and we denote it by $L(\theta_t)$:

$$L(\theta_t) = \theta_t A - \frac{\gamma}{2}(B\theta_t^2 + 2C\theta_t), \tag{C.84}$$

where we define:

$$A = (\mu_t - r)e^{r(T-t)}, \tag{C.85}$$

$$B = \sigma_t^2 e^{2r(T-t)}, \tag{C.86}$$

$$C = \sigma_t e^{r(T-t)}\left(\sigma_t S_t \frac{\partial f_t}{\partial S_t} + \rho\nu_t \frac{\partial f_t}{\partial X_t}\right). \tag{C.87}$$

We obtain the optimal $\theta_t^*$ by solving the first-order condition:

$$\frac{dL}{d\theta_t} = A - \gamma(B\theta_t + C) = 0. \tag{C.88}$$

Solving for $\theta_t^*$:

$$\theta_t^* = \frac{A - \gamma C}{\gamma B}. \tag{C.89}$$

Substituting expressions (C.85)–(C.87) into (C.89):

$$\theta_t^* = \frac{(\mu_t - r)e^{r(T-t)} - \gamma\sigma_t e^{r(T-t)}\left(\sigma_t S_t \frac{\partial f_t}{\partial S_t} + \rho\nu_t \frac{\partial f_t}{\partial X_t}\right)}{\gamma\sigma_t^2 e^{2r(T-t)}}. \tag{C.90}$$

Cancelling out $e^{r(T-t)}$ from the numerator and denominator gives us the optimal stock investment policy provided in Basak and Chabakauri (2010).

$$\theta_t^* = \frac{\mu_t - r}{\gamma\sigma_t^2} e^{-r(T-t)} - \left(S_t \frac{\partial f_t}{\partial S_t} + \frac{\rho\nu_t}{\sigma_t}\frac{\partial f_t}{\partial X_t}\right) e^{-r(T-t)}. \tag{C.91}$$

Isolating the hedging demand (the second term in Equation (C.91)), $\theta_H$, we can derive an alternative representation in terms of covariance, allowing for deeper intuition on its role.

We now compute the instantaneous covariance between $\mathrm{d}S_t/S_t$ and $\mathrm{d}f_t$. Isolating the stochastic terms from Equations (C.76) and (3.1) we obtain:

$$\mathrm{cov}_t\left(\frac{\mathrm{d}S_t}{S_t}, \mathrm{d}f_t\right) = \mathrm{cov}_t\left(\sigma_t\,\mathrm{d}w_t, \sigma_t S_t \frac{\partial f_t}{\partial S_t}\mathrm{d}w_t + \nu_t \frac{\partial f_t}{\partial X_t}\mathrm{d}w_{X,t}\right). \tag{C.92}$$

Using the standard properties:

$$\mathrm{cov}_t(\mathrm{d}w_t, \mathrm{d}w_t) = \mathrm{d}t, \quad \mathrm{cov}_t(\mathrm{d}w_t, \mathrm{d}w_{X,t}) = \rho\,\mathrm{d}t, \tag{C.93}$$

we obtain:

$$\mathrm{cov}_t\left(\frac{\mathrm{d}S_t}{S_t}, \mathrm{d}f_t\right) = \sigma_t^2 S_t \frac{\partial f_t}{\partial S_t}\,\mathrm{d}t + \rho\sigma_t\nu_t \frac{\partial f_t}{\partial X_t}\,\mathrm{d}t. \tag{C.94}$$

Dividing both sides of (C.94) by $\sigma_t^2\mathrm{d}t$ and multiplying by $-e^{-r(T-t)}$:

$$\theta_{Ht} = -\frac{\mathrm{cov}_t\left(\frac{\mathrm{d}S_t}{S_t}, \mathrm{d}f_t\right)}{\sigma_t^2\,\mathrm{d}t} e^{-r(T-t)} = -\left(S_t \frac{\partial f_t}{\partial S_t} + \frac{\rho\nu_t}{\sigma_t}\frac{\partial f_t}{\partial X_t}\right) e^{-r(T-t)}. \tag{C.95}$$

## C.4 Derivation of the Explicit Expression for the Optimal Policy Using the Hedge-Neutral Measure

We aim to derive an explicit expression for the optimal stock policy $\theta_t^*$ in terms of the model's exogenous parameters. In its current form, Equation (C.91) contains $f$, which depends on the entire future path of the optimal policy. This path-dependence makes the expression implicit and prevents a closed-form analytical solution. To address this, we introduce a hedge-neutral measure $\mathbb{P}^*$ under which the intertemporal hedging demand vanishes, thereby removing the dependence on future policy paths and allowing the problem to be solved analytically.

The hedge-neutral measure $\mathbb{P}^*$ is constructed via Girsanov's theorem by adjusting the drift of the Brownian motion driving the risky asset. Under $\mathbb{P}^*$, the risky asset earns the risk-free rate, and the state variable dynamics are adjusted via their correlation with the risky asset. This transformation eliminates the motive to hedge against changes in the investment opportunity set, isolating the myopic component of the optimal policy.

#### Step 1: Dynamics under the hedge-neutral measure

We define a new Brownian motion $w_t^*$ under $\mathbb{P}^*$ as:

$$dw_t^* := dw_t + \frac{\mu_t - r}{\sigma_t}\, dt, \tag{C.96}$$

which eliminates the risk premium from the risky asset dynamics. The resulting transformed dynamics are:

$$\frac{dS_t}{S_t} = r\, dt + \sigma_t\, dw_t^*, \tag{C.97}$$

$$dw_{X,t}^* = dw_{X,t} + \rho\, \frac{\mu_t - r}{\sigma_t}\, dt, \tag{C.98}$$

$$dX_t = \left( m_t - \rho\, \nu_t\, \frac{\mu_t - r}{\sigma_t} \right) dt + \nu_t\, dw_{X,t}^*. \tag{C.99}$$

The corresponding Radon–Nikodym derivative from $\mathbb{P}$ to $\mathbb{P}^*$ is:

$$\frac{d\mathbb{P}^*}{d\mathbb{P}}\bigg|_{\mathcal{F}_t} = \exp\left( -\int_0^t \frac{\mu_s - r}{\sigma_s}\, dw_s - \frac{1}{2} \int_0^t \left( \frac{\mu_s - r}{\sigma_s} \right)^2 ds \right). \tag{C.100}$$

Since the market is incomplete due to the non-traded state variable $X_t$, the hedge-neutral measure $\mathbb{P}^*$ is not unique. Our choice eliminates the intertemporal hedging demand term in the policy expression.

#### Step 2: PDE for $f$ under $\mathbb{P}^*$

Let $f(S_t, X_t, t)$ (from Equation (3.14)) denote the anticipated future portfolio gains from time $t$ to $T$. Under $\mathbb{P}^*$, applying the infinitesimal generator $\mathcal{L}$ of $(S_t, X_t)$ to $f$ gives:

$$\begin{aligned}\mathcal{L} f_t = r S_t \frac{\partial f_t}{\partial S_t} + \left( m_t - \rho\, \nu_t\, \frac{\mu_t - r}{\sigma_t} \right) \frac{\partial f_t}{\partial X_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2} + \frac{1}{2} \nu_t^2 \frac{\partial^2 f_t}{\partial X_t^2} \\ + \rho\, \nu_t\, \sigma_t\, S_t \frac{\partial^2 f_t}{\partial S_t \partial X_t}.\end{aligned} \tag{C.101}$$

Since $f$ is defined in Equation (3.14) as the conditional expectation of a deterministic integral functional of $(S_t, X_t)$, the Feynman–Kac theorem (Karatzas and Shreve, 1991) implies that $f$ satisfies the backward parabolic PDE:

$$\frac{\partial f_t}{\partial t} + \mathcal{L} f_t + \frac{1}{\gamma} \left( \frac{\mu_t - r}{\sigma_t} \right)^2 = 0, \quad f(S_T, X_T, T) = 0, \tag{C.102}$$

or explicitly:

$$\begin{aligned}\frac{\partial f_t}{\partial t} + r S_t \frac{\partial f_t}{\partial S_t} + \left( m_t - \rho\, \nu_t\, \frac{\mu_t - r}{\sigma_t} \right) \frac{\partial f_t}{\partial X_t} + \frac{1}{2} \sigma_t^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2} + \frac{1}{2} \nu_t^2 \frac{\partial^2 f_t}{\partial X_t^2} \\ + \rho\, \nu_t\, \sigma_t\, S_t \frac{\partial^2 f_t}{\partial S_t \partial X_t} + \frac{1}{\gamma} \left( \frac{\mu_t - r}{\sigma_t} \right)^2 = 0.\end{aligned} \tag{C.103}$$

#### Step 3: Feynman–Kac representation

By the Feynman–Kac theorem (Karatzas and Shreve, 1991), the unique classical solution to Equation (C.103) is:

$$f(S_t, X_t, t) = \mathbb{E}_t^{\mathbb{P}^*} \left[ \int_t^T \frac{1}{\gamma} \left( \frac{\mu_s - r}{\sigma_s} \right)^2 ds \right], \tag{C.104}$$

where $\mathbb{E}_t^{\mathbb{P}^*}[\cdot]$ denotes conditional expectation under the hedge-neutral measure.

#### Step 4: Substitution into the optimal policy

Substituting Equation (C.104) into the earlier expression for $\theta_t^*$ (Equation (C.91)) yields:

$$\theta_t^* = \frac{\mu_t - r}{\gamma\sigma_t^2} e^{-r(T-t)}$$
$$- \frac{1}{\gamma}\left( S_t \frac{\partial}{\partial S_t} \mathbb{E}_t^{\mathbb{P}^*}\left[ \int_t^T \left( \frac{\mu_s - r}{\sigma_s} \right)^2 ds \right] + \rho \frac{\nu_t}{\sigma_t} \frac{\partial}{\partial X_t} \mathbb{E}_t^{\mathbb{P}^*}\left[ \int_t^T \left( \frac{\mu_s - r}{\sigma_s} \right)^2 ds \right] \right) e^{-r(T-t)}. \tag{C.105}$$

# D Continuous-Time Mean-Variance Optimisation with Trade-Speed Transaction Cost Penalty

This appendix provides the full derivation for the alternative trade-speed formulation of quadratic transaction costs, as referenced in Section 3.2, where the trading speed becomes the control variable instead of the allocation itself.

## D.1 Economic Setup

We retain the multiple-risky-asset framework introduced in Section 3.1.3, based on the generalisation of Basak and Chabakauri (2010), with $D$ risky assets and $K$ state variables. The dynamics of $\mathbf{S}_t$ and $\mathbf{X}_t$ remain as in Equations (3.28)–(3.29), and $\boldsymbol{\theta}_t \in \mathbb{R}^D$ denotes the allocation to risky assets at time $t$.

To incorporate a trade-speed penalty, we assume that $\boldsymbol{\theta}_t$ is absolutely continuous:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_0 + \int_0^t \dot{\boldsymbol{\theta}}_u \, du, \tag{D.106}$$

with $\dot{\boldsymbol{\theta}} \in L^2([0,T];\mathbb{R}^D)$. This ensures that $\dot{\boldsymbol{\theta}}_t$ is well defined, so that

$$d\boldsymbol{\theta}_t = \dot{\boldsymbol{\theta}}_t \, dt, \tag{D.107}$$

and $\boldsymbol{\theta}_t$ itself must be treated as part of the state vector, while $\dot{\boldsymbol{\theta}}_t$ becomes the control variable.

The investor's wealth dynamics are then modified to

$$dW_t = \left[ rW_t + \boldsymbol{\theta}_t^\top (\boldsymbol{\mu}_t - r\mathbf{1}) \right] dt + \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \, d\mathbf{w}_t - \lambda \|\dot{\boldsymbol{\theta}}_t\|^2 dt, \tag{D.108}$$

where the quadratic term $\lambda\|\dot{\boldsymbol{\theta}}_t\|^2$ penalises rapid rebalancing, with $\lambda > 0$ controlling the severity.

## D.2 Value Function and HJB Derivation

We define the value function as

$$J_t(W_t, \boldsymbol{\theta}_t, \mathbf{S}_t, \mathbf{X}_t, t) = \mathbb{E}[W_T \mid \mathcal{F}_t] - \frac{\gamma}{2}\mathrm{Var}[W_T \mid \mathcal{F}_t], \tag{D.109}$$

which reflects the dynamic mean–variance criterion under the information set $\mathcal{F}_t$.

Expanding explicitly,

$$J_t = \mathbb{E}[W_T \mid \mathcal{F}_t] - \frac{\gamma}{2}\left( \mathbb{E}[W_T^2 \mid \mathcal{F}_t] - (\mathbb{E}[W_T \mid \mathcal{F}_t])^2 \right). \tag{D.110}$$

Applying Itô's lemma to $J(W_t, \boldsymbol{\theta}_t, \mathbf{S}_t, \mathbf{X}_t, t)$, using dynamics (D.108), (D.106), and (3.29), gives

$$dJ = \frac{\partial J}{\partial t}\,dt + \frac{\partial J}{\partial W}\,dW_t + \frac{\partial J}{\partial \boldsymbol{\theta}}^{\top} d\boldsymbol{\theta}_t + \sum_{j=1}^{K} \frac{\partial J}{\partial X_j}\,dX_{jt}$$
$$+ \tfrac{1}{2}\frac{\partial^2 J}{\partial W^2}\,\langle dW_t, dW_t\rangle + \tfrac{1}{2}\sum_{i,j=1}^{K} \frac{\partial^2 J}{\partial X_i \partial X_j}\,\langle dX_{it}, dX_{jt}\rangle + \sum_{j=1}^{K} \frac{\partial^2 J}{\partial W \partial X_j}\,\langle dW_t, dX_{jt}\rangle, \quad \text{(D.111)}$$

since all quadratic variation terms involving $d\boldsymbol{\theta}_t$ vanish (as it is of finite variation).

The quadratic variations are

$$\langle dW_t, dW_t\rangle = \boldsymbol{\theta}_t^{\top} \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^{\top} \boldsymbol{\theta}_t\,dt, \quad \text{(D.112)}$$
$$\langle dX_{it}, dX_{jt}\rangle = \boldsymbol{\nu}_i^{\top} \boldsymbol{\nu}_j\,dt, \quad i, j = 1, \dots, K, \quad \text{(D.113)}$$
$$\langle dW_t, dX_{jt}\rangle = \boldsymbol{\theta}_t^{\top} \boldsymbol{\sigma}_t \rho_j \boldsymbol{\nu}_j\,dt. \quad \text{(D.114)}$$

Substituting into (D.111) and collecting drift terms yields

$$\mathbb{E}[dJ \mid \mathcal{F}_t] = \frac{\partial J}{\partial t} + \Big[rW_t + \boldsymbol{\theta}_t^{\top}(\boldsymbol{\mu}_t - r\mathbf{1}) - \lambda\|\dot{\boldsymbol{\theta}}_t\|^2\Big]\frac{\partial J}{\partial W} + \frac{\partial J}{\partial \boldsymbol{\theta}}^{\top}\dot{\boldsymbol{\theta}}_t$$
$$+ \sum_{j=1}^{K} m_j(\mathbf{X}_t, t)\frac{\partial J}{\partial X_j} + \tfrac{1}{2}\boldsymbol{\theta}_t^{\top}\boldsymbol{\sigma}_t\boldsymbol{\sigma}_t^{\top}\boldsymbol{\theta}_t\,\frac{\partial^2 J}{\partial W^2} + \tfrac{1}{2}\sum_{i,j=1}^{K} \boldsymbol{\nu}_i^{\top}\boldsymbol{\nu}_j\frac{\partial^2 J}{\partial X_i \partial X_j}$$
$$+ \sum_{j=1}^{K} \boldsymbol{\theta}_t^{\top}\boldsymbol{\sigma}_t\rho_j\boldsymbol{\nu}_j\frac{\partial^2 J}{\partial W \partial X_j}. \quad \text{(D.115)}$$

By the dynamic programming principle,

$$\sup_{\dot{\boldsymbol{\theta}}_t} \mathbb{E}[dJ \mid \mathcal{F}_t] = 0. \quad \text{(D.116)}$$

Thus the HJB equation is

$$0 = \frac{\partial J}{\partial t} + \sup_{\dot{\boldsymbol{\theta}}_t}\Bigg\{\Big[rW_t + \boldsymbol{\theta}_t^{\top}(\boldsymbol{\mu}_t - r\mathbf{1}) - \lambda\|\dot{\boldsymbol{\theta}}_t\|^2\Big]\frac{\partial J}{\partial W} + \frac{\partial J}{\partial \boldsymbol{\theta}}^{\top}\dot{\boldsymbol{\theta}}_t + \sum_{j=1}^{K} m_j(\mathbf{X}_t, t)\frac{\partial J}{\partial X_j}$$
$$+ \tfrac{1}{2}\boldsymbol{\theta}_t^{\top}\boldsymbol{\sigma}_t\boldsymbol{\sigma}_t^{\top}\boldsymbol{\theta}_t\,\frac{\partial^2 J}{\partial W^2} + \tfrac{1}{2}\sum_{i,j=1}^{K} \boldsymbol{\nu}_i^{\top}\boldsymbol{\nu}_j\frac{\partial^2 J}{\partial X_i \partial X_j} + \sum_{j=1}^{K} \boldsymbol{\theta}_t^{\top}\boldsymbol{\sigma}_t\rho_j\boldsymbol{\nu}_j\frac{\partial^2 J}{\partial W \partial X_j}\Bigg\},$$
$$\text{(D.117)}$$

with terminal condition

$$J_T(W_T) = W_T - \tfrac{\gamma}{2}W_T^2. \quad \text{(D.118)}$$

**First-Order Condition.** The terms in (D.117) involving $\dot{\boldsymbol{\theta}}_t$ are

$$-\lambda\|\dot{\boldsymbol{\theta}}_t\|^2\,\frac{\partial J}{\partial W} + \frac{\partial J}{\partial \boldsymbol{\theta}}^{\top}\dot{\boldsymbol{\theta}}_t,$$

a concave quadratic with FOC

$$-2\lambda\frac{\partial J}{\partial W}\dot{\boldsymbol{\theta}}_t + \frac{\partial J}{\partial \boldsymbol{\theta}} = 0,$$

yielding the closed-form optimal trading speed

$$\dot{\boldsymbol{\theta}}_t^{*} = \frac{\frac{\partial J}{\partial \boldsymbol{\theta}}}{2\lambda\,\frac{\partial J}{\partial W}}. \quad \text{(D.119)}$$

## D.3 Numerical Solution via Deep 2BSDE

### Forward–Backward 2BSDE System

The optimal trading speed (D.119) enters the dynamics through both the forward and backward components of the 2BSDE system. The portfolio allocations evolve according to

$$d\boldsymbol{\theta}_t = \dot{\boldsymbol{\theta}}_t^* \, dt, \tag{D.120}$$

linking the control directly to the state dynamics.

**Forward Dynamics.** Wealth evolves under the forward SDE in Equation (D.108).

**Backward Dynamics.** To approximate the value function, we define

$$Y_t := J(W_t, \boldsymbol{\theta}_t, \mathbf{S}_t, \mathbf{X}_t, t), \quad Z_t := \frac{\partial J}{\partial W}, \quad \mathbf{U}_t := \frac{\partial J}{\partial \boldsymbol{\theta}}, \quad \Gamma_t := \frac{\partial^2 J}{\partial W^2}. \tag{D.121}$$

The 2BSDE system then reads

$$-dY_t = g(W_t, \mathbf{S}_t, \mathbf{X}_t, \boldsymbol{\theta}_t, Z_t, \Gamma_t, \mathbf{U}_t, t) \, dt - Z_t^\top \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t d\mathbf{w}_t, \tag{D.122}$$

$$dZ_t = -\Gamma_t^\top \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t d\mathbf{w}_t, \tag{D.123}$$

with terminal condition $Y_T = J_T(W_T)$. The generator $g$ corresponds to the HJB (D.117) drift restricted to the wealth dimension

$$g(W_t, \mathbf{S}_t, \mathbf{X}_t, \boldsymbol{\theta}_t, Z_t, \Gamma_t, \mathbf{U}_t, t) = \Big[ rW_t + \boldsymbol{\theta}_t^\top (\boldsymbol{\mu}_t - r\mathbf{1}) \Big] Z_t + \tfrac{1}{2} \boldsymbol{\theta}_t^\top \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top \boldsymbol{\theta}_t \Gamma_t - \lambda \|\dot{\boldsymbol{\theta}}_t^*\|^2 Z_t. \tag{D.124}$$

**Neural Network Parameterisation.** Neural networks approximate the derivatives

$$Z_t \approx \frac{\partial J}{\partial W}, \quad \mathbf{U}_t \approx \frac{\partial J}{\partial \boldsymbol{\theta}}, \quad \Gamma_t \approx \frac{\partial^2 J}{\partial W^2}. \tag{D.125}$$

The optimal trading speed is recovered at each step from the first-order condition:

$$\dot{\boldsymbol{\theta}}_t^* = \frac{\mathbf{U}_t}{2\lambda Z_t}, \tag{D.126}$$

and used to update $\boldsymbol{\theta}_t$ in the forward simulation. This ensures the control remains consistent with the HJB structure while preserving full differentiability of the 2BSDE scheme.

# E Proofs

## E.1 Proof of Ratio Product Summation Identity

We now prove the identity used in Equation (B.62):

$$\sum_{t=0}^{T-1} B_t \cdot \prod_{k=t+1}^{T-1} (1 - B_k)^{-1} = 1 - \prod_{t=0}^{T-1} (1 - B_t). \tag{E.127}$$

Define

$$M_t = \prod_{k=t}^{T-1} (1 - B_k), \tag{E.128}$$

with the convention $M_T = 1$.

Clearly,

$$M_t = (1 - B_t)\, M_{t+1}, \tag{E.129}$$

so that

$$M_{t+1} = \frac{M_t}{1 - B_t}. \tag{E.130}$$

Consider the sum:

$$S = \sum_{t=0}^{T-1} B_t \cdot \prod_{k=t+1}^{T-1} \left(1 - B_k\right)^{-1}. \tag{E.131}$$

Noting that

$$\prod_{k=t+1}^{T-1} \left(1 - B_k\right)^{-1} = \frac{1}{M_{t+1}}, \tag{E.132}$$

we obtain

$$S = \sum_{t=0}^{T-1} \frac{B_t}{M_{t+1}}. \tag{E.133}$$

From (E.129),

$$B_t = 1 - \frac{M_{t+1}}{M_t}, \tag{E.134}$$

so that

$$\frac{B_t}{M_{t+1}} = \frac{1}{M_{t+1}} - \frac{1}{M_t}. \tag{E.135}$$

Substituting into (E.133) gives

$$S = \sum_{t=0}^{T-1} \left( \frac{1}{M_{t+1}} - \frac{1}{M_t} \right). \tag{E.136}$$

This telescopes to

$$S = \frac{1}{M_T} - \frac{1}{M_0}. \tag{E.137}$$

Since $M_T = 1$,

$$S = 1 - \frac{1}{M_0}. \tag{E.138}$$

But

$$M_0 = \prod_{t=0}^{T-1} \left(1 - B_t\right). \tag{E.139}$$

Therefore,

$$S = 1 - \prod_{t=0}^{T-1} \left(1 - B_t\right), \tag{E.140}$$

which completes the proof.

## E.2 Proof that the Ratio Product Equals $1/\prod s_k$

We prove the identity

$$\prod_{k=t+1}^{T-1} \frac{1-2B_k}{s_k(1-B_k)} = \prod_{k=t+1}^{T-1} \frac{1}{s_k}. \tag{E.141}$$

From the backward recursions (B.49)–(B.51), we have

$$\alpha_t = \alpha_{t+1}\, s_t^2(1-B_t), \tag{E.142}$$
$$\beta_t = \beta_{t+1}\, s_t(1-2B_t). \tag{E.143}$$

Hence, their ratio evolves as

$$\frac{\beta_t}{\alpha_t} = \frac{\beta_{t+1}}{\alpha_{t+1}} \cdot \frac{1-2B_t}{s_t(1-B_t)}. \tag{E.144}$$

Iterating (E.144) from $t+1$ to $T-1$ gives

$$\frac{\beta_{t+1}}{\alpha_{t+1}} = \frac{\beta_T}{\alpha_T} \cdot \prod_{k=t+1}^{T-1} \frac{1-2B_k}{s_k(1-B_k)}. \tag{E.145}$$

At terminal time $T$, the boundary conditions imply

$$\frac{\beta_T}{\alpha_T} = w\gamma^*. \tag{E.146}$$

Substituting (E.146) into (E.145) yields

$$\frac{\beta_{t+1}}{\alpha_{t+1}} = w\gamma^* \cdot \prod_{k=t+1}^{T-1} \frac{1-2B_k}{s_k(1-B_k)}. \tag{E.147}$$

On the other hand, from the optimal policy formula (A.32) we also have:

$$\frac{\gamma^*}{2} = \frac{\beta_{t+1}}{2w\alpha_{t+1}} \cdot \left[ \prod_{k=t+1}^{T-1} \frac{s_k(1-B_k)}{1-2B_k} \right]. \tag{E.148}$$

Equating (E.147) and (E.148) and cancelling $w\gamma^*$ gives

$$\prod_{k=t+1}^{T-1} \frac{1-2B_k}{1-B_k} = 1. \tag{E.149}$$

Therefore,

$$\prod_{k=t+1}^{T-1} \frac{1-2B_k}{s_k(1-B_k)} = \prod_{k=t+1}^{T-1} \frac{1}{s_k}, \tag{E.150}$$

which proves the identity (E.141).

# F Numerical Implementation of the Discrete-Time Model (Precommitment, Li and Ng (2000))

This appendix provides the numerical implementation details for the discrete-time precommitment model of Li and Ng (2000), introduced in Appendix A. As this model is included for completeness rather than as part of the main contribution, we summarise only the essential implementation steps. The strategy is applied to daily financial data (market setup as described in Section 4.1) with rebalancing at the same frequency. Parameters are estimated on a rolling basis from historical observations to reflect evolving market conditions.

## F.1 Rolling Parameter Estimation

Given adjusted close prices $\{\mathbf{S}_t\}$ for $n$ risky assets, daily gross returns are

$$\mathbf{R}_t = \frac{\mathbf{S}_t}{\mathbf{S}_{t-1}}. \tag{F.151}$$

Excess returns relative to the daily gross risk-free rate $s_t$ are

$$\mathbf{P}_t = \mathbf{R}_t - s_t \cdot \mathbf{1}, \tag{F.152}$$

where

$$s_t = \exp\left(\frac{r}{252}\right), \tag{F.153}$$

with $r$ denoting the annualised risk-free rate.

Parameters are estimated on a rolling window of one trading year ($L = 252$). At each time $t$,

$$\boldsymbol{\mu}_t \approx \frac{1}{L} \sum_{j=t-L+1}^{t} \mathbf{P}_j, \tag{F.154}$$

$$\boldsymbol{\Sigma}_t \approx \frac{1}{L-1} \sum_{j=t-L+1}^{t} \big(\mathbf{P}_j - \boldsymbol{\mu}_t\big)\big(\mathbf{P}_j - \boldsymbol{\mu}_t\big)^\top. \tag{F.155}$$

These rolling estimates correspond directly to the quantities used in the computation of $\mathbf{Q}_t$, $\mathbf{K}_t$, and $B_t$ in Equations (A.23), (A.19), and (A.24) of our mathematical formulation.

## F.2 Computation of Model Coefficients

At each time step $t$, the rolling estimates $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ are used to compute the model coefficients defined in Section A, namely $\mathbf{Q}_t$, $\mathbf{K}_t$, and $B_t$ (Equations (A.23), (A.19), and (A.24)).

$$\mathbf{Q}_t = \mathrm{Cov}_t(\mathbf{P}_t) + \mathbb{E}_t[\mathbf{P}_t]\,\mathbb{E}_t[\mathbf{P}_t]^\top \;\approx\; \boldsymbol{\Sigma}_t + \boldsymbol{\mu}_t\,\boldsymbol{\mu}_t^\top, \tag{F.156}$$

$$\mathbf{K}_t = \mathbf{Q}_t^{-1}\,\mathbb{E}_t[\mathbf{P}_t] \;\approx\; \mathbf{Q}_t^{-1}\,\boldsymbol{\mu}_t, \tag{F.157}$$

$$B_t = \mathbb{E}_t[\mathbf{P}_t]^\top\,\mathbf{K}_t \;\approx\; \boldsymbol{\mu}_t^\top\,\mathbf{K}_t. \tag{F.158}$$

The inversion of $\mathbf{Q}_t$ can be unstable when assets are highly correlated or the rolling window is short. To improve conditioning, we regularise:

$$\mathbf{Q}_t \;\leftarrow\; \mathbf{Q}_t + \varepsilon\mathbf{I}, \tag{F.159}$$

where $\varepsilon > 0$ shifts the eigenvalues of $\mathbf{Q}_t$ away from zero. This is ridge (Tikhonov) regularisation with $B = \mathbf{I}$ (Golub and Loan, 2013, Section 6.1.4). In practice we set $\varepsilon = 10^{-3}$, which ensures stable inversion without materially altering the estimates.

The coefficients $\mathbf{Q}_t$, $\mathbf{K}_t$, and $B_t$ are then supplied to the backward recursions that determine the optimal portfolio policy.

## F.3 Backward Recursions

We compute the sequences $\{\alpha_t\}$ and $\{\beta_t\}$ together with the scalar $\gamma^*$, which fully determine the dynamic mean–variance optimal policy. Their recursive definitions and terminal conditions are given in Equations (A.21) and (A.22), and the closed-form expression for $\gamma^*$ in Equation (A.31).

The entire paths $\{\alpha_t\}_{t=0}^{T}$ and $\{\beta_t\}_{t=0}^{T}$ are obtained by backward iteration from the terminal conditions (as described in Algorithm 9).

**Algorithm 9** Backward computation of $\alpha_t$, $\gamma^*$, and $\beta_t$

Set $\alpha_T \leftarrow 1$
**for** $t = T-1$ down to $0$ **do**
    $\alpha_t \leftarrow \alpha_{t+1} \cdot s_t^2 \cdot \big(1 - B_t\big)$
**end for**
Compute $\gamma^*$ as:

$$\gamma^* \leftarrow 2 \cdot \left(\prod_{t=0}^{T-1} s_t\right) \cdot x_0 + \frac{1}{w \cdot \prod_{t=0}^{T-1} (1 - B_t)}.$$

Set $\beta_T \leftarrow w \cdot \gamma^*$
**for** $t = T-1$ down to $0$ **do**
    $\beta_t \leftarrow \beta_{t+1} \cdot s_t \cdot \big(1 - 2B_t\big)$
**end for**

These precomputed paths are then used in the computation of the optimal policy in the next section.

## F.4 Optimal Policy Computation

Given the coefficients, the optimal allocation at time $t$ takes the affine form (Appendix A)

$$\mathbf{u}_t^* = -\, s_t \mathbf{K}_t \, x_t \; + \; \mathbf{v}_t, \tag{F.160}$$

where the intertemporal hedging component is

$$\mathbf{v}_t = \frac{\beta_{t+1}}{2w\alpha_{t+1}} \mathbf{K}_t. \tag{F.161}$$

At each time step, $\mathbf{u}_t^*$ is evaluated using the precomputed sequences $\{\alpha_t\}$ and $\{\beta_t\}$, the rolling estimate $\mathbf{K}_t$, and current wealth $x_t$. The resulting allocation is applied in the wealth update during backtesting.

The coefficient $w$ controls risk aversion in this discrete-time setting by scaling only the hedging demand in (F.161), while the myopic term in (F.160) remains unaffected. Its role is therefore analogous to the continuous-time parameter $\gamma$, though the two are not directly comparable. In practice, increasing $w$ beyond moderate values produces little further reduction in wealth variability unless trading constraints are imposed. We set $w = 5$, which yields a volatility profile comparable to the $\gamma$-calibrated strategies in the main text while avoiding excessively dampened hedging responses.

## F.5 Rolling Backtesting Framework

We summarise the numerical implementation of the discrete-time mean–variance model as a rolling backtest.

**Algorithm 10** Rolling Backtest of Discrete-Time Precommitment Mean–Variance Strategy

**Input:** Market data $\{\mathbf{S}_t\}$, initial wealth $x_0$, risk aversion $w$, risk-free rate $r$, transaction cost parameter $\lambda_{\text{day}}$, rolling window size $L$
Compute daily risk-free return $s_t$ via Equation (F.153)
Initialise $\mathbf{u}^*_{t-1} \leftarrow \mathbf{0}$
**for** each window of length $L$ ending at $t$ **do**
    Estimate $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ from excess returns
    Compute $\mathbf{Q}_t$ (Equation (F.156)), regularise as in Equation (F.159)
    Compute $\mathbf{K}_t$ (Equation (F.157)), $B_t$ (Equation (F.158))
**end for**
Compute $\{\alpha_t\}_{t=0}^{T}$ and $\{\beta_t\}_{t=0}^{T}$, and $\gamma^*$ via Algorithm 9
**for** each trading day $t$ **do**
    Compute $\mathbf{v}_t$ (Equation (F.161))
    Compute $\mathbf{u}^*_t$ (Equation (F.160))
    Update wealth:

$$x_{t+1} = \left(\frac{\mathbf{u}^*_t}{\mathbf{S}_t}\right)^{\top} \mathbf{S}_{t+1} + \left(x_t - \mathbf{u}^{*\top}_t \mathbf{1}\right) s_t - \lambda_{\text{day}} \|\mathbf{u}^*_t - \mathbf{u}^*_{t-1}\|^2$$

    Save $x_{t+1}$ and $\mathbf{u}^*_t$; set $\mathbf{u}^*_{t-1} \leftarrow \mathbf{u}^*_t$
**end for**
**Output:** Wealth path $\{x_t\}_{t=0}^{T}$, allocations $\{\mathbf{u}^*_t\}_{t=0}^{T-1}$

Constraints are imposed to ensure numerical stability and prevent extreme behaviour. Positions $\mathbf{u}^*_t$ are clipped within $\pm 3 \cdot x_0$, allocation changes are similarly capped, leverage is limited to 1.5x, and a wealth floor prevents ruin. These safeguards are not part of the theoretical model but are necessary because the precommitment framework can generate highly leveraged, unstable allocations. In theory, the strategy assumes preferences and parameters remain fixed over the horizon and cannot adapt to shocks. In practice, parameters are re-estimated on a rolling basis, but the policy within each window is still precommitment-based and does not dynamically adjust to changing market conditions between recalibrations.

While these constraints highlight the impracticality of the precommitment model for direct deployment, they allow us to study its dynamics under rolling implementation without results being dominated by instability or model failure.

# G Global Loss Training Algorithm

For completeness, we include the global loss method of Beck et al. (2019), referenced in Section 4.3.3 but not adopted in our main experiments. The scheme optimises the value process only at terminal time, which often leads to instability under transaction costs or high dimensions.

**Algorithm 11** Global Loss Training (Deep BSDE)

1: Initialise networks $\mathrm{NN}_\phi, \mathrm{NN}_\psi, \mathrm{NN}_\xi$ and learnable scalar $Y_0$
2: **for** each epoch **do**
3: Simulate forward paths $\{W_t, \mathbf{S}_t, \mathbf{X}_t, \Delta\mathbf{w}_t\}$ using Algorithm 5
4: Set $Y \leftarrow Y_0$
5: **for** each $t_n = 0, \dots, T-1$ **do**
6: Evaluate neural networks and scale allocation:

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_{t_n} &\leftarrow \mathrm{NN}_\phi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}) \\
\boldsymbol{\theta}_{t_n} &\leftarrow W_{t_n}\,\tilde{\boldsymbol{\theta}}_{t_n} \\
Z_{t_n} &\leftarrow \mathrm{NN}_\psi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n}) \\
\Gamma_{t_n} &\leftarrow \mathrm{NN}_\xi(t_n, W_{t_n}, \mathbf{S}_{t_n}, \mathbf{X}_{t_n})
\end{aligned}$$

7: Update value:

$$\begin{aligned}
Y_{t_{n+1}} = Y_{t_n} + \Bigg( & \left[ rW_{t_n} + \boldsymbol{\theta}_{t_n}^\top(\boldsymbol{\mu}_{t_n} - r\mathbf{1}) - \lambda \left\| \tfrac{\boldsymbol{\theta}_{t_n} - \boldsymbol{\theta}_{t_{n-1}}}{\Delta t} \right\|^2 \right] Z_{t_n} \\
& + \tfrac{1}{2}\,\Gamma_{t_n}\,\boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n} \boldsymbol{\sigma}_{t_n}^\top \boldsymbol{\theta}_{t_n} \Bigg) \Delta t - Z_{t_n}\,\boldsymbol{\theta}_{t_n}^\top \boldsymbol{\sigma}_{t_n}\,\Delta\mathbf{w}_{t_n}.
\end{aligned}$$

8: **end for**
9: Compute terminal loss:

$$\mathcal{L}_{\text{global}} = \mathbb{E}\left[ \left| Y_T - (W_T - \tfrac{\gamma}{2} W_T^2) \right|^2 \right]$$

10: Update parameters by backpropagation
11: **end for**

# H Additional Numerical Results

## H.1 Discrete-Time Model Historical Backtest Results

This appendix reports historical backtest outcomes for the discrete-time mean–variance model, complementing the implementation details given earlier in Appendix F.

| Metric | Value |
|---|---|
| Terminal Wealth (£) | 50.97 |
| CAGR (%) | -6.53 |
| Annual Volatility (%) | 24.46 |
| Sharpe Ratio | -0.37 |
| Sortino Ratio | -0.54 |
| Calmar Ratio | -0.09 |
| Maximum Drawdown (%) | 69.95 |
| Average Drawdown (%) | 6.61 |
| Average Recovery Time (days) | 22.89 |
| VaR 95% (%) | 2.41 |
| Expected Shortfall 95% (%) | 3.52 |
| Sharpe Difference t-stat | -36.54 |
| Sharpe Difference p-value | 1.0000 |
| Returns Difference t-stat | -2.11 |
| Returns Difference p-value | 0.9826 |
| Average Leverage | 1.31 |
| Maximum Position Size (£) | 122.63 |
| Average Turnover per Step (%) | 9.71 |
| Annual Turnover (%) | 2447.09 |
| Profit per Turnover (£) | -0.00 |
| Standard Deviation of Portfolio Weights (%) | 47.98 |
| Average Absolute Change in Portfolio Weights (%) | 9.63 |
| Total Transaction Costs (£) | 25.22 |

Table 1: Performance Metrics (discrete-time precommitment model, historical).


Figure 2: Wealth (discrete-time precommitment model, historical).

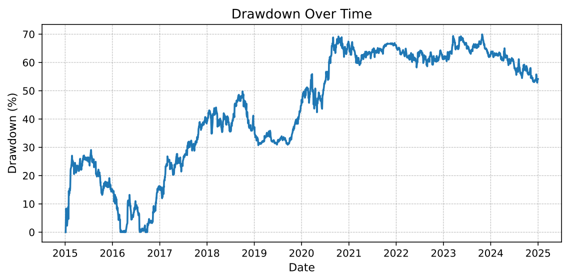



Figure 3: Drawdowns (discrete-time precommitment model, historical).

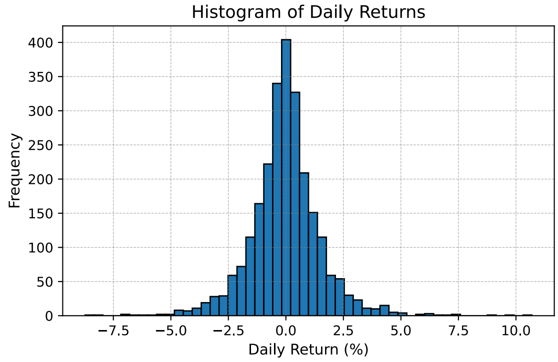



Figure 4: Return distribution (discrete-time precommitment model, historical).

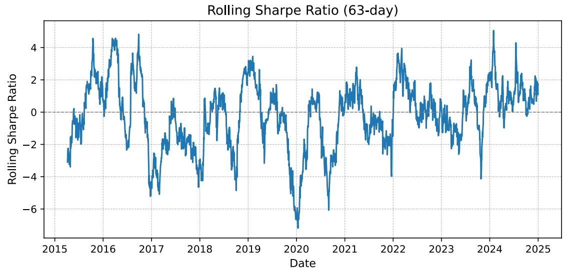



Figure 5: Rolling 63-day Sharpe ratio (discrete-time precommitment model, historical).

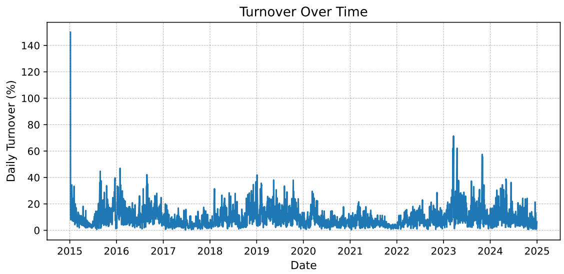



Figure 6: Daily turnover (discrete-time precommitment model, historical).

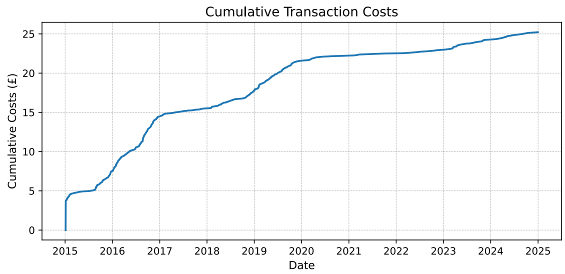



Figure 7: Cumulative costs (discrete-time precommitment model, historical).

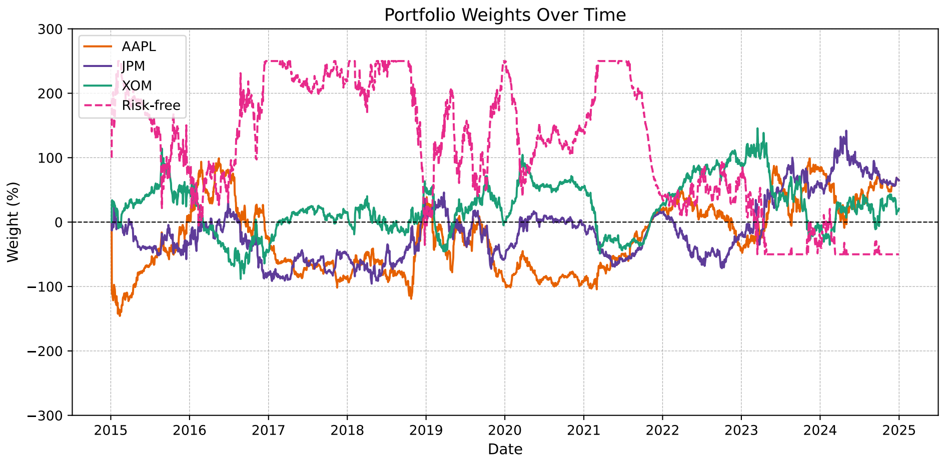



Figure 8: Portfolio weights (discrete-time precommitment model, historical).

## H.2 Discrete-Time Model Parametric Bootstrap Results

This appendix presents parametric bootstrap results for the discrete-time mean-variance model, providing distributional evidence on performance and robustness across resampled market paths.

| Metric | Historical | Bootstrap Median | 5th %ile | 95th %ile | p-value | Prob. of Outperformance (%) |
|---|---|---|---|---|---|---|
| Terminal Wealth (£) | 50.97 | 38.63 | 16.72 | 90.51 | 0.628 | 32.00 |
| CAGR (%) | -6.53 | -8.79 | -15.88 | -0.96 | 0.644 | 33.00 |
| Annual Volatility (%) | 24.46 | 26.03 | 23.27 | 29.45 | 0.402 | 84.00 |
| Sharpe Ratio | -0.37 | -0.43 | -0.70 | -0.13 | 0.748 | 38.00 |
| Sortino Ratio | -0.54 | -0.63 | -1.07 | -0.20 | 0.724 | 37.00 |
| Calmar Ratio | -0.09 | -0.11 | -0.18 | -0.01 | 0.819 | 38.00 |
| Maximum Drawdown (%) | 69.95 | 84.36 | 73.17 | 92.03 | 0.028 | 96.00 |
| Average Drawdown (%) | 6.61 | 9.13 | 3.38 | 54.14 | 0.878 | 66.00 |
| VaR 95% (%) | 2.41 | 2.61 | 2.34 | 2.88 | 0.208 | 86.00 |
| Expected Shortfall 95% (%) | 3.52 | 3.69 | 3.18 | 4.26 | 0.600 | 71.00 |
| Annual Turnover (%) | 2447.09 | 2424.82 | 1819.81 | 2904.67 | 0.946 | 48.00 |
| Average Absolute Change in Portfolio Weights (%) | 9.63 | 9.20 | 6.56 | 11.28 | 0.769 | 37.00 |
| Total Transaction Costs (£) | 25.22 | 26.42 | 8.51 | 62.92 | 0.952 | 53.00 |

Table 2: Performance metrics (discrete-time precommitment model, bootstrap).

| Metric | Mean Difference | t-stat | p-value | Probability > Benchmark (%) |
|---|---|---|---|---|
| Sharpe Ratio Difference | -1.2220 | -34.78 | 1.0000 | 0.00% |
| Returns Difference (%) | -27.39 | -32.11 | 1.0000 | 0.00% |

Table 3: Statistical performance metrics (discrete-time precommitment model, bootstrap).

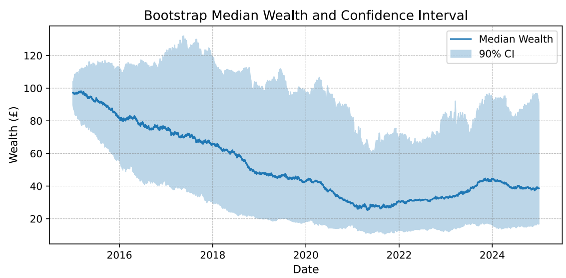



Figure 9: Wealth paths with 90% confidence interval (discrete-time precommitment model, bootstrap)

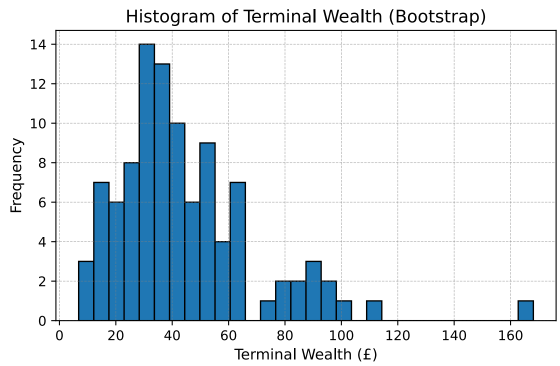



Figure 10: Distribution of terminal wealth (discrete-time precommitment model, bootstrap)

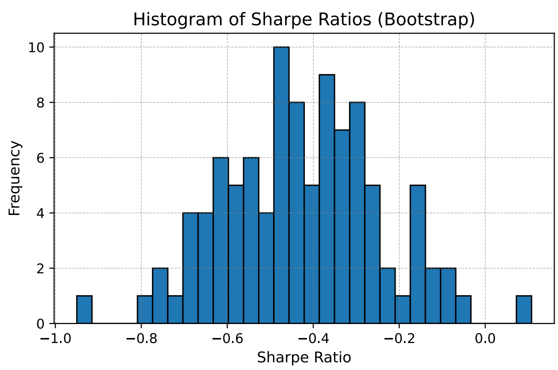



Figure 11: Distribution of Sharpe ratios (discrete-time precommitment model, bootstrap)

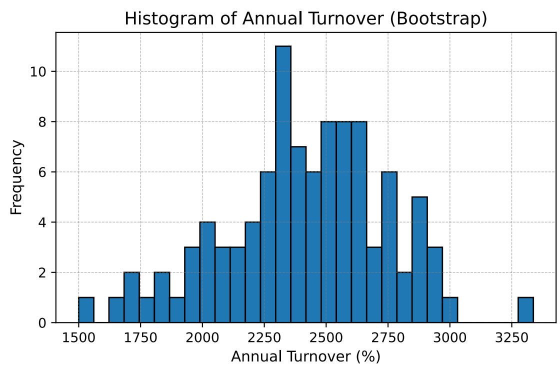



Figure 12: Distribution of annual turnover (discrete-time precommitment model, bootstrap).

## H.3 Cost-Unaware Continuous-Time Model Parametric Bootstrap Results

This appendix presents bootstrap results for the cost-unaware model referenced in Section 5.3, included here for transparency though they add little insight beyond the historical backtest.

| Metric | Historical | Bootstrap Median | 5th %ile | 95th %ile | p-value | Prob. of Outperformance (%) |
|---|---|---|---|---|---|---|
| Terminal Wealth (£) | 136.15 | 134.65 | 102.81 | 171.82 | 0.947 | 49.00 |
| CAGR (%) | 3.18 | 2.95 | 0.27 | 5.43 | 0.891 | 48.00 |
| Annual Volatility (%) | 3.35 | 5.01 | 3.41 | 6.89 | 0.159 | 95.00 |
| Sharpe Ratio | 0.20 | 0.11 | -0.39 | 0.69 | 0.775 | 43.00 |
| Sortino Ratio | 0.25 | 0.13 | -0.42 | 0.88 | 0.738 | 42.00 |
| Calmar Ratio | 0.48 | 0.31 | 0.02 | 0.79 | 0.539 | 27.00 |
| Maximum Drawdown (%) | 6.64 | 10.66 | 5.65 | 27.06 | 0.572 | 84.00 |
| Average Drawdown (%) | 0.20 | 0.38 | 0.23 | 0.98 | 0.853 | 98.00 |
| VaR 95% (%) | 0.23 | 0.37 | 0.25 | 0.52 | 0.129 | 98.00 |
| Expected Shortfall 95 (%) | 0.46 | 0.75 | 0.48 | 1.09 | 0.134 | 98.00 |
| Average Absolute Change in Portfolio Weights (%) | 18.53 | 24.98 | 16.61 | 34.66 | 0.260 | 83.00 |
| Annual Turnover (%) | 4671.89 | 6291.95 | 4185.15 | 8726.29 | 0.261 | 83.00 |
| Total Transaction Costs (£) | 321.25 | 573.80 | 270.83 | 1,247.33 | 0.445 | 88.00 |

Table 4: Performance Metrics (cost-unaware model, bootstrap).

| Metric | Mean Difference | t-stat | p-value | Probability > Benchmark (%) |
|---|---|---|---|---|
| Sharpe Ratio Difference | -0.6988 | -4.78 | 1.0000 | 0.00% |
| Returns Difference (%) | -13.22 | -4.67 | 1.0000 | 0.00% |

Table 5: Statistical performance metrics (cost-unaware model, bootstrap).

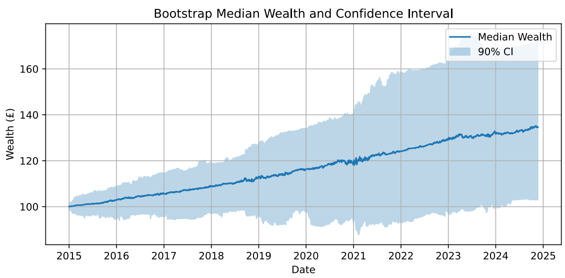



Figure 13: Wealth paths with 90% confidence interval (cost-unaware model, bootstrap).

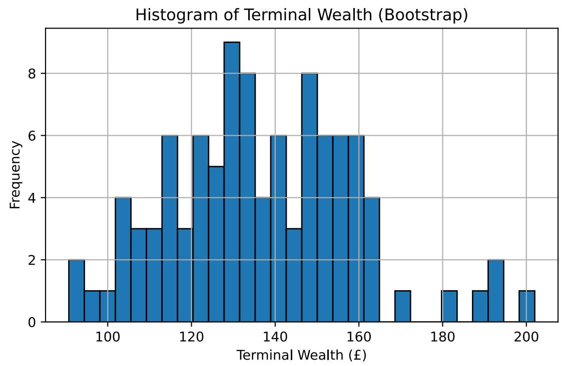



Figure 14: Distribution of terminal wealth (cost-unaware model, bootstrap).

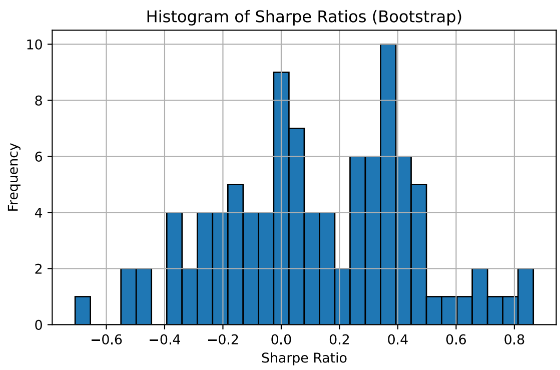



Figure 15: Distribution of Sharpe ratios (cost-unaware model, bootstrap).

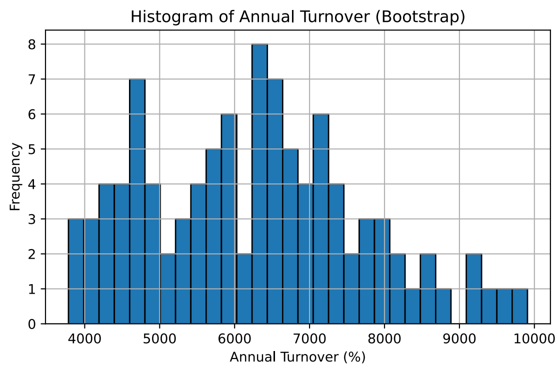



Figure 16: Distribution of annual turnover (cost-unaware model, bootstrap).